



FITTER-EU

D3.2

Report on the identification of DGs



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Abbreviations

List of abbreviations and acronyms featured throughout the document:

CAES: Spanish Energy Saving Certificates

CAP: Common Agricultural Policy

DGs: Disadvantaged Groups

DE: Germany

DoA: Description of Action

ES: Spain

ETS: Emission Trading System

ETUI: European Trade Union Institute

EU: European Union

EUT: Eurecat

EV: Electric Vehicles

HU: Hungary

IE: Ireland

IT: Italy

IZ: Inclusionary Zoning

KPIs: Key Performance Indicators

MaaS: Mobility-as-a-service

MLP: Multi-Level Perspective

NA: North Africa

OECD: Organisation for Economic Co-operation and Development

PESTLE: Political, Economic, Social, Technological, Legal and Environmental

PT: Portugal

RE: Renewable Energy

SDGs: Sustainable Development Goals (SDGs).

TJ: Terajoules

TT: Twin Transition

Summary

Deliverable D3.2 presents the results of an extensive foresight exercise designed to explore plausible futures for the twin transition in four key sectors: agriculture and food, energy, transport and mobility, and housing and the built environment. The work aims to inform policy and innovation strategies that ensure these transformations are both sustainable and socially inclusive, with particular attention to the impacts on disadvantaged groups.

The methodological approach combined the Multi-Level Perspective (MLP) framework with a participatory, multi-method process involving active consulting of stakeholders, semi-structured interviews, Delphi surveys, and analysis of national strategy plans. The MLP allowed transitions to be examined across three interrelated levels: niche innovations, dominant regime structures, and broader landscape trends, enabling a holistic view of how multiple layers of changes interact over time.

The resulting scenarios describe sector-specific 2050 futures that integrate environmental imperatives, digital transformation, market forces, and social dynamics. Across all sectors, common themes emerge: the critical role of policy in guiding transitions, the interplay between technological innovation and accessibility, the need for workforce adaptation, and the persistent risk of socio-economic stratification if equity measures are not integrated in transition planning. The analysis also identifies disadvantaged groups in each sector (such as low-income households, rural communities, and workers in vulnerable occupations) and outlines how their exposure to risks may evolve.

This deliverable is a key pillar for subsequent FITTER tasks and work packages, which, thanks to the tools for reflection established through scenarios, allow stakeholders to anticipate systemic changes and design targeted interventions, which will be key to feed the project's digital platform.

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1 Introduction

1.1 Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. Through the appliance of a co-creation methodological approach, the ecosystem aims to enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curve potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem incorporates wide platform connectivity capability, which helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group (CG). The group includes key national and pan-European members such as policymakers and civil society organizations.

1.2 Purpose and scope of the Deliverable

The present deliverable (D3.2) outlines the scenario-building process and preliminary identification of Disadvantaged Groups (DGs) in the context of the Twin Transition (TT). Elaborated under T3.2, its main purpose is to present a first set of exploratory future scenarios for the year 2050, developed across four key sectors: agriculture and food, energy, transport and mobility, and housing and built environment. These scenarios were co-designed using a methodology that combines literature review, semi-structured expert interviews and Delphi surveys.

The deliverable builds upon the work conducted in D2.1 and D3.1 and operationalises a scenario-building methodology guided by the Multi-Level Perspective (MLP) framework and focused on anticipating how structural changes, driven by the green and/or digital transitions common policy objectives, may reshape sectoral systems and affect different population groups unequally.

Therefore, the scope of D3.2 is twofold:

1. First, to document the creation of sector-specific 2050 scenarios grounded in both expert knowledge and consensus. These scenarios are not predictive but exploratory and aim to capture a range of plausible futures shaped by key drivers and uncertainties around the fulfilment of a policy objective.
2. Second, to provide an initial identification of population groups that may be structurally disadvantaged under each sector-specific 2050 scenario.

The insights from D3.2 will feed into the design of future project tasks including T3.3, T4.1 and T4.2. D3.3 will follow the future scenarios description with the characterization of Disadvantaged Groups (DGs), hazards and negative impacts.

1.3 Structure of the Deliverable

This deliverable is structured as follows:

- Section 2 describes the methodological framework, including an overview of scenario-building methodologies, the Multi-Level Perspective (MLP), and the definition of a common policy objective across sectors for the 6 FITTER-EU member states.
- Section 3 details the semi-structured interview process: the script, the expert nomination process, the sample description, and interview format.
- Section 4 explains the Delphi survey design, its structure and the expert sample during the various rounds.
- Section 5 presents the procedure for analysis, describing the 6-steps approach including the different interview and Delphi survey rounds and how the conceptualisation of future scenarios was elaborated.
- Section 6 outlines the scenario building procedure results, with separate subsections for each sector: Housing and Built Environment, Agriculture and Food, Energy, and Mobility and Transport. Each includes the output from interviews, Delphi surveys, and analysis of results.
- Section 7 presents the scenario building results for each sector following the methodology and procedure for analysis described. For each sector, 3 different scenarios have been made.
- Section 8 discusses the limitations of the work conducted.
- Section 9 summarises the key conclusions and implications of this deliverable for the FITTER-EU future work.
- Section 10 contains annexes with supplementary material such as full interview and Delphi survey scripts.

2 Methodological framework

2.4 Introduction to scenario-building methodologies

Scenario-building is a structured approach aimed at exploring how the future may unfold under conditions of uncertainty and complexity. Rather than attempting to predict what will happen, scenarios are used to anticipate a range of plausible futures, allowing decision-makers to test assumptions, prepare for changes and enhance long-term strategic thinking. This methodology is particularly relevant when addressing systemic transformations, such as those associated with the green and digital transitions, where conventional forecasting tools often fall short due to the interaction of numerous dynamic variables.

As the OECD (2025) highlights in the *Foresight toolkit for resilient public policy*, scenario-building helps stakeholders to move beyond linear projections by constructing narratives that examine how different disruptions and trends might interact. This approach fosters ‘resilience’ by enabling the design of public policies that are robust under diverse future conditions, considering for instance the consequences of policy impact to Disadvantage Groups (DGs).

In this context, scenarios are best understood not as singular predictions, but as internally consistent, contrasting stories that describe how the world might evolve from the present to a specific time horizon (e.g., 2030, 2040 or 2050). Each scenario is shaped by driving forces and assumptions and serves as a reference point for strategic reflection.

From a methodological standpoint, van den Berg et al. (2021) propose a multi-layered framework for keeping scenarios up to date over time. They distinguish between four layers: framework, storylines, industry-specific fundamentals, and numbers, and advocate for systematically integrating new information according to its level of uncertainty and structural impact. This dynamic updating process helps preserve the relevance and plausibility of scenarios in rapidly changing environments.

Similarly, Meinert (2014) under the European Trade Union Institute (ETUI) describes scenario-building as a participatory and narrative-driven learning process, grounded in systemic thinking and stakeholder engagement. Scenarios are built around critical uncertainties and reflect diverse perspectives, enabling shared understanding and collective anticipation. The method typically involves steps such as defining the focal question and time horizon, identifying major drivers and uncertainties, clustering them into scenario logics, and elaborating coherent narratives. These narratives then serve as the foundation for strategic discussion and policy design.

Overall, scenario-building enables actors to expand their perception of possible futures, challenge dominant assumptions, and co-create strategic responses. This methodology strengthens the capacity to act under uncertainty, not by eliminating it, but by making it more visible and manageable.

2.5 Multi-Level Perspective (MLP)

The methodological approach selected in FITTER-EU is the **Multi-Level Perspective (MLP)**. This framework conceptualises socio-technical transitions as dynamic, multi-dimensional processes unfolding across three interrelated levels: niches (micro-level innovation spaces), regimes (meso-level, dominant institutions, technologies, and norms) and landscapes (macro-level structural trends and shocks) (Rip & Kemp, 1997; Geels, 2002, 2005).

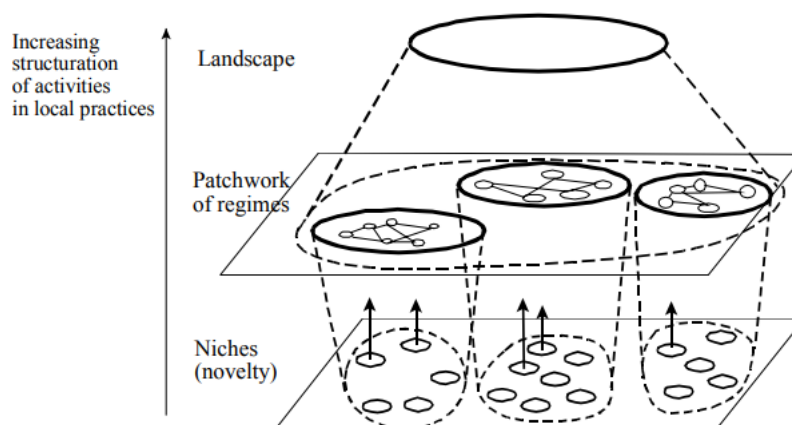


Figure 1: Hierarchy (micro-meso-macro) of MLP elements. Source: Geels (2002)

By structuring transitions along these levels, the MLP enables a deeper understanding of how disruptive innovations (e.g., smart energy systems, green hydrogen, AI-enabled mobility), existing institutional frameworks, and broader contextual forces (e.g., climate crisis, demographic change) may interact in shaping possible futures. It also helps anticipate how these transitions might reproduce or transform structural inequalities. This methodology has been employed in the sectors targeted by the FITTER-EU project, such as agriculture and food (El Bilali, 2019) and mobility and transport (Geels, 2012).

In scenario-building, the MLP may serve as a diagnostic and analytical tool to construct exploratory scenarios, that is, plausible stories that explore a range of future developments without prescribing preferred outcomes (Börjeson et al., 2006). This type of scenario is particularly useful in anticipating new configurations of risk or opportunity that might emerge in a decarbonising and digitalising Europe (twin transition).

The key elements to define are in the first place, the **landscape**, referring to environmental, social and economic global trends that influence and pressure the regime. In the second place, the **regime**, that refers to the mainstream society supported by social norms and already integrated systems (preserving the status quo) and finally, the **niches**, that consists of new ideas and innovations that can challenge the preexisting regime (El Bilali, 2019).

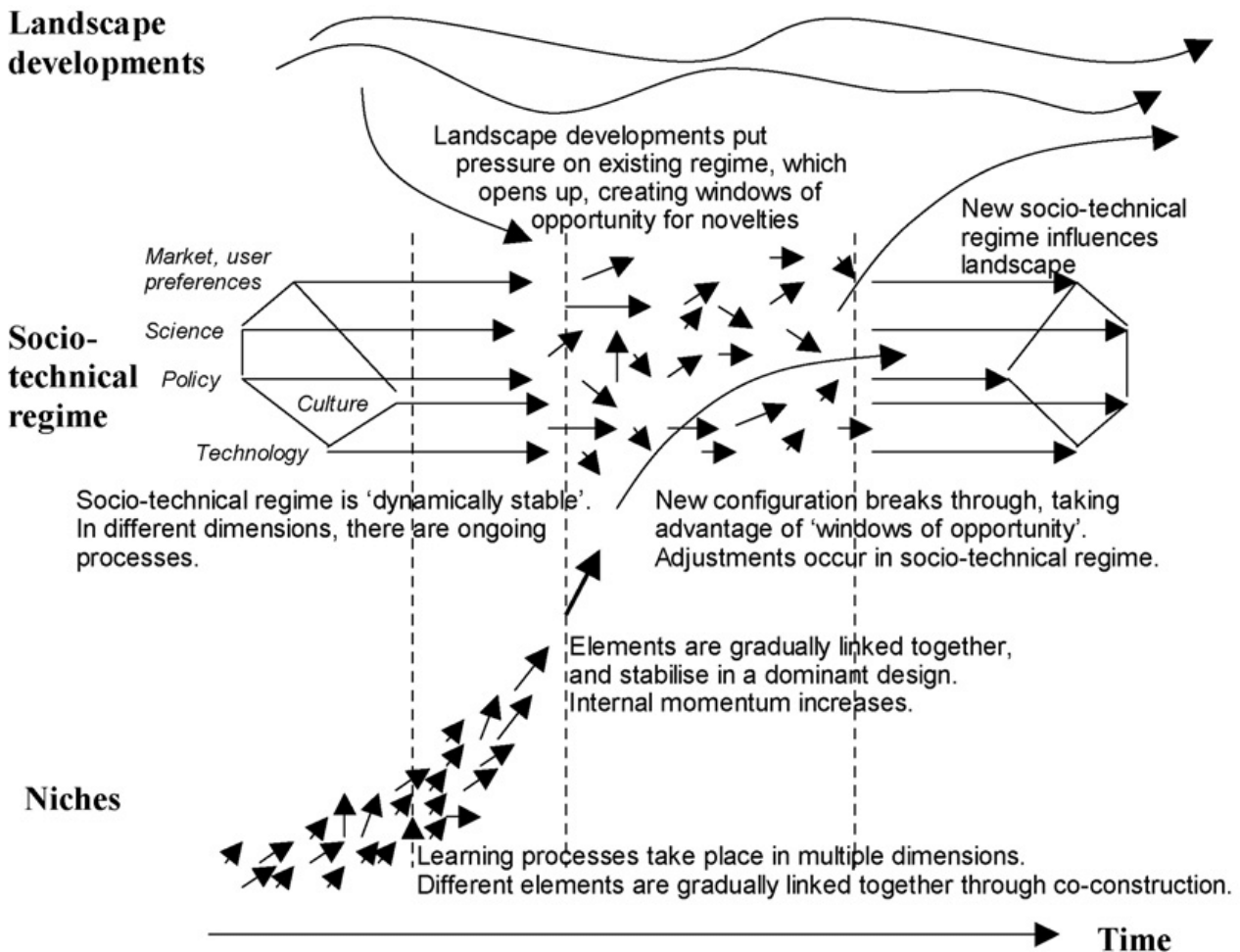


Figure 2: A dynamic multi-level perspective on system innovations. Source: Genus & Coles (2008), adapted from Geels (2004)

Therefore, exploratory scenarios based on MLP allow for examining:

1. Whether niche innovations (such as community-owned energy systems) can scale up and challenge dominant regimes.
2. How regime-level policies might reinforce access barriers for DGs. Regimes' nature is socio-technical, referring to the rules established from current engineering practices, product characteristics and skills, and the interlinkage to social groups, from the same engineers to other social groups, as may be DGs receiving the impact of the regimes (Geels, 2006). Normally, regimes change incrementally and rarely undergo transformation or reconfiguration (El Bilali, 2019).
3. Or how landscape shocks, like geopolitical crises or extreme weather, might destabilise existing systems in ways that disproportionately affect DGs. However, it may create opportunities too in some regimes for some social groups (El Bilali, 2019).

As Andreescu et al. (2013) argue, such scenarios foster reflexivity among policymakers by creating complex futures in which vulnerabilities can evolve or become newly visible. According to El Bilali (2019), within the context of MLP, we need to understand transition as the shift from one to another socio-technical regimes, as a result of the interaction between landscape-regime-niche.

This makes the MLP a particularly appropriate lens for addressing the objectives of FITTER-EU: to anticipate how structural inequalities may shift under future twin transition pathways and to support forward-looking policy design that leaves no one behind.

2.6 Common policy objective within sectors

In order to create scenarios on each one of FITTER sectors (housing and building, mobility and transport, agriculture and food and energy) it was necessary to narrow the scope of the analysis. From an MLP approach, the niche-innovations that could lead to twin transition on each one of those sectors, alongside the evolution of each one of the socio-technical regimes, could tend to an enormous number that could suppose serious limitations for the FITTER analysis. For instance, analyzing each one of the potential innovations that could exist in Europe in energy by 2050 could suppose a whole separate project and research; and therefore, the Consortium agreed that this could not be plausible to do for the four sectors of analysis.

With the objective of narrowing the scope of analysis, first we divided the sectors among countries. Therefore, the Consortium decided the following division:

1. Energy: All participant countries
2. Mobility and transport: Hungary and Spain
3. Building and housing: Germany and Portugal
4. Agriculture and food: Ireland and Italy

Thus, meaning that each country had to analyze 2 sectors. Then, in strict coordination with T3.1, we developed a complete overview of each one of the national strategy plans that each FITTER participant country has for each of their sectors of analysis. The goal was to find line of action that met 3 requisites:

1. The line of action was in line with European twin transition macro-strategies
2. The line of action was present in the sector national strategy plan of Country A
3. The line of action was present in the sector national strategy plan of Country B

After running this analysis, the Consortium found the following lines of action:

Countries	Sector	Line of action	Rationale
All	Energy	Transition towards Renewable Energy	D3.1, pp.8-9
Hungary and Spain	Mobility and transport	Transition to electric vehicles	D3.1, p.41
Germany and Portugal	Housing and built environment	Adaptation of housing to climate change	D3.1, pp.59-61
Ireland and Italy	Agriculture and food	Digitalization of agricultura	D3.1 pp.87-91

Table 1: Countries, sectors and lines of action

3 Semi-structured interviews

3.1 Script and structure

Built upon the Multi-Level Perspective (MLP), the script designed for the semi-structured interviews aimed to explore how socio-technical transitions unfold across three levels (niches, regimes and landscape trends) under the scope of a policy goal in a sector.

The interviews were structured in three main thematic blocks aligned with the logic of exploratory scenario-building: Horizon scanning (What?), scenario exploration (How?) and implications (Who?) script.

These blocks enable experts to identify key drivers and uncertainties, reflect on possible future developments, and assess how these might differently affect population groups across time horizons (2030, 2040, 2050).

The design of the script was informed by established foresight methodologies, particularly the OECD Foresight Toolkit (2025) and the ETUI scenario-building field manual (Meinert, 2014). The questions were open-ended to stimulate forward-looking and divergent thinking, going beyond linear projections to uncover alternative development paths within sectors.

Each interview begins with a brief introduction to the FITTER-EU project and the goals of the discussion, followed by questions tailored to the specific sector of expertise (agriculture, energy, transport & mobility, or housing and building). For each sector, the structure is as follows (note that the questions for all the sectors are included in the subsection 0 of the Annex:

1. **Horizon scanning** focuses on identifying key structural assumptions (givens) and potential disruptions (uncertainties). Experts are prompted to consider technological, political, economic, social, and environmental (PESTLE) factors that may influence the sector up to 2050. Particular attention is paid to which trends can be considered predictable and which remain deeply uncertain. The addressed questions for the sector of agriculture are the following:
 - a. *Taking the EU current technological context into account – can you list please - what factors may be still important with regard to the digitalisation of agriculture up to 2050?*
 - b. *Taking the EU current political context into account - what factors are important with regard to the digitalisation of agriculture up to 2050?*
 - c. *Taking the EU current economic context into account - what factors may be still important with regard to the digitalisation of agriculture up to 2050?*
 - d. *Taking the EU current environmental context into account - what factors may be still important with regard to the digitalisation of agriculture up to 2050?*
 - e. *Taking the EU current social context into account – what factors may be still important with regard to the digitalisation of agriculture up to 2050?*
 - f. *Of all of the mentioned factors - which do you believe that have a high influence on the digitalisation of agriculture and we can be reasonably certain they will occur or develop in predictable ways by 2050?*
2. **Scenario exploration** seeks to map the interactions between drivers and uncertainties, inviting participants to speculate on how certain trends might amplify each other or generate new risks or opportunities. This stage connects the micro (niche-level innovation), meso (regime-level

structure), and macro (landscape-level disruption) components of MLP thinking. The questions within the agriculture sector were the following:

- a. *Among the remaining factors mentioned, or any new ones you wish to introduce, which do you consider most uncertain for the digitalisation of agriculture? (DRIVERS)*
- b. *In other words, what assumptions for the digitalisation of agriculture might no longer be valid in 2050*

3. **Scenario implications** asks experts to reflect on who might “win or lose” under different futures. Interviewees are encouraged to consider differentiated impacts on structurally vulnerable population groups, such as low-income households, rural communities, or ageing populations. This final block also explores potential policy responses or institutional innovations required to ensure socially just transitions. The two questions for the agriculture experts were:

- a. *Who would be the winners and losers in a highly digitalized agricultural future (in other words; looking at multiple intersected socio-economic characteristics, which population cohorts will benefit the most and which ones will be negatively impacted)?*
- b. *What specific policies, programs or initiatives would need to be newly created and/or adjusted if this scenario was to play out?*

3.2 Nomination of experts and sample description

The nomination process for the semi-structured interview experts was led by Eurecat (EUT), in close collaboration with the national case study leaders. EUT was responsible for communicating the required sectoral profiles and areas of expertise needed for the interviews, based on the methodological objectives defined under Task 3.2 and the scenario-building framework guiding the project.

Each national case study team was asked to identify and nominate experts with proven experience in one of the four sectors under analysis and with knowledge of the dynamics and challenges posed by the green and digital transitions. To facilitate outreach, EUT provided standardised email templates that national teams could use to introduce the project and explain the purpose of the interviews. In many cases, national partners shared the direct contact details of the nominated experts with EUT, who subsequently took the lead in coordinating the interviews and arranging the online sessions.

This collaborative and decentralised approach made it possible to account for both national and sector-specific particularities, while ensuring consistency across the overall scenario-building process. A shared tracking document accessible to all project partners was used to monitor the progress of expert outreach and the fulfilment of key performance indicators.

Table 2 presents the 24 experts interviewed for this study. While names and surnames have been removed to ensure privacy, a pseudonymization process has been applied, preserving information on role, gender, organization, and geographical scope to maintain rigor. The last column (#code) indicates the interview reference number, which will be used in section 6 to summarise the interviews. This numbering is non-sequential, as it reflects the coding assigned during the analytical process for each round rather than the chronological order of the interviews

Housing and built environment				
<i>Gender</i>	<i>Role</i>	<i>Organization</i>	<i>Geographical scope</i>	<i># Code</i>
Female	Senior researcher and architect	RMIT Europe	EU	1

Male	Business developer – construction sector	Eurecat	ES	2
Male	Researcher – Climate-Adaptive Construction	Technical University of Munich	DE	20
Agriculture				
<i>Name and surnames</i>	<i>Role</i>	<i>Organization</i>	<i>Geographical scope</i>	<i># Code</i>
Male	Senior researcher and farmer	Eurecat	ES	3
Male	Regional startup manager	EIT Food	EU	4
Male	Director	BETA Tech center	ES	5
Male	Researcher	University of Lleida	ES	11
Male	Technology & Innovation Executive	Irish Farmers' Association	IE	12
Female	Research officer	Teagasc Climate Centre	IE	13
Male	Farmer, researcher & Head of energetic use at the Chair of Marketing and Management of Biogenic Resources	University of Applied Sciences Weihenstephan-Triesdorf	DE	14
Male	Researcher and president	Institute of Agrifood Research and Technology / Central Board of Users of Baix Ter	ES	23
Energy				
Male	Business developer – energy sector	Eurecat	ES	6
Female	Public policy & ICT regulation advisor	Catalan regional government	ES	7
Male	Senior Research Scholar	Energy Systems Modelling Analytics (ESMA)	IE	17
Male	Researcher & Head of energetic use at the Chair of Marketing and Management of Biogenic Resources	University of Applied Sciences Weihenstephan-Triesdorf	DE	18

Male	Senior Postdoctoral Fellow	University College Cork	IE	19
Female	Senior Project manager	Netzwerk Energiewende Jetzt e.V.	DE	21
Female	Coordinador, expert on photovoltaic	Posteo	DE	22
Mobility & transport				
Male	Business developer – mobility sector	Eurecat	ES	8
Laia Garriga	Business developer – mobility sector	Eurecat	ES	9
Male	Leadership manager	EIT Mobility	EU	10
Male	Coordinator	Ecomod association	PT	15
Male	Researcher	Politecnico di Torino	IT	16
Male	CEO	Nemi	ES	24

Table 2: First round of interviews - Nomination of experts

Table 3 summarises the distribution of the 24 interviews (80% of the target) conducted by country and sector, providing an aggregated view of the geographical and thematic coverage achieved. Spain recorded the highest participation with 11 interviews (46.15% of the total), covering all four sectors. Germany followed with 5 interviews (19.23%) distributed across Agriculture, Housing and Built Environment, and Energy, while Ireland contributed 4 interviews (15.38%) focused on Agriculture and Energy. Italy and the EU category each accounted for 2 interviews (7.69%), and Portugal contributed 1 interview (3.85%) in the Mobility & Transport sector.

From a sectoral perspective, Agriculture was the most represented, with 8 interviews (33.33%), followed by Energy with 7 interviews (29.17%), Mobility & Transport with 6 interviews (25%), and Housing and Built Environment with 3 interviews (12.5%). This distribution shows a relatively balanced coverage across sectors, while also reflecting the stronger representation of certain countries (particularly Spain) in the sample. Despite not reaching the KPI of 30 interviews, the results provide valuable insights to detect trends and identify relevant variables for the scenario-building process.

Overall, the sample included 6 female experts (25%) and 17 male experts (70.83%) across the four sectors.

Country/ Sectors	Number of experts (interviews)				Total per country
	Mobility & Transport	Agriculture	Housing and built environment	Energy	
ES	3	4	2	2	11 (46.15%)
IT	1			0	1 (7.69%)
HU					0

IE	2		2	4 (15.38%)
DE	1	1	3	5 (19.23%)
PT	1			1 (3.85%)
EU	1	1		2 (7.69%)
Total per sector	6 (25%)	8 (33.33%)	3 (12.5%)	7 (29.17%)
				24

Table 3: Experts per country and sector (Interviews)

3.3 Interviews format

Each interview lasted approximately 60 minutes and was conducted in an online format to ensure flexibility and cross-country participation. All interviews followed a semi-structured protocol and were carried out by members of the EUT research team in English language. Exceptionally, interviews were conducted in Spanish or Catalan when both interviewers and interviewees shared the same native language.

The interviews were conducted by two interviewers, each of whom specialised in two sectors, ensuring thematic coherence and depth of understanding during the sessions.

All contributions shown in the results section are anonymised to protect the identity of the participants and to encourage open and forward-looking dialogue. The information collected during the interviews was used to inform both the development of the sectoral scenarios and the formulation of the Delphi survey.

4 Delphi survey

The **Delphi method** is a structured, iterative technique for eliciting and refining expert judgment, particularly useful in contexts where evidence is incomplete and ambiguity is high. This is the reality for future foresight, where there is deep uncertainty. Originally formulated in the late 1940s-early 1950s (Khodyakov et al., 2023) and refined by Linstone and Turoff (1975) the method has evolved into a foundational tool relevant for policy foresight and anticipatory governance. It allows for the systematic collection of expert knowledge through multiple rounds of consultation, ensuring anonymity-controlled feedback and statistical aggregation of responses (Belton et al., 2019).

In contrast to traditional group-based discussions, which can be biased by dominant voices or institutional hierarchies, the Delphi process supports the emergence of collective intelligence by suppressing interpersonal dynamics and enabling participants to reflect and revise their positions in response to group feedback. Through this process, the method fosters what Rowe and Wright (2001, cited in Schmalz, Spinler & Ringbeck, 2020) describe as “group response”, offering a robust alternative to unstructured consultation practices.

Delphi surveys are particularly well suited for the construction and validation of exploratory scenarios (van den Berg et al., 2021). According to Nowack, Endrikat & Guenther (2011), it is a powerful complement to scenario building as it fosters structured creativity and imaginative thinking “thinking the unthinkable”. Very importantly, it allows for controlled convergence of expert judgments, while still making space for persistent dissensus, which can be just as informative in scenarios addressing future uncertainties.

Within the scope of WP3, the Delphi survey plays a central methodological role. It is applied in a sector-specific format, meaning that each round is deployed independently for experts in the four target sectors

of FITTER-EU (e.g., energy experts only answer energy questions). A total of two rounds were carried out, adjusted by three rounds of interviews.

4.1 Design and structure

The design of the first-round Delphi survey was directly informed by insights gathered through 10 semi-structured expert interviews, which explored disruptive trends, critical uncertainties, and socio-technical dynamics relevant to each domain. These interviews followed the MLP-based scenario logic and served as the foundation for formulating the forward-looking statements evaluated by Delphi participants. The software used for the Delphi survey was [qualtrics](#) and the language was English. Only under national use case leader petition, it was translated to Spanish and Hungarian aiming at improving engagement rates.

Following the 10 interviews for all sectors, the EUT researchers grouped the main trends, the potential advantaged groups 2050 and the disadvantaged groups.

All statements in the Delphi survey were formulated in the future tense and written as plausible, self-contained projections of what might happen by 2050 within each sector. This stylistic choice responds to the exploratory nature of scenario building and invites participants to assess unfolding trajectories rather than current conditions. The phrasing avoids conditional or speculative language (e.g., “might” or “could”) and instead uses affirmative constructions (e.g., “By 2050, X will...” or “In 2050, Y will be...”) to prompt more decisive expert reflection.

Each statement is assessed according to three dimensions:

1. Likelihood of the scenario occurring in Europe by 2050: on a 1-to-5 scale where 1 was very low and 5 very high.
2. Impact of the scenario, particularly in terms of its effects on the EU sector: on a 1-to-5 scale where 1 was very low and 5 very high.
3. Qualitative comment (optional), where respondents can nuance their answer, suggest conditions, or highlight contextual considerations.

4.2 Sample of experts

Table 4 presents the distribution of the expert sample consulted in the Delphi survey, classified by country and sector. A total of 62 experts participated, exceeding the initial target of 50. The largest share of experts were from Spain (22; 35.48%), followed by Portugal (12; 19.35%) and Italy (10; 16.13%). In terms of sectoral representation, the energy sector accounted for the highest number of specialists (20; 32.26%), followed by mobility & transport (19; 30.65%), agriculture (12; 19.35%) and housing and built environment (11; 17.74%).

The sample also included experts with an EU-wide scope (2; 3.23%), providing cross-cutting perspectives.

Regarding gender distribution, 33 participants identified as male (55.00%), 26 as female (43.33%), and 1 preferred not to disclose this information (1.67%).

<i>Country/ Sectors</i>	<i>Number of experts (interviews)</i>				
	Mobility & Transport	Agriculture	Housing and built environment	Energy	Total per country
<i>ES</i>	8	4	5	5	22 (35.48%)
<i>IT</i>	1	2	1	6	10 (16.13%)

<i>HU</i>	1	0	0	1	2 (3.23%)
<i>IE</i>	3	2	0	1	6 (9.68%)
<i>DE</i>	3	1	2	2	8 (12.90%)
<i>PT</i>	1	3	3	5	12 (19.35%)
<i>EU</i>	2	0	0	0	2 (3.23%)
<i>Total per sector</i>	19 (30.65%)	12 (19.35%)	11 (17.74%)	20 (32.26%)	62

Table 4: Experts per country and sector (Delphi survey)

5 Procedure for analysis

The scenario-building analysis followed a six-step process integrating both quantitative and qualitative evidence to produce coherent and forward-looking narratives for 2050 (Figure 3):

1. First round of interviews
2. First round of Delphi survey
3. Second round of interviews
4. Second round of Delphi survey
5. Third round of interviews
6. Elaboration of future scenarios

Each step is described in detail in the following subsections (5.1–5.6). The process combined deductive coding of expert interviews, based on categories from the MLP model (niche, regime, landscape) and the FITTER conceptual framework (vulnerability dimensions and disadvantaged groups), with statistical analysis of Delphi survey results, ensuring both depth and robustness in the resulting scenarios

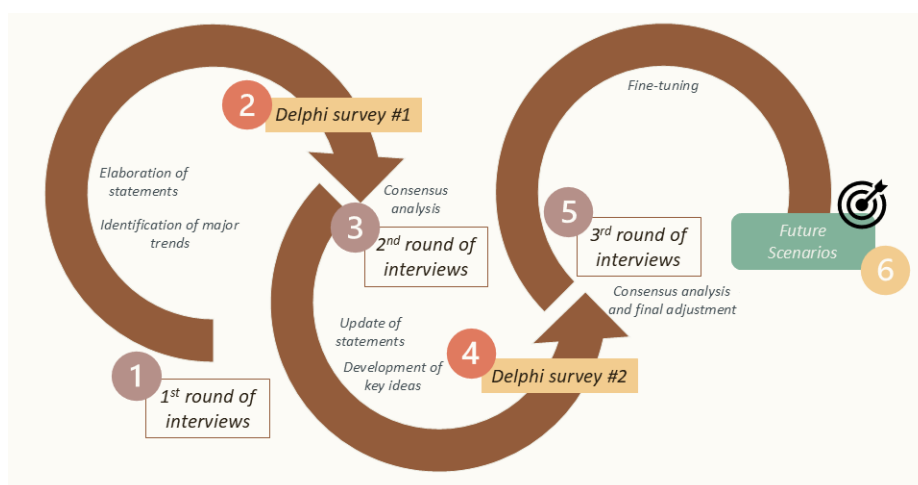


Figure 3: Scenario building procedure of analysis

A total of 86 experts contributed to the process through interviews and/or the Delphi survey. Table 5 presents the aggregated distribution of experts by country and sector.

At the country level, Spain recorded the highest participation, with 33 experts (38.37%) covering all four sectors: mobility & transport (12), agriculture (8), housing and built environment (6) and energy (7). Italy contributed 11 experts (12.79%), mainly in energy (7), followed by mobility & transport (2), agriculture (2) and housing (1). Germany and Portugal each contributed 13 experts (15.12%), with representation across all sectors. Ireland accounted for 10 experts (11.63%), distributed across agriculture (4), mobility & transport (3) and energy (3). Hungary contributed 2 experts (2.33%), one in mobility & transport and one in energy. Finally, the EU category included 4 experts (4.65%), concentrated in mobility & transport (3) and agriculture (1).

From a sectoral perspective, energy was the most represented, with 28 experts (32.56%), followed by mobility & transport with 26 (30.23%), agriculture with 20 (23.26%) and housing and built environment with 13 (15.12%). This distribution shows a relatively balanced coverage, with a stronger representation in the energy and mobility sectors.

Regarding gender distribution in this total sample, 33 participants identified as male (55.00%), 26 as female (43.33%), and 1 (1.67%) preferred not to disclose this information.

<i>Country/ Sectors</i>	<i>Number of experts (Delphi survey and interviews)</i>					
	Mobility & Transport	Agriculture	Housing and built environment	Energy	Total	per country
<i>ES</i>	12	8	6	7	33	(38.37%)
<i>IT</i>	2	2	1	7	11	(12.79%)
<i>HU</i>	1	0	0	1	2	(2.33%)
<i>IE</i>	3	4	0	3	10	(11.63%)
<i>DE</i>	3	2	3	5	13	(15.12%)
<i>PT</i>	2	3	3	5	13	(15.12%)
<i>EU</i>	3	1	0	0	4	(4.65%)
<i>Total per sector</i>	26 (30.23%)	20 (23.26%)	13 (15.12%)	28 (32.56%)	86	

Table 5: Total experts per country and sector (Delphi survey and interviews)

5.1 First round of interviews

The organization of 11 interviews was carried out with experts in order to gather key trends and variables to create the first statements of the Delphi survey. The qualitative analysis of expert interviews conducted within the FITTER-EU project was based on a deductive coding approach, using pre-established analytical categories derived from both the MLP model (niche, regime and landscape) and the project's conceptual framework (vulnerability dimensions and DGs).

The interviews were analysed using the ATLAS.ti software¹, which enabled systematic coding and retrieval of insights across transcripts. Literal quotes from participants were tagged and included in the results sections, using a parenthesis () to indicate the expert highlighting a specific one.

¹https://atlasti.com/?x-source=pmax&x-campaign=pmax-en&x-id=21759486841&x-term=pmax-01&utm_source=google&utm_medium=pmax&utm_campaign=21759486841&utm_term=&utm_content=&utm_ad_group=&device=c&placement=&matchtype=&network=x&gad_source=1&gad_campaignid=21759491233&gbraid=0

Based on the insights from the interviews, Delphi survey statements were proposed and iterated in following rounds.

5.2 First round of Delphi survey

In order to evaluate whether a Delphi statement has reached expert consensus, a quantitative threshold based on the distribution of responses was applied. As mentioned for each statement, respondents evaluate two dimensions, likelihood and impact, using a 5-point ordinal scale (*Very low, low, neutral, high, very high*).

To determine consensus for each dimension:

1. A consensus threshold of 60% is applied for the first round: if 60% or more of responses fall within the top two categories (“High” or “Very High”), the statement is considered to have reached positive consensus on that dimension.
2. This is calculated relative to the total number of responses per item and per round.
3. If no single response cluster reaches this 60% threshold, the statement is classified as having no consensus.
4. The same logic applies independently to likelihood and impact, and both are analysed for each statement.

This approach allows the research to be rigorous. It avoids overinterpreting majority trends while still recognising collective alignment when it emerges. Importantly, the method also records the degree of divergence, allowing the project team to detect areas of controversy, which are highly valuable in exploratory scenario building.

An additional indicator, “% Very high among those who agreed”, captures the proportion of highly confident responses within the subgroup that agrees (i.e. those who selected “high” or “very high”). This helps assess not only convergence but also the strength of agreement within the expert group.

Statements that reach consensus in the likelihood dimension are prioritised as robust signals, even if consensus is not reached on impact.

During the second round of consultation, additional inputs were gathered from other experts, and statements that did not reach consensus in the first round were replaced by new ones, in order to reassess the level of expert agreement.

For each question in the Delphi survey, in addition to the quantitative rating, experts were also invited to provide a brief qualitative comment for every statement. These comments allowed respondents to justify their answers, provide additional context or nuance their positions. The qualitative feedback has been analysed and included in the subsections of the Delphi survey per each round and sector in the scenario building procedure (section 6).

5.3 Second round of interviews

Following the same process than the same round, 10 additional interviews were conducted to experts. The results were transcribed and analysis using the same deductive coding procedure. The results are presented in section 3, under the respective subsection.

The rates of engagement by sector were dependent on the availability and willingness of participating of experts. That is why the rates are different.

5.4 Second round of Delphi survey

Statements from the first round with $\geq 60\%$ consensus were reassessed, and those reaching $\geq 75\%$ consensus in this round were considered to show strong agreement. Statements were categorised by their expected effect on the twin transition: positive, negative, or mixed trends, depending on the context.

Statements were then classified according to their expected effect on the twin transition: positive trends supporting both the green and digital dimensions, negative trends potentially hindering them, and in some cases, mixed trends with potential to act in both directions depending on context.

5.5 Third round of interviews

This round aimed to deepen the understanding of perspectives emerging from the second Delphi round. However, in the housing and built environment sector, no third-round interviews could be conducted due to the lack of available experts. This limited the depth of qualitative insights for that sector, and findings should therefore be interpreted with caution.

5.6 Conceptualisation of scenarios

Once the data collection process finished, the next step integrated the selected and interpreted statements into three internally coherent scenario pathways: optimistic, neutral, and pessimistic. Each scenario was written as a stand-alone narrative, covering the interactions between the MLP elements dimensions.

The interpretation focused on describing the developments in a narrative, style. In this stage, evidence from each sectors' interviews was used to nuance the interpretation of the trends, ensuring that the scenarios reflected both the agreed likelihood and impact of each statement and the contextual factors that could influence its realisation.

The optimistic scenario includes only positive trends with strong consensus, describing a future where the twin transition is inclusive and sustainable. The neutral scenario mixes positive trends with evidence of partial or uneven adoption, showing coexistence of advanced and traditional systems. The pessimistic scenario focuses on negative trends with high consensus, highlighting landscape shocks and structural barriers that hinder progress.

A key part of the analysis involved assessing the differentiated effects of each trend on specific population subgroups that may experience disadvantages in certain contexts. Instead of treating disadvantaged groups as a single category, the analysis considered the implications for low-income households, rural communities, renters in inefficient housing, migrants, older adults, people with disabilities, former fossil fuel workers, and digitally excluded citizens. For each subgroup, the analysis explored potential impacts in terms of the three key vulnerability dimensions for FITTER-EU: finance, skills and employment.

The final narratives therefore combine high-consensus trends and contextual qualitative insights to produce realistic and evidence-based foresight visions for each sector. This procedure not only grounds the

scenarios in robust evidence but also ensures they reflect the complexities of the twin transition and its varied impacts across European society.

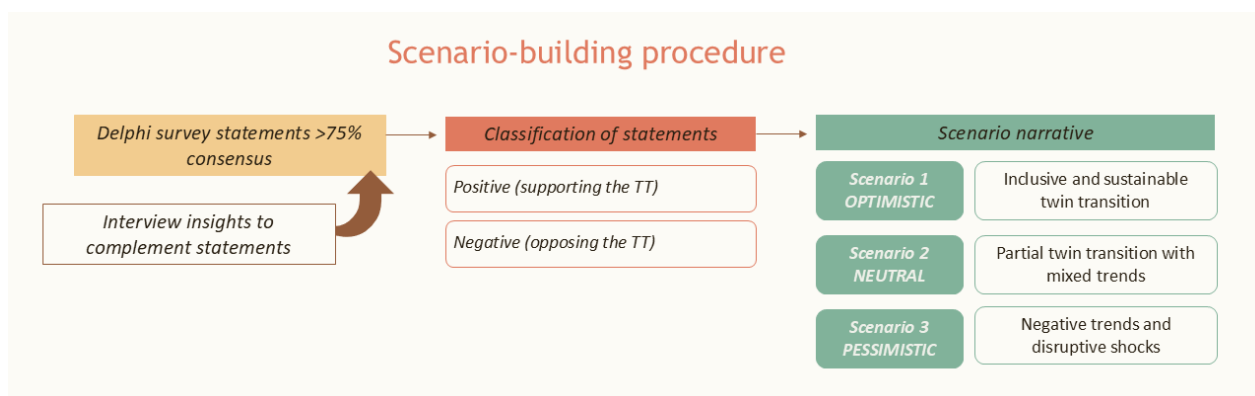


Figure 4: Step 6 – Scenario building procedure

6 Scenario building procedure

For this section, the results of the first 5 steps of the scenario building methodology will be compiled. In the case of interviews, the tables describing the results will be structured in tables. These tables contain the “summary of description” column providing an aggregated overview of the expert’s input, while the “literal citations” column include experts’ quotes in quotation marks and italics, with a number in parentheses at the end (X) referring to the number of interview (See subsection 3.2). Each table includes subcategories coded with an underscore (_) and, in parentheses, the number of citations corresponding to each.

For each round of the Delphi, an elaboration of statements subsection will be included. These were based on the insights gathered from interviews. To ensure clarity and traceability for the reader, the names of the deductive coding categories used, following the MLP framework are included in brackets.

The statements aim to build consensus among experts by questioning regime-related aspects highlighted during the interviews, introducing counter-statements for the niches mentioned, rephrasing key elements, or broadening certain topics to include related issues. In some cases, the same idea is intentionally expressed in different ways to capture nuances and encourage thoughtful responses.

Each Delphi round also includes a subsection analysing the results. The results tables combine quantitative and qualitative dimensions. They present the statement under consideration, followed by the distribution of responses in terms of likelihood and impact, showing both the absolute number of responses and the percentage they represent over the total. This is followed by an indication of whether consensus was reached, along with the exact degree of agreement, based on the threshold defined for that round. Finally, the table includes a synthesis of the qualitative comments provided by experts, which serve to contextualise or nuance the quantitative scores.

6.1 Housing and built environment

6.1.1 First round of interviews

6.1.1.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [12]	Emerging economic niche around building retrofitting and sustainable materials, though its scalability is still uncertain. The idea of massive rehabilitation is seen as a potential future trend for 2050, driven by stricter regulation and incentives. Technified materials (e.g., engineered wood, modular systems) are expected to have more demand if current prices decrease or local supply chains develop. However, current cost barriers make this niche fragile. There's also speculation about future carbon penalties that could shift economic incentives, as currently, greenwashing dominates the discourse of many private construction companies. Innovative economic ideas for 2050 mentioned include: 1. Widespread economic pressure to retrofit due to climate and legal frameworks. 2. Wood as a viable alternative to cement if production becomes localized and modular. 3. Future economies of scale for sustainable materials, currently constrained by cost. 4. Possible emergence of financial penalties or bonus-malus systems to push change.	<i>"Fines will come sooner or later... the private sector is going to move" (2)</i> <i>"There's a pressure context that could make rehabilitation profitable, but it's not yet the case" (2)</i> <i>"The public sector won't be able to cover all retrofitting, someone has to see business in it" (2)</i> <i>"Technified solutions might allow future economies of scale, but this is very complicated"(2)</i>
Niche_Governance [13]	The interviews show a clear emergence of governance innovation in the housing sector, shaped by the growing influence of European frameworks, such as the New European Bauhaus and long-term climate objectives for 2030 and 2050. These new governance approaches are not yet dominant, but experts see them as seeds of transformation. A key idea arising is the need to move away from purely individual responsibility and toward collective governance models. The current overload of administrative procedures and lack of incentives for the private sector show the limitations of the existing system. Experts suggest that by 2050, governance will need to be more participatory, decentralized and locally embedded. This includes proximity mechanisms such as subsidy support, green offices and/or points of energy advisory who act as mediators between citizens and complex policies. Interviewees also stress that new governance models must be more inclusive and stable across political cycles. There is a perceived need to "depoliticize" social governance so that long-term agendas can be maintained regardless of electoral changes.	<i>"You can't leave it (retrofitting) to individuals, that leads to paralysis. We need collective mechanisms (1)"</i> <i>"By 2050, people will know their contact person in the neighbourhood, and political agendas will be more stable" (1)</i> <i>"Governance models need to change (even if it sounds like a cliché): green offices aim to do this... There must be touchpoints with citizens that shift their perspective."(1)</i>

	At the same time, regulatory innovations are already starting to shape the field. These include new laws reducing the percentage of neighbour agreement required for building renovation, the use of sustainability criteria in public procurement, and urban strategies to deconstruct or relocate buildings in high-risk zones. While implementation remains uneven, these developments are seen as precursors of a more mission-oriented governance system for the built environment.	
Niche_Social [7]	Interviewees point to a growing demand from society for radical change in how housing is designed, lived in and understood. There is an emerging social niche rooted in greater citizen awareness and openness to new forms of energy-efficient and climate-adaptive living. However, this momentum still needs to be translated into behavioural change, and experts agree that “social conviction” will be essential if these practices are to become mainstream by 2050. The industrialisation of construction is seen not only as a technological change, but also as a socially transformative process. It is perceived as a way to diversify talent, attract younger generations and increase inclusion in a sector still marked by rigid labour dynamics. This shift is opening the door for new profiles, including women and people with lower physical demands, to participate in building the future of housing. Nevertheless, interviewees highlight a critical gap between technological solutions and user experience. Smart homes, apps, and gamified systems are often too complex or short-lived in their impact. There is a strong call for more intuitive, low-tech or even analogue solutions that meet the needs of different age and interest groups. Training and user support emerge as key drivers of social change. The vision for 2050 includes neighbourhood-based education and hands-on onboarding for new residents, bridging the gap between digital tools and real-world usage. These grassroots forms of engagement may be the key to activating sustained behavioural change culturally rooted in housing practices.	<i>“There’s a missing bridge between design, implementation and users... there’s too much friction between technologies and needs...” (1)</i> <i>“People want in-person training. It works well when new residents receive orientation. Some like apps and gamification, but the effect doesn’t last unless they have intrinsic motivation”(1)</i> <i>“Industrialisation can attract more talent and make the sector more inclusive” (2)</i>
Niche_Technological [12]	Experts describe a dynamic and evolving landscape of technological innovation in the housing sector. New materials and construction techniques are increasingly viewed as promising alternatives to traditional cement-based methods. Wood and ceramics are seen as lower-impact options gaining traction, with engineered wood in particular positioned as a viable substitute that emits half as	<i>“We will not invent new materials, but we will make better use of them and technify more” (2)</i> <i>“Wood is a potential alternative to cement – more engineered, trendier, and with</i>

	much as concrete. While concrete remains dominant, there is an expectation that by 2050 it will become less polluting due to decarbonisation efforts. There is widespread agreement that a massive rehabilitation wave is needed to address the inefficiency of the existing building stock. Some estimates suggest that up to 8 million homes would need retrofitting, yet the process is currently slowed by financial, organizational and labor barriers. Experts envision the industrialization of rehabilitation as a key innovation area, potentially leading to more inclusive and efficient systems, though they also highlight a lack of skilled workforce and user-friendly solutions. Technological niches are also emerging in smart energy management systems, adaptive materials, and hybrid design. Innovations such as decentralized heat pumps, home management systems (HMS), and smart windows are already being tested under EU programmes. However, interviewees note that user interaction with such technologies is inconsistent. Many people misuse or ignore them due to lack of training, maintenance habits or overly complex interfaces. The future of technological change will likely depend on stronger integration between designers, users and system operators. Low-tech innovations are not dismissed: some experts point to analogue solutions like low-energy heaters as powerful in delivering comfort without complexity. Overall, the trajectory of technological innovation in housing hinges on usability, scalability, and systemic coordination.	<i>lower environmental impact” (2)</i> <i>“People only look at the manual when something goes wrong; technologies don’t work well if maintenance is neglected” (1)</i> <i>“Whether retrofitting will work depends on the social use of buildings and not just the technology” (1)</i>
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6.1.1.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [1]	Currently, public housing projects can apply more ambitious sustainability criteria than private developments. Political institutions sometimes use public procurement to test or advance stricter environmental standards.	<i>“In public promotion they dare to do more than in private. Example: social housing with well-defined criteria.” (2)</i>
Regime _ Economic [2]	The construction sector remains highly dependent on cement, which is responsible for a significant share of emissions. Although decarbonisation efforts are beginning, current practices are still	<i>“The construction sector is responsible for 36% of total emissions; 7–8% of global construction emissions come from cement (roads and buildings)”. (2)</i>

	shaped by cost-efficiency and large-scale material use.	"Cement companies are doing decarbonisation by injecting it..." (2)
Regime _Social [1]	Users maintain old habits even after building retrofits, and this is a problem, as energy rehabilitation requires not only new technologies but also behavioural change, which current social practices often resist.	"People keep living as they did before, even if the building has changed. Energy rehabilitation needs not only technology, but also a change in behaviour" (1).
Regime _Technological [2]	Technological change is occurring within the dominant regime, especially through decarbonisation efforts in cement production and large-scale renovation programmes in Europe. These projects are improving building conditions and reducing energy costs, but they operate within existing systems rather than transforming them.	"Many European projects are retrofitting housing stock: no damp issues, lower energy bills, clear user satisfaction" (1) "Cement companies are trying..." (2)

6.1.1.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [5]	Experts highlight that political instability, due to fossil fuel dependence or shifting geopolitical alliances may generate external shocks that affect regulation, investment and housing policy. While not guaranteed, these disruptions could reshape the policy environment. European directives might evolve reactively in response to future crises.	"Each revision of European regulation is shaped by war or geopolitical issues" (2)
Landscape _Economic [2]	Interviewees mention that a possible rise in material costs and energy prices, driven by resource scarcity or external market instability could challenge the affordability of sustainable housing. If such a trend consolidates, it may affect construction dynamics and delay innovation, but it is not certain and will depend on global economic trajectories.	"It would be a threat to allow again the freedom to build with no restrictions" (2) "Materials may get scarcer and more expensive... this would increase prices of sustainable housing" (2)
Landscape _Social [2]	Potential social resistance to climate action, including denialist narratives could undermine European directives and undo all the progress made. Another possible pressure is the persistence of individualist culture, which could	"Climate change denialism would dismantle European directives" (2)

	weaken collective responses needed for large-scale retrofitting.	
Landscape _Environmental [3]	The Mediterranean region could experience growing exposure to climate change impacts. This may create external pressure to improve building efficiency and climate comfort, especially in areas more vulnerable to flooding or heat. However, the degree and timing of these impacts remain uncertain and unevenly distributed.	<i>"Relevant climate change: in the Mediterranean, we haven't cared enough about energy efficiency or climate comfort" (2)</i> <i>"Climate risk vulnerability depends on location (e.g., floodplains, neighbourhoods)" (2)</i> <i>"Materials are scarcer and more expensive... this raises prices (e.g., heating/cooling)" (2)</i>

6.1.1.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [1]	Construction workers are currently in a structurally vulnerable position due to exposure to climate risks, poor ergonomic conditions, and the sector's low attractiveness for new talent. This limits their resilience and adaptability in the face of sectoral transformation.	"Construction workers are DGs, exposed to climate risks, poor ergonomics, low talent attraction" (2)
Vulnerability_Finance [6]	Financial vulnerability is one of the most critical barriers to sustainable housing. Low-income groups cannot afford efficient buildings or rehabilitation, and there is insufficient supply of social housing. Retrofitting costs are high (20k–40k per household), and many buildings from before the 1980s lack basic insulation. Rising energy costs deepen the gap.	"Vulnerable groups can't access social housing; there's not enough supply (2)."
Vulnerability_Skills [3]	There is a lack of skilled labour in construction, especially among older professionals who are retiring. Moreover, many people (especially older adults) struggle with new technologies due to cultural habits, digital illiteracy, or lack of accessible guidance, creating barriers to effective use of smart or energy-efficient systems.	"The sector is losing skilled labour; people don't know how to install ceramics, concrete is less precise." (2) "Older people are vulnerable to new technologies." (1) "There's no workforce to carry out the rehabilitation" (2).

6.1.1.5 Disadvantaged groups (DGs)

According to the interviews, the DGs are the ones at risk of exclusion from climate-adaptive housing. The experts point out that the most vulnerable individuals are those unable to afford sustainable buildings or access public housing. A significant policy gap exists due to the insufficient volume of social housing compared to the EU average. Low-income individuals, especially those living in flood-prone urban areas built during past expansion phases are highly vulnerable. The construction workforce itself is also identified as a disadvantaged group due to precarious working conditions, low resilience to climate risks, and low attractiveness for new talent.

Disadvantaged groups (DGs) [9]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Older people and/or with low tech skills	Individuals with limited digital literacy and adaptability to new technologies in sustainable housing.	Unable to use or benefit fully from energy-saving technologies like smart systems or digital apps for building management.	<i>"Older people are overwhelmed by the amount of technology; they don't read manuals or know that apps exist." (1)</i>
Social housing applicants	People eligible for protected housing who cannot access it due to insufficient supply	Lack of access to public housing that typically integrates better sustainability standards.	<i>"There is not enough public housing: the EU average is 13–15%, while in Spain we don't even reach 2%." (2)</i>
Renters	People who rent rather than own, often with limited power to make structural changes to improve sustainability in their homes.	Depend on landlords to carry out renovations; excluded from subsidies and long-term adaptation planning.	<i>"There's a big issue with tenants; they can't apply for subsidies or carry out retrofits unless the owner agrees." (2)</i>
Undocumented or non-registered in the municipal census	Vulnerable groups not registered in the local census, often migrants.	Cannot access subsidies or formal support schemes for housing adaptation, though they may have other pressing needs such as health and nutrition.	<i>"Vulnerable groups, those not registered, cannot access these subsidies, but there are other things more relevant to them: health, nutrition... maybe you need to start there." (2)</i>
Low-income individuals	People with limited financial resources who cannot afford climate-adaptive renovations or access social housing. Their vulnerability is aggravated in urban areas with old, thermally inefficient housing stock.	Unavailable to adapt their houses to the climate change and experiencing the consequences of losing thermic comfort	<i>"There will be poorer people who will be excluded from sustainable buildings and who won't be able to rehabilitate them because they don't have the money to do so" (2)</i>
Construction workforce	Workers in the building sector, often operating under precarious conditions,	Precarious working conditions and exposition to climate hazards	<i>"Currently, construction workers are a disadvantaged group, exposed to climate risks,</i>

	with limited protection from climate-related risks and low access to training or career progression.		<i>poor ergonomics, and low attractiveness for talent and conditions” (2)</i>
People living in flood-prone areas	Residents in areas developed during expansive urban planning phases, often with low economic capacity and inadequate building regulations.	Increased vulnerability to climate-related hazards like floods, with buildings that cannot be adapted and should be relocated.	<i>“Large-scale urban expansion led to construction in flood-prone areas.”(2)</i>

6.1.2 First round of Delphi survey

6.1.2.1 Elaboration of statements

1. By 2050, retrofitting subsidies will require high bureaucratic knowledge and financial literacy. **[Vulnerability_Skills]**
2. Public housing will lead on the adoption of sustainable building materials and design compared to the private sector **[Regime_Political]**
3. Buildings highly vulnerable to climate events (e.g., flood-prone zones) will increasingly be abandoned or relocated by 2050. **[Regime_Political]**
4. By 2050, concrete will remain the dominant construction material **[Niche_Technological]**
5. Technological complexity in smart housing systems (e.g., windows, energy dashboards) will create usability gaps among populations older than 55 years old and/or non-tech savvy population. **[Vulnerability_Skills] [Disadvantaged_Groups]**
6. By 2050, the construction sector will be dominated by assembly and automation roles. **[Vulnerability_Skills] [Vulnerability_Employment]**
7. The demand for traditional low-skill labor (e.g. construction workers) will decrease by 2050. **[Vulnerability_Skills]**
8. By 2050, sustainable materials (e.g., ceramics, wood) will represent a significant share of materials used in construction. **[Niche_Technological]**
9. Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households. **[Niche_Political]**
10. The shift to zero-emission buildings will be primarily driven by rising energy prices rather than climate awareness. **[Landscape_Economic] [Landscape_Environmental]**
11. The majority of citizens of EU countries will be unable to participate in retrofitting initiatives. **[Vulnerability_Skills] [Disadvantaged_Groups]**
12. Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050. **[Vulnerability_Finance] [Disadvantaged_Groups]**

13. The New European Bauhaus framework from the European Commission will significantly influence urban regeneration and social housing design. **[Niche_Governance]**
14. Energy prices will significantly influence incentives for adopting sustainable housing solutions. **[Landscape_Economic]**
15. Cultural attitudes (e.g. having the windows opened during cold times of the year) towards climate-adaptive housing will result in maladapted behavior, such as forcing opening the windows in automatic and self-regulated smart windows systems. **[Niche_Social]**
[Niche_Technological]
16. Sustainable architecture will be more scalable and affordable by 2050. **[Niche_Technological]**
[Niche_Economic]
17. Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies. **[Niche_Technological]**
18. The construction workforce will increasingly include profiles with less physical strength and more technical training, enabling greater inclusion of women and people with disabilities. **[Vulnerability_Skills]** **[Vulnerability_Employment]** **[Disadvantaged_Groups]**
19. By 2050, climate-adaptive housing technologies will be underutilized by residents. **[Niche_Social]**
[Niche_Technological]
20. Urban development plans will increasingly include climate relocation strategies for housing built in high-risk areas where adaptation is no longer feasible. **[Niche_Governance]**
21. Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation. **[Niche_Technological]**

6.1.2.2 Analysis of Delphi survey results

By 2050, retrofitting subsidies will require high bureaucratic knowledge and financial literacy		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	50%

Summary of qualitative comments	Over time and with growing collective experience, the steep learning curve around subsidy applications can be mitigated, but the current system is seen as “very exclusive,” effectively locking out anyone who lacks administrative savvy or financial know-how.
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Public housing will lead on the adoption of sustainable building materials and design compared to the private sector		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	0	0%
Low	3	75%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	0	0%
Low	2	50%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	It would be inspiring and politically powerful if public-sector landlords could pioneer green construction and sustainable retrofits, but that will only happen if national housing authorities embrace genuine innovation. In some countries we already see pilot projects, but too often budgets and procurement rules harden into traditional concrete-and-brick deliveries. If public housing agencies commit to binding environmental standards and allocate upfront funding for R&D, they could shift entire supply chains. However, for the moment, the private sector builds way more.	

Buildings highly vulnerable to climate events (e.g., flood-prone zones) will increasingly be abandoned or relocated by 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total

Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Abandonment is “already happening” , and there is a limit on what can be done to prevent. Each case must be analyzed individually but some areas will have to be reorganized or renaturalized.	

By 2050, concrete will remain the dominant construction material		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Progress is really slow on this. Concrete is very difficult to substitute. We can expect more hybrid constructions, using materials such as (especially) wood and ceramics. Still, concrete will have to be adapted to be more sustainable	

Technological complexity in smart housing systems (e.g, windows, energy dashboards) will create usability gaps among populations older than 55 years old and/or non-tech savvy population		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%

Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	This is already happening and getting worse; it is a form of exclusions, and we need not to assume everyone is at the same level of capabilities. However, there are views that the systems are going to be improved, especially from a usability perspective, and they will be easier to use.	

By 2050, the construction sector will be dominated by assembly and automation roles		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Experts agree, arguing that this is the pathway we are on right now.	

The demand for traditional low-skill labor (e.g. construction workers) will decrease by 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	0	0%
Low	1	25%
Very low	1	25%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	25%
Neutral	0	0%
Low	2	25%

Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	The experts claim that we need to work on alternative careers for those affected, in case it happened	

By 2050, sustainable materials (e.g, ceramics, wood) will represent a significant share of materials used in construction)		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	2	50%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	The experts argue that the impact of this would be significant, but the current trajectory is not encouraging	

Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%

Summary of qualitative comments	We are already seeing this; the impact is concentrated on those with low-incomes and in disadvantaged regions of the world.
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The shift to zero-emission buildings will be primarily driven by rising energy prices rather than climate awareness		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0
Impact	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	2	50%
Low	1	25%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Ultimately citizens react to what affects them, and there is a generational shift towards caring more about the environment. However those in very limited financial situations are undoubtedly motivated by cost.	

The majority of citizens of EU countries will be unable to participate in retrofitting initiatives		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	3	75%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	25%

Summary of qualitative comments	Depending on the level of retrofitting. Smaller things like changing windows are more accessible to more people. Measures like external wall insulation are more costly and difficult to implement.
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Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050		
Likelihood	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Experts point out that this is the case, and that this undermines the purpose of a just transition	

The New European Bauhaus framework from the European Commission will significantly influence urban regeneration and social housing design		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	3	75%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	1	25%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	Implementation depends on Member States but having this instruction from the EU does have an impact, even if limited; furthermore, if effective, this would have a huge impact.	

Energy prices will significantly influence incentives for adopting sustainable housing solutions		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	No qualitative comments were included by the respondents.	

Cultural attitudes (e.g. having the windows opened during cold times of the year) towards climate-adaptative housing will result in maladapted behavior, such as forcing opening the windows in automatic and self-regulated smart windows systems		
Likelihood	Likelihood	Likelihood
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Impact	Impact	Impact
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Education can help overcome it. Installers can have a role in educating people, and we need to place this responsibility more firmly at their door. At current there are no standards for the level of information imparted but this must change.	

Sustainable architecture will be more scalable and affordable by 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	100%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	This will bring important changes in accessibility, currently sustainable architecture is out of reach for many. As the industry develops, it will evolve.	

Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	0%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	This should be the case, but experts are not sure we are moving fast enough here. If it is achieved, this would drive needed changes in construction practices	

The construction workforce will increasingly include profiles with less physical strength and more technical training, enabling greater inclusion of women and people with disabilities		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	0	0%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	0	0%
Low	1	25%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Industrialization will integrate people that were usually excluded for different reasons, helping them enter the sector.	

By 2050, climate-adaptive housing technologies will be underutilized by residents		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	25%
Neutral	2	50%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	Currently people are not well informed enough and there is a big gap to address. If achieved, this could have a high impact for both householders and for carbon neutrality goals.	

Urban development plans will increasingly include climate relocation strategies for housing built in high-risk areas where adaptation is no longer feasible		
Likelihood	Number of answers	% over the total

Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	This will become increasingly necessary, but it will place additional strain on housing markets which are already stressed.	

Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	We seem stuck in the statu quo. It is a sensible and impactful change. Cost savings, both for sustainable material and to make housing affordable would be a game changer. Tackling the exclusive nature of the real estate market is also part of a just transition.	

6.1.3 Second round of interviews

6.1.3.1 Niches

Niches code	Summary of description	Literal citations
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<i>*Number of citations per category in brackets [X]</i>		<i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [11]	Economic innovations in the housing sector revolve around balancing affordability, renovation support, and climate resilience. The expert insights stress the role of targeted subsidies, especially for single-family home upgrades, as well as the need to shift from a focus on upfront costs to lifecycle costs (including energy, maintenance, and emissions). However, widespread uncertainty about financial return continues to deter building owners from investing in retrofits. Public housing is seen as a key lever to pilot and scale sustainable solutions, using procurement to drive demand for low-carbon materials. Yet, questions remain about whether these alternatives can overcome cost-driven inertia in favour of concrete and other traditional materials. Climate shocks and regional inequalities may amplify demand for affordable, climate-resilient housing in safer zones, further highlighting the need for strategic investment. Regional equity (including investments in low-resource areas) is viewed as critical to ensure that innovative housing models reach those most in need.	<i>"Uncertainty about costs and return on investment deters building owners from renovation."</i> (20) <i>"Operational and lifecycle costs... versus upfront retrofitting or construction expenses."</i> (20) <i>"Housing affordability and access to renovation support... are critical."</i> (20) <i>"Rising operational costs and growing financial significance of energy efficiency."</i> (20) <i>"Will low-carbon materials become widely available, or will concrete dominate due to cost and inertia?"</i> (20) <i>"Expand public housing to pilot and scale sustainable building approaches."</i> (20)
Niche_Governance [13]	Governance innovation in the housing sector focuses on enabling large-scale transformation through regulatory adaptation, better data, and inclusive policy design. Interviewee insights stress the need for flexible zoning and planning laws, along with clear criteria and systematic data to assess and retrofit building stock. The absence of standardized solutions is seen as a bottleneck for refurbishment at scale. Lifecycle-based policies and aggressive regulation are seen as potential accelerators for emissions reduction, but there is concern that current governance tools lag behind technological development. Simplifying subsidy access (particularly for vulnerable or less tech-savvy groups) and mandating inclusive design in smart systems are identified as key priorities. Public housing is again positioned as a strategic governance lever: both to experiment with sustainable practices and to guide markets via procurement. Governance models that promote regional equity are also essential to ensure transformation reaches low-resource areas.	<i>"Clear criteria for assessing building stock potential are necessary."</i> (20) <i>"Regulatory adaptation... zoning and planning law flexibility."</i> (20) <i>"Absence of standardized solutions and systematic data inhibits efficient refurbishment."</i> (20) <i>"Holistic, lifecycle-based policies are still lacking."</i> (20) <i>"Will regulations and incentives keep pace with technology, or lag behind market needs?"</i> (20) <i>"Simplify and target subsidy access... low-bureaucracy support for vulnerable and less tech-savvy groups."</i> (20) <i>"Mandate inclusive design in smart building systems."</i> (20) <i>"Aggressive regulation and rapid material innovation could accelerate emissions reductions."</i> (20) <i>"Expansion of regulatory frameworks targeting energy and emissions."</i> (20) <i>"Promote regional equity: direct investment and policy"</i>

		<i>support to at-risk/low-resource areas.” (20)</i> <i>“Increasing scarcity of resources and stricter climate policies.” (20)</i>
Niche_Social [12]	The social dimension of innovation in housing is shaped by diverging abilities to adapt to digitalisation, automation, and climate-driven displacement. Interviewee insights highlight how prefabrication and smart systems create opportunities for highly skilled workers while reducing demand for traditional low-skilled roles. Without proactive reskilling and inclusive design, these shifts risk deepening existing inequalities; particularly among older adults and low-income populations. Smart housing systems may not be equally accessible to all, given persistent disparities in digital literacy and comfort expectations. Climate vulnerability and regional inequalities may also force population shifts, increasing demand for climate-resilient, affordable housing in safer areas. On the positive side, the rise of green construction brings potential for new job creation and improved living standards, if paired with workforce support and inclusive infrastructure.	<i>“Automation and prefabrication reduce demand for low-skilled labor while increasing the need for digital and technical skills.” (20)</i> <i>“Comfort expectations, willingness to adopt sustainable practices...” (20)</i> <i>“Regional inequality and climate vulnerability may force population shifts, disproportionately affecting low-income groups.” (20)</i> <i>“Persisting disparities in digital literacy by age and socioeconomic status.” (20)</i> <i>“Will all demographic groups be able to adapt to and benefit from smart building systems?” (20)</i> <i>“Lack of public acceptance... could slow down material innovation and emissions reduction.” (20)</i> <i>“New challenges may include digital divides, new forms of housing exclusion, and shifting urban/rural dynamics.” (20)</i> <i>“Opportunities involve new jobs in green building sectors, reduced emissions, and improved living standards.” (20)</i>
Niche_Technological [17]	Technological innovation in the housing sector is advancing through digitalisation, smart systems, and circular construction practices. Serial prefabrication and automation are increasing efficiency and quality control, while innovations in materials focus on lowering emissions and enabling reuse. Circular economy principles (such as design for disassembly, material passports, and life-cycle assessments) are central to this shift. Smart building technologies (e.g., energy dashboards and control systems) are becoming standard, but their complexity can exclude less tech-savvy users unless inclusive design is prioritised. Data gaps and lack of standardisation continue to obstruct large-scale retrofitting. Meanwhile, the sector is also under pressure to ensure buildings are resilient to future climate extremes, not just current conditions.	<i>“Automated, digitalized processes and serial prefabrication in construction are central.” (20)</i> <i>“Emphasis on materials with low emissions and high reuse potential.” (20)</i> <i>“Design for disassembly, recyclable materials, and robust life-cycle assessment drive sustainability.” (20)</i> <i>“Integration of technical building systems... potentially creates barriers for certain user groups.” (20)</i> <i>“Absence of standardized solutions and systematic data... inhibits efficient</i>

	The success of these technological niches depends on sustained funding, supportive regulation, and labour market adaptation. Without reskilling, automation and digitalisation risk reinforcing inequalities. Public housing is seen as a key testing ground for sustainable and inclusive tech solutions.	<i>refurbishment.” (20)</i> <i>“Construction methods impact biodiversity, land use, water, waste and pollution.” (20)</i> <i>“Buildings need to be resilient to future climatic extremes, not just energy efficient.” (20)</i> <i>“Will low-carbon materials become widely available, or will concrete dominate due to cost?” (20)</i> <i>“Will expected rebound effects undermine efficiency gains?” (20)</i>
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6.1.3.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [1]	Government strategies in the housing sector are guided by the “Efficiency First” principle, which prioritises reducing energy consumption through stringent regulatory standards. This doctrine shapes national frameworks such as the <i>Effizienzhaus 55</i> standard, embedding efficiency into the political fabric of housing policies. While effective in setting benchmarks, its success depends on enforcement, feasibility, and public buy-in.	<i>“The ‘Efficiency First’ doctrine underpins government strategies... focusing on reducing energy consumption through stringent standards.” (20)</i>
Regime _ Economic [1]	Economic constraints within the existing housing regime continue to deter renovation and retrofitting. Uncertainty around investment payback timelines (particularly regarding energy efficiency upgrades) discourages building owners from acting. This reflects a structural disincentive embedded in the current economic model, where short-term costs outweigh long-term benefits in decision-making processes.	<i>“Uncertainty about costs and return on investment deters building owners from renovation.” (20)</i>
Regime _Social [1]	Social inequalities are embedded in the current housing regime through the growing reliance on digital systems. As housing solutions become increasingly tech-driven (from smart controls to digital subsidy access) there is a rising risk of excluding vulnerable populations who lack digital skills or access. Without corrective measures, the social dimension of the regime may reinforce, rather than reduce, inequality.	<i>“Digitally-driven solutions risk excluding vulnerable populations.” (20)</i>
Regime_Technological [3]	The existing technological regime in housing is being challenged by environmental and climate demands. Traditional construction methods still dominate, despite their significant ecological impacts (including land consumption, water use, and pollution). Technological standards largely focus on current energy efficiency, but fail to fully incorporate resilience to future climatic extremes like heatwaves or flooding.	<i>“Construction methods impact on biodiversity, land consumption, water use, waste and pollution; environmentally responsive urban planning is essential.” (20)</i> <i>“Buildings need to be resilient to future climatic extremes...”</i>

	At the same time, the regime's growing reliance on digital solutions (from smart controls to management systems) raises concerns about inclusivity. Without adjustments for accessibility and usability, these innovations may leave behind digitally excluded groups.	<i>not just energy efficient under current conditions.” (20)</i> <i>“Digitally-driven solutions risk excluding vulnerable populations.” (20)</i>
Regime_Environmental [1]	Environmental concerns are increasingly relevant within the current housing regime. Construction practices continue to exert significant pressure on ecosystems, affecting biodiversity, land use, water resources and waste generation. The interview highlights the need for a shift toward environmentally responsive urban planning, indicating that mainstream approaches still fall short in addressing these broader ecological impacts.	<i>“Construction methods impact on biodiversity, land consumption, water use, waste and pollution; environmentally responsive urban planning is essential.” (20)</i>

6.1.3.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [2]	The housing sector operates within a volatile political landscape shaped by climate policy ambitions and societal expectations. Stricter emission targets are driving regulatory pressure, compelling the sector to minimise lifecycle emissions and address resource limitations. However, the stability of supportive policies — such as subsidies or digital infrastructure investment — is not guaranteed. Interviewee insights caution that shifts in political will, economic crises, or public resistance could derail current assumptions, slowing down material innovation and undermining long-term housing transition strategies.	<i>“Assumptions such as universal access to digitalized systems, constant public support for subsidies, and smooth material innovation may become invalid if economic crises, policy reversals, or social resistance occur.” (20)</i>
Landscape _Economic [2]	Economic uncertainty poses a significant risk to long-term housing transformation. Interviewee insights highlight that current strategies rely on assumptions (such as stable funding for retrofitting, universal digital access, and ongoing public support) that may not hold through future economic crises or shifting policy priorities. The sustainability of material innovation and subsidy schemes is especially vulnerable to downturns and political reversals. These macroeconomic pressures could undermine efforts to scale low-carbon housing solutions, particularly in regions already facing affordability constraints or social resistance.	<i>“Will funding for retrofitting and innovation remain robust through economic cycles?” (20)</i>
Landscape _Social [1]	Social resistance (whether due to digital divides, trust issues, or unmet expectations) could disrupt transformation efforts, especially in times of economic or political instability.	<i>“Assumptions such as universal access to digitalized systems, constant public support for subsidies, and</i>

		<i>smooth material innovation may become invalid if economic crises, policy reversals, or social resistance occur.” (20)</i>
Landscape _Environmental [2]	Environmental pressures at the landscape level are reshaping the housing sector. Stricter climate targets and resource scarcity demand urgent action to reduce lifecycle emissions and optimise material use. At the same time, intensified climate events (such as heatwaves or flooding) are likely to increase the displacement of vulnerable populations, creating new demands for affordable, climate-resilient housing in safer regions. These dynamics amplify the urgency of integrating sustainability, equity and resilience into mainstream housing policy and practice.	<i>“The sector must minimize lifecycle emissions, address resource limitations, and adapt to stricter climate targets.” (20)</i> <i>“Intensified climate events and regional economic gaps could compound displacement of vulnerable populations, increasing demand for climate-resilient, affordable housing in safer zones.” (20)</i>

6.1.3.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [3]	Labour market shifts driven by automation and prefabrication are creating new vulnerabilities within the housing sector. As digital and technical skills become more essential, the demand for low-skilled construction workers is declining. Without targeted reskilling strategies, workers currently employed in traditional roles may be left behind. At the same time, the transition also brings new employment opportunities in green construction and smart building systems. However, whether these benefits reach vulnerable labour groups will depend on the inclusivity of training, recruitment, and technological design.	<i>“Automation and prefabrication reduce demand for low-skilled labor while increasing the need for digital and technical skills.” (20)</i> <i>“Low-skill construction labor [is at risk] due to automation and prefab.” (20)</i> <i>“Opportunities involve new jobs in green building sectors, reduced emissions, and improved living standards through smarter and more adaptive building stock.” (20)</i>
Vulnerability_Finance [2]	Financial vulnerability in the housing sector is shaped by uneven access to subsidies and investment. While targeted subsidies exist (particularly for single-family home upgrades) regional disparities in public investment leave some areas structurally disadvantaged. Regions exposed to climate risks but lacking adequate funding are particularly vulnerable, as they face both environmental and economic precarity. Addressing this requires a deliberate focus on regional equity, ensuring that low-resource areas receive the investment and policy support needed to participate in the housing transition.	<i>“Targeted subsidies (especially for single-family home upgrades).” (20)</i> <i>“Regions exposed to climate risk and minimal public investment.” (20)</i> <i>“Promote regional equity: direct investment and policy support to at-risk/low-resource areas.” (20)</i>
Vulnerability_Skills [10]	The growing digitalisation and automation of the housing sector is exposing deep skill-based vulnerabilities. Interviewee insights highlight	<i>“Digitally-driven solutions risk excluding vulnerable populations.” (20)</i>

	that low-skilled and digitally excluded populations (including older adults and people with lower socioeconomic status) may struggle to navigate smart technologies, access subsidies, or adapt to new job market demands. Without targeted support, these groups risk exclusion from both the benefits and opportunities of the transition. As smart systems and digital controls become more widespread, inclusive design and low-bureaucracy assistance will be critical. At the labour market level, the shift toward high-skilled digital and green construction roles intensifies the urgency of reskilling initiatives to ensure that vulnerable workers are not left behind.	<i>"Digital literacy among residents... older adults facing challenges with smart technologies."</i> (20) <i>"Persisting disparities in digital literacy by age and socioeconomic status."</i> (20) (20) <i>"New challenges may include digital divides."</i> (20) <i>"Digitally excluded... unable to navigate complex subsidy systems or smart controls."</i> (20) <i>"Simplify and target subsidy access, with hands-on, low-bureaucracy support for vulnerable and less tech-savvy groups."</i> (20) <i>"Mandate inclusive design in smart building systems."</i> (20) <i>"Support workforce reskilling toward digital and green construction skill."</i> (20)
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6.1.3.5 Disadvantaged groups (DGs)

The interview highlights multiple groups at risk of exclusion from the housing transition. These include low-income individuals, older adults, rural residents, digitally excluded populations, low-skilled construction workers, and residents of underinvested regions. Without targeted investment and inclusive design, these groups may struggle to access subsidies, adapt to smart systems, or benefit from green construction jobs. Climate vulnerability and digital divides further compound these disadvantages, reinforcing patterns of housing exclusion and displacement. Addressing these disparities will require proactive policy support, inclusive technological design, and dedicated reskilling programmes.

Disadvantaged groups (DGs) [13]	Summary of description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Low-income households	Limited capacity to access renovations or adapt to rising housing costs	Greater risk of exclusion from retrofitting benefits; vulnerable to displacement	<i>"Losers: vulnerable groups — low-income..."</i> (20)
Digitally excluded individuals and/or people with low digital literacy	Unable to navigate smart systems or digital subsidy platforms	Excluded from subsidy access and digital housing innovations	<i>"Digitally excluded (e.g., unable to navigate complex subsidy systems or smart controls)."</i> (20) <i>"Digitally-driven solutions risk excluding vulnerable populations."</i> (20)

Elderly population group	More likely to face digital literacy challenges and usability issues with smart systems	Excluded from benefits of smart homes and automation; at risk of housing marginalisation	<i>“Digital literacy among residents... older adults facing challenges.” (20)</i>
Rural populations	Often underserved in terms of infrastructure and subsidy targeting	Limited access to support and higher vulnerability to climate impacts	<i>“Promote regional equity... to at-risk/low-resource areas.” (20)</i>
Low-skilled construction workers	At risk due to automation and prefabrication reducing demand for traditional labour	Job displacement without access to reskilling	<i>“Low-skill construction labor (due to automation and prefab).” (20)</i>

6.1.4 Second round of Delphi survey

6.1.4.1 Elaboration of statements

22. By 2050, no publicly funded housing will remain in high-risk flood or wildfire zones without certified adaptation measures or financed relocation plans. **[Niche_Governance]**
[Landscape_Environmental]
23. All smart-home control systems will include an independently certified “easy mode” accessible to users aged 65 + and those with low digital literacy. **[Disadvantaged_Groups]**
[Vulnerability_Skills]
24. More than the half of construction workers and/or companies will hold an accredited certificate in climate-adaptive building techniques. **[Disadvantaged_Groups]** **[Vulnerability_Skills]**
25. The majority of public retrofit subsidies will be paid before construction starts, removing liquidity barriers for households. **[Niche_Governance]** **[Niche_Economic]**
26. By 2050, deep-retrofit uptake will be similar among households regardless of income level.
27. By 2050, digital-twin modelling will enable residents of historic buildings to explore and approve retrofit options virtually, reducing resistance and speeding up decision-making.
[Niche_Technological] **[Niche_Governance]**
28. By 2050, national renovation strategies will offer a graduated menu of upgrade bundles—from quick window replacement to full-envelope insulation—so households can choose a depth of retrofit that matches their means. **[Niche_Governance]** **[Vulnerability_Finance]**
29. By 2050, time-of-use tariffs will automatically reward homes that achieve zero-emission performance, making energy savings a stronger driver than up-front subsidies. **[Niche_Technological]**
30. By 2050, climate-aware behaviour programmes will be embedded in schools and community centers, shifting everyday practices—such as habitual winter window-opening—toward energy-efficient norms. **[Niche_Social]**

6.1.5 Analysis of results

Buildings highly vulnerable to climate events (e.g, flood-prone zones) will increasingly be abandoned or relocated by 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	40%
Neutral	1	20%

Low	1	20%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	0	0%
Low	0	0%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	While some expect this to happen, others note that current policies do not seem to be moving in that direction. Examples from reconstruction efforts in flood-affected regions show that rebuilding in vulnerable areas continues. Abandonment would enhance safety but might cause overcrowding in other zones, creating new social and infrastructure challenges.	

By 2050, concrete will remain the dominant construction material		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	4	80%
Neutral	0	0%
Low	0	0%
Very low	0	0
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	0	0%
Low	0	0%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Concrete's dominance is expected to persist due to its low cost, entrenched industry practices, and familiarity within the sector. However, alternatives like light steel framing (LSF) and modular housing are growing in market share. Advocates for these alternatives highlight their faster construction times and better thermal and acoustic performance.	

Technological complexity in smart housing systems (e.g., windows, energy dashboards) will create usability gaps among populations older than 55 years old and/or non-tech savvy population		
Likelihood	Number of answers	% over the total

Very high	1	20%
High	1	20%
Neutral	0	0%
Low	3	60%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	0	0%
Low	2	40%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Views differ on the extent of this risk. Many expect that by 2050, future elderly populations (today's younger generations) will be more comfortable with technology, reducing exclusion. Nevertheless, people with low digital literacy may still face barriers, and overly complex systems could be challenging unless designed with user-friendly interfaces.	

By 2050, the construction sector will be dominated by assembly and automation roles		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	3	60%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments	Some see this as a logical outcome of workforce shortages, cost pressures, and technological advances. Prefabrication can reduce labor needs and speed up construction. Others note the sector is already moving in this direction, with potential for even faster adoption.	

Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	3	60%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	3	60%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments	Several respondents believe this is already occurring, particularly in city centers and other areas under growing pressure from climate-related risks. Climate change is expected to make such displacement more common, especially in vulnerable regions. While some note that certain natural disasters, like floods, can affect populations across income levels, other climate phenomena may disproportionately impact low-income communities. Displacement could contribute to overcrowding in safer urban areas, exacerbating housing shortages and social inequality.	

Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050		
Likelihood	Number of answers	% over the total
Very high	3	60%
High	2	40%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	0	0%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%

Summary of qualitative comments	Participants widely agree this pattern is already evident: cheaper housing options tend to be of lower quality and less resilient to climate risks. Wealthier households will be able to afford better-adapted properties, although some note that they will not be entirely shielded from indirect impacts such as pollution originating from less-protected areas. The issue is fundamentally linked to persistent social class divisions, and without substantial policy intervention, disparities in protection are expected to widen over time.
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Energy prices will significantly influence incentives for adopting sustainable housing solutions		
Likelihood	Number of answers	% over the total
Very high	3	60%
High	1	20%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	60%
High	1	20%
Neutral	1	20%
Low	0	0%
Very low	0	0
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments	Respondents generally agree that energy prices play a decisive role in motivating or deterring the adoption of sustainable housing measures. High costs for renewable energy can be prohibitive for much of the population, making affordability a critical barrier. While some see the effect as already evident, others suggest there is a threshold beyond which price changes may no longer drive further adoption, particularly if the energy transition stalls. Education and awareness are also viewed as important factors, influencing whether households act on price signals. If prices remain high without adequate support, fewer people will choose sustainable options, potentially limiting the housing sector's contribution to emissions reductions.	

Sustainable architecture will be more scalable and affordable by 2050		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	3	60%

Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Participants believe the evolution will depend heavily on political support and cost structures. Advances in technology are expected to expand the range of sustainable solutions, making them more efficient, widely available, and financially accessible. However, affordability will remain linked to market conditions and policy incentives. If achieved, greater scalability would deliver significant environmental benefits while also reducing costs for households, enhancing both ecological and economic outcomes.	

Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	3	60%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	1	20%
Low	0	0%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Respondents largely agree this shift will occur, especially if mandated by regulation. Legislative requirements could embed climate-adaptation criteria into procurement, creating a competitive advantage for companies able to meet them. While some point out that large firms currently dominate tenders (potentially limiting opportunities for smaller companies) others note that this trend is already visible in some markets. Education	

	and capacity-building in climate-adaptive design will be necessary to ensure broader industry participation.
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The construction workforce will increasingly include profiles with less physical strength and more technical training, enabling greater inclusion of women and people with disabilities		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	0	0%
Low	2	40%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Respondents see that there is shortage of trained personnel and an increasing digitalisation and automation of construction processes. The integration of robotics and technology will reduce the physical demands of many roles, broadening accessibility. This shift will diversify skill sets within the sector and open opportunities for underrepresented groups. However, some anticipate challenges in achieving this transition, suggesting that adaptation may be uneven across regions and companies.	

Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	0	0%
Low	2	40%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	0	0%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree

	No, remove	60%
Summary of qualitative comments	Opinions point to a gradual shift toward these approaches, given their potential to shorten build times and lower costs. However, the pace of adoption may be slowed by low innovation interest within parts of the construction industry, where traditional methods remain profitable. Where adopted, prefabricated and modular techniques are expected to deliver faster implementation and more affordable retrofits, increasing accessibility for both public and private projects.	

By 2050, no publicly funded housing will remain high-risk flood or wildfire zones without certified adaptation measures or financed relocation plans		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	40%
Neutral	1	20%
Low	1	20%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Opinions are mixed on the feasibility of this target. While stronger enforcement and oversight in construction zones could help, political realities may undermine relocation or adaptation efforts. Historical examples from flood-prone regions like the Arntal and the Danube areas in Bavaria illustrate the persistence of high-risk settlements despite hazards. In some cases, adaptation may be more practical than relocation, especially where moving large populations (e.g., millions living near Mount Vesuvius) is unrealistic. If implemented, the policy could lead to the depopulation or “desertification” of some areas, though alternative approaches could achieve similar risk-reduction goals	

All smart-home control systems will include an independently certified “easy mode” accessible to users aged 65+ and those with low digital literacy		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	60%
Neutral	1	20%
Low	1	20%

Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	1	20%
Low	2	40%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	60%
Summary of qualitative comments	While some respondents believe such a feature is likely to be introduced, others question whether smart-home systems themselves will remain dominant, suggesting a shift toward “sustainable technologies” better matched to users’ capabilities. The implementation of an easy mode is seen as a way to enhance autonomy and safety for older adults and those with limited digital skills. However, broader improvements in digital literacy could reduce reliance on such adaptations over time.	

More than half of construction workers and/or companies will hold an accredited certificate in climate-adaptive building techniques		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	60%
Summary of qualitative comments	Respondents see accredited certification in climate-adaptive techniques as a potential emerging standard in the industry. Such qualifications are viewed as beneficial for all stakeholders, enhancing both the quality and resilience of construction. The anticipated growth in certification uptake reflects increasing policy emphasis on climate adaptation and the competitive advantage it offers companies that invest in these skills	

The majority of public retrofit subsidies will be paid before construction starts, removing liquidity barriers for households		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	1	20%
Low	3	60%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	0	0%
Low	1	20%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	0%
Summary of qualitative comments	Most respondents are skeptical about the likelihood of this change, noting that governments may be reluctant to take on the associated financial risk. Feasibility is seen as highly dependent on national context, with some countries, such as Portugal, perceived as unlikely to adopt such a system. Existing barriers include bureaucratic complexity, difficulties in accessing subsidies, and inconsistent application of available support. While paying subsidies upfront could ease liquidity constraints for households, some caution it might also contribute to price inflation in the retrofit market.	

By 2050, deep-retrofit uptake will be similar among households regardless of income level		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	1	20%
Low	2	40%
Very low	2	40%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	0%

Summary of qualitative comments	Most respondents doubt that deep-retrofit adoption will reach full parity across income groups. They point to persistent differences in household earnings, which affect both the ability to afford retrofitting and ongoing maintenance. Funding availability is a central concern. without substantial and well-targeted financial Wsupport, lower-income households are likely to lag behind. Some stress that ignoring these disparities risks making sustainability policies a driver of increased inequality and poverty. Others suggest that while equality in uptake is desirable, alternative measures might be needed to achieve similar environmental and social impacts.
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By 2050, digital-twin modelling will enable residents of historic buildings to explore and approve retrofit options virtually, reducing resistance and speeding up decision-making		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	0	0%
Neutral	2	40%
Low	2	40%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	60%
Summary of qualitative comments	Some participants expect technological advances to potentially make this feasible, with some noting that early forms of such tools are already in use. Digital-twin models could facilitate more informed decision-making and help residents visualise the benefits of retrofits, making it easier to approve changes. However, financial constraints will remain a deciding factor. Those with sufficient resources will be more likely to proceed with renovations, while less affluent owners may still face barriers despite improved decision-making tools	

By 2050, national renovation strategies will offer a graduated menu of upgrade bundles – from quick window replacement to full-envelope insulation – so households can choose a depth of retrofit that matches their means		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	0	0%
Low	2	40%

Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	0	0%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	60%
Summary of qualitative comments	Some participants note that similar options already exist in certain markets, enabling households to choose among different renovation levels. While improved technology and materials could enhance quality and affordability, doubts remain about the availability of financial support, which would be critical for broader adoption. Such a graduated approach could expand choice and make retrofits more accessible, but without subsidies or incentives, uptake among lower-income households may still be limited. The model has the potential to deliver environmental benefits while matching solutions to household budgets.	

By 2050, time-of-use tariffs will automatically reward homes that achieve zero-emission performance, making energy savings a stronger driver than up-front subsidies		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	0	0%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Some view this outcome as desirable in the medium to long term. Automatic tariff incentives could encourage households to adopt measures that make them energy self-sufficient, improving environmental performance while reducing reliance on subsidies. The approach is expected to directly link financial benefits to	

	energy efficiency, reinforcing behavioural change toward zero-emission housing.
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By 2050, climate-aware behaviour programmes will be embedded in schools and community centers, shifting everyday practices – such as habitual winter window-opening – toward energy-efficient norms		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	20%
Neutral	2	40%
Low	2	40%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	1	20%
Neutral	3	60%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	20%
Summary of qualitative comments	Respondents see these programmes as a potential driver of mindset change, embedding energy efficiency and climate awareness into everyday routines from an early age. However, there are doubts about societal readiness to fully embrace such shifts, with some pointing out that certain current practices, like winter window-opening, are linked to health considerations (e.g., preventing mould or Legionella) and may be difficult to change entirely. Successful implementation will require balancing behavioural adaptation with health and safety needs, ensuring energy savings do not come at the expense of well-being.	

6.1.6 Third round of interviews

For the case of housing and built environment, due to the difficulties of finding sectorial experts. No third round of interviews was conducted.

6.2 Agriculture and food

6.2.1 First round of interviews

6.2.1.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [15]	Interviewees identify contrasting but coexisting economic tendencies in agriculture. On one hand, large-scale farming is expanding through land concentration and investment from large companies.	"Open data spaces are used to give value to digitalisation in small farms" (3)

	On the other hand, many emerging niches point to alternative and potentially transformative economic models. A strong example is the push for data valorisation via open data spaces, where farmers retain ownership of their information and can negotiate its use, framing it as a new "data economy." Small producers will also innovate through short value chains, local wineries, ecological horticulture and community-supported agriculture. These models aim to improve food quality, reduce pollution and promote generational renewal, while preserving the margins of the fine food sector. New types of agricultural business models are also emerging, including farmers as providers of ecosystem services, not just food. Examples include carbon farming practices and biodiversity-oriented production, which are still incipient but could gain traction with the development of carbon credit markets. External support from actors like EIT Food helps de-risk innovation through pilot programmes, with some farmers acting as early adopters. In order to happen, institutional frameworks support land access, fair data governance and ecosystem service monetisation would need to be prioritised. Regarding different regulations between countries (EU and NA), mechanisms like carbon border adjustment taxes are seen as necessary to reflect the true environmental cost of imported products	<i>"Horticultural business may be simpler, with less infrastructure and more people entering" (3)</i> <i>"There will be increase in small projects (e.g. wineries) – diversification and generational renewal through shorter value chains."(3)</i> <i>"There will be more presence of large companies" (3)</i> <i>"We should develop alternative economic systems in rural areas" (5)</i> <i>"Farmers should see agriculture as a provider of ecosystem services beyond food" (4)</i> <i>"Large-scale farming is growing" (5)</i> <i>"Carbon border adjustments are needed to offset the environmental cost of products coming from Morocco" (5)</i>
Niche_Governance [12]	The experts suggest that future governance models in agriculture will need to become more localised, interoperable and inclusive, especially for small and medium farmers. A key concern is that current legal and regulatory frameworks are too generalised and fail to address territorial specificities such as microclimates or local risks. Efforts are emerging to build shared data infrastructures (e.g., data lakes) and improve communication across regional and national authorities (multi-level). These could enable better coordination, precision farming and support for decision-making. However, political fragmentation each region wanting to manage its own data, remains a barrier. Interviewees also warn that digitalisation is reinforcing the power of large corporate actors unless governance structures are deliberately designed to empower smaller ones. Other innovations include proposals for digital governance hubs, more transparent value chains, and cooperative frameworks (e.g., dairy cooperatives	<i>"There won't be farmers, only big companies; medium-sized ones will survive thanks to subsidies"</i> <i>"Data communication between authorities exists, but each region wants to control its own "</i> <i>"It's not just about sensors, but about reaching consensus on what to measure and why"</i> <i>"Administrative management must be local, current phytosanitary recommendations are too general; precision agriculture should be more targeted"</i> <i>"Agrarian management only reaches regional governments;</i>

	supporting transitions). There is also demand for capacity-building and agreement on what indicators matter most for sustainable farming. Some expect a decline in globalisation and a shift toward more territorialised governance, especially if geopolitical instability continues.	<i>there's duplication of data, a common data lake is being developed."</i>
Niche_Social [12]	Interviewees identify a better perception of proximity agriculture, increased awareness about food origin, and a potential cultural transformation toward valuing local production. However, they also note resistance to change, especially among certain age groups and a broader lack of motivation to work in the agricultural sector. The digitalisation of rural life is fostering unexpected dynamics, such as urban remote workers moving to the countryside, potentially triggering a "revival" of some agricultural areas. New rural job opportunities could emerge linked to digital services, attracting more qualified and tech-savvy professionals. Still, the problem of depopulation remains a critical concern, often exacerbated by a lack of social incentives and territorial policies. Agricultural communities are also mobilising politically, demanding equal regulatory conditions with third countries, while younger farmers adopt technology more selectively. Interviewees stress that agriculture is increasingly recognised as a complex system requiring adaptation and social innovation.	<i>"To enhance rural growth, we need cultural change" (3)</i> <i>"Digitalisation is generating a 'revival' in agricultural areas, urbanites move out, cities push them away in search of calm" (4)</i> <i>"Farmers are mobilising so African products meet the same standards as Europe, there's resistance to change, younger farmers adopt technology partially" (3)</i> <i>"Fruit is globalised, it depends on whether people value local products" (3)</i> <i>"We may attract talent of tech-savvy professionals" (5)</i>
Niche_Technological [23]	Automation, predictive monitoring and precision agriculture are rapidly expanding, supported by satellite data, image-based diagnostics and platforms for anticipating pests, managing water use or reducing fertiliser needs. These solutions align with the goals of climate adaptation and disaster mitigation. However, several interviewees express critical views on regenerative agriculture. While its communicative appeal is growing, its productivity and impact are seen as limited or uncertain by many. There's also ambiguity between different labels (ecological vs regenerative vs carbon farming), and a perception that consumer expectations may not match reality. Significant hopes are placed in biotechnological advances, such as carbon-capture microbes, natural fertilisers and AI-based crop modelling. Some initiatives aim to develop soil supplements, diversified energy sources or improve the efficiency of water technologies such as desalinization. Also, cross-sectoral are also expected: agriculture and	<i>"The digital transition will happen due to efficiency and economic viability; the green transition is not the same. It may be more costly to do things right. They have different late motivs" (5)</i> <i>"There will be more local facilities that, from the fields themselves, can create biofertilizers, moving away from large industrial parks far from the territory..."(4)</i>

	biofertilizers, pharmaceutical and livestock farming (e.g., slaughterhouse), beekeeping, AI, biotech... In parallel, digitalisation is expected to enhance data for policymakers, make business models more competitive, and reduce inefficiencies. Still, the environmental cost of digital transitions is also acknowledged.	
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6.2.1.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [6]	The political regime is shaped by the EU's Common Agricultural Policy (CAP), which dominates the regulatory framework through rules on herbicides and phytosanitary products. This top-down approach has generated significant bureaucratic demands. While some actors, like cooperatives, are better positioned to handle EU requirements, there's criticism of overregulation and lack of social insight by regulators. Nevertheless, Europe is seen as a stable and forward-thinking global reference, capable of maintaining sustainable economic growth through solid policy frameworks.	<i>"Regulators have all the maps and many data, but need to step the ground, as they lack understanding of the sector"</i> (5) <i>"We could be the global regulatory benchmark. Europe has always been a step ahead... it has a more stable strategy..."</i> (4) <i>"Europe is very good at regulating, maybe too much, we have 270 labels"</i> (4)
Regime _ Economic [10]	The interview data reveals an economic regime shaped by globalisation, business concentration, and growing tensions around the cost of sustainability. The European agricultural sector is described as heavily externalised and lacking food sovereignty, with key inputs such as fertilisers being imported. This structure is perceived as fragile, especially in light of current geopolitical and environmental pressures. Interviewees highlight that strict sustainability regulations in Europe raise production costs, making it harder to compete with countries that have laxer standards. A core tension emerges between dominant actors embracing digitalisation and the persistence of more traditional or localised models. Although proximity-based agriculture may survive, it often relies on public support or niche consumer trends. Meanwhile, contradictory signals from consumers demanding both fair trade and low	<i>"Europe has outsourced too much, we don't have food sovereignty or fertilisers"</i> (5) <i>"Consultants help save money"</i> (3) <i>"The problem is intermediaries"</i> (3) <i>"Consumers go crazy over proximity, fair trade, vegan products, but then it's costly"</i> (3) <i>"Proximity and seasonality mean seasonal products are always expensive"</i> (4)

	prices reflect deeper mismatches in the current economic regime.	
Regime _Social [12]	The social regime of agriculture is characterised by structural limitations in workforce renewal, cultural resistance to change and a disconnection between agricultural practices and broader societal recognition. Farmers are perceived as working with their hearts rather than business logic, and many lack digital tools or strategic planning. The average age of farmers is over 55, which creates a major barrier to adaptation and innovation, especially in a context where intergenerational renewal is weak or absent. Several interviewees describe a generalised ignorance about the agricultural world from outside stakeholders and institutions. This gap exacerbates misunderstandings and contributes to the stigmatisation of the sector. At the same time, the narrative surrounding agriculture is shifting, sometimes criminalising its environmental footprint rather than valuing its social and cultural role. Risk aversion remains high, especially among those with precarious livelihoods, reinforcing a conservative mindset that limits experimentation.	<i>“Small farmers work with their heart, not their head; no one spends a dime on software” (3)</i> <i>“There is a lack of available workforce, currently scarce and mostly migrant” (5)</i> <i>“The average age of farmers is over 55, which creates barriers to adaptation” (5)</i> <i>“-It’s always been done this way - there is aversion to risk in agriculture. If they lose two harvests, they lose the whole crop” (4)</i>
Regime _Technological [1]	The current technological regime in agriculture remains largely influenced by the introduction of agrochemicals from the 1970s. These approaches are still widespread and reflect a legacy of productivity-driven logic that often disregards long-term soil health and ecological balance. Despite the rise of alternative or regenerative approaches in niches, dominant practices continue.	<i>“Traditional agriculture that started in the 70s introduced agrochemicals, which are not taking care of the soil, is a problem” (4)</i>
Regime _Environmental [3]	The current environmental regime in agriculture continues to rely on agrochemical systems that contribute to soil degradation and dependence on fossil fuels. Despite increasing awareness of their ecological impact, especially under the pressure of climate change, shifts toward environmentally sustainable practices remain limited within dominant structures. Existing practices are still misaligned with ecological regeneration and resilient soil care, suggesting inertia within the environmental regime of agriculture.	<i>“Agrochemicals have been refined, but soils are increasingly degraded. They are aware, but they are not yet making the change” (4)</i>

6.2.1.3 Landscapes

Landscapes *Number of citations per category in brackets [X]	Summary of description	Literal citations *Number of interview in parenthesis (), see section 3.2
Landscape _Political [9]	Participants foresee future disruptions derived from geopolitical conflicts and regulatory instability. Political uncertainty, protectionist debates and the	<i>“After the conflict in Ukraine, the EU realised its vulnerability</i>

	perceived disconnect between EU-level regulations and local realities are seen as external pressures. The war in Ukraine is viewed as a trigger for shifts toward energy independence and food autarky. These broader political shocks may lead to a move away from global interdependence toward more self-sufficient models. Although some stakeholders advocate for more localised policies, the European Union remains highly interventionist and bureaucratically dense, often perceived as unaware of on-the-ground impact.	<i>and is returning to self-sufficiency” (4)</i> <i>“We need to raise awareness and adopt political measures to protect our products” (3)</i> <i>“Political uncertainty is present” (3)</i>
Landscape _ Economic [1]	Some experts underline that the broader economic context is expected to act as a barrier rather than an enabler of transformation. Structural costs, especially those related to inputs, are anticipated to rise significantly. This evolving context described as economically adverse could have a critical effect on the feasibility of sustainable or innovative transitions in the agricultural sector.	<i>“The economic context will not help is everything is more expensive” (5)</i>
Landscape _Social [1]	Some experts warn that increasing social tensions and the acceleration of rural depopulation may act as significant external pressures on the sector. These dynamics, while not certain, could exacerbate existing vulnerabilities in rural areas.	<i>“There will be many social conflicts and depopulation” (5)</i>
Landscape _Environmental [2]	Interviewees pointed to environmental risks, such as the potential for major plagues (e.g. locusts or flies) or climate catastrophes in key global exporters like Brazil or Argentina, as possible external disruptions to the food system. These landscape-level dynamics would be difficult to anticipate or control from within the sector, but could fundamentally reshape availability, pricing, and the urgency of adaptation.	<i>“Potential plagues could disrupt everything” (4)</i> <i>“Any climate disaster affecting major importers and exporters such as Brazil and Argentina could have a substantial impact on us” (4)</i>

6.2.1.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [3]	The sector shows signs of increasing fragility in terms of job continuity and generational replacement. Many farmers are closing down due to economic unsustainability, and the ageing of current land owners threatens business continuity and pension security.	<i>“Agricultural economic activity (for a small farmer) is no longer viable, they are closing” (5)</i> <i>“Without replacement, you must close and the retirement is at risk” (4)</i> <i>“There is no generational replacement.” (5)</i>
Vulnerability_Finance [4]	Financial vulnerability is particularly evident among small and medium farmers who rely on public subsidies to stay afloat. Rising costs,	<i>“There is a financial overburden on self-employed workers” (5)</i>

	overburdening of the self-employed, and lack of effective economic models are key concerns. Also, poor households that are not able to face the increase of prices due to inflation, climate change and geopolitical reasons.	<i>"People do not want subsidies, they want a purpose. Bread for today is hunger for tomorrow". (5)</i> <i>"Small and medium land-owners will survive from subsidies" (3)</i>
Vulnerability_Skills [7]	The lack of technological readiness, aging workforce and digital exclusion are seen as significant obstacles to adaptation. There's a widespread recognition of a generational and capacity gap that limits innovation.	<i>"Skills gap: surely there is" (4)</i> <i>"The countryside is prioritized beyond strategy and digitalization... Farmers should realize they will be left out..." (3)</i> <i>"There has been a change on required skills (with digitalisation)" (5)</i>

6.2.1.5 Disadvantaged groups (DGs)

According to the interviews, DGs in the agricultural sector are those most at risk of being excluded from the transition toward more sustainable, digitized and climate-resilient farming. These include small and medium farmers relying on subsidies, older farmers with low digital skills, isolated rural workers and migrant labourers concentrated in precarious roles (e.g., seasonal farm workers that arrive in summer from fruit harvest).

Experts highlight that many smallholder farmers face financial dependence on public aid and struggle to adapt due to gaps in digitalization and innovation. Aging profiles, lack of succession from parents to sons and daughters and insufficient generational renewal are also major barriers. Moreover, migrant workers, though essential for the sector, often face social and territorial marginalization, with limited mobility or prospects for long-term integration. The absence of territorial incentives and the erosion of labor attractiveness further deepen these vulnerabilities.

Disadvantaged groups (DGs) [12]	Summary of description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Farmers with small or medium fields	Dependent on subsidies, low investment capacity, risk of exclusion from digital and sustainable farming.	Risk of being left behind in the twin transition due to low financial autonomy, making them highly dependent on public support and unable to adopt innovations.	<i>"Small and medium farmers will survive thanks to subsidies. They are SMEs, clear innovation gap" (3)</i>

Older farmers	Generational barriers to adaptation, resistance to digital tools and innovation.	Difficulty in adopting new technologies and adapting to changes due to aging and lack of digital or strategic renewal, reinforcing structural stagnation in the sector.	<i>“The average age of farmers and livestock keepers is over 55” (5)</i>
Isolated rural workers	Precarious work cycles, seasonal concentration, lack of territorial incentives.	Exposure to precarious and short-term employment, high turnover, and weak territorial retention, leading to reduced social cohesion and depopulation.	<i>“There are no territorial incentives to be in the rural areas” (5) “Some people do not go outside their ‘niche’” (5)</i>
Migrant laborers	Overrepresentation in low-value roles, limited upward mobility, long-term social risk	Social vulnerability due to structural dependence on unstable roles, with limited integration and growing risk of exclusion in scenarios of sectoral decline or automation.	<i>“What will happen to established migrant population without employment in the rural areas? A big social risk” (5)</i>
Low-income households	Households with low purchasing power who are highly sensitive to food price volatility	People facing obstacles to access healthy, local or sustainable products due to the increase of prices.	<i>“The economic context may not help to poor households neither, products will be more expensive” (5)</i>

6.2.2 First round of Delphi survey

6.2.2.1 Elaboration of statements

32. By 2050, fewer than 30% of small farms (bellow 10 hectares) will be able to independently adopt precision agriculture technologies without external support.
[Vulnerability_Skills] [Niche_Governance] [Niche_Economic]

33. The twin transition will accelerate land concentration, reducing the number of family farms across Europe by 2050. **[Niche_Economic] [Niche_Social] [Niche_Governance]**
34. Data collected by farmers will increasingly be monetized by third parties unless regulatory frameworks ensure data ownership and value sharing. **[Niche_Economic] [Niche_Technological] [Landscape_Political]**
35. Blockchain technology will significantly increase transparency and trust in agricultural supply chains by 2050. **[Niche_Technological] [Niche_Economic] [Niche_Social] [Niche_Economic]**
36. By 2050, rural depopulation will be a reality in the EU. **[Niche_Social]**
37. Technological investment in agriculture (AI, biotech) will primarily benefit large agribusinesses. (i.e. non-adapted to local context) **[Niche_Economic] [Niche_Technological]**
38. Generalist digitalization policies (i.e. non-adapted to local context) in agriculture will be a paradigm by 2050. **[Niche_Governance] [Landscape_Politics]**
39. Data-driven agriculture will be a reality in 2050, with efficient and interoperable data lakes and data hubs as key components of modern farming. **[Niche_Economic] [Niche_Technological]**
40. Small farmers will increasingly shift to fine food, local production and proximity-based distribution to stay competitive. **[Niche_Economic]**
41. Large investment funds will own more than 65% of the fields in Europe by 2050. **[Niche_Economic]**
42. Urban consumers' demand for high-quality, traceable food will be a major driver of small farmers' adoption of new business models. **[Niche_Economic] [Niche_Social]**
43. Artificial intelligence will be widely used in decision-making processes for crop and livestock management in Europe by 2050. **[Niche_Technological]**
44. Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems. **[Landscape_Political]**
45. Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labor. **[Niche_Economic] [Niche_Social] [Vulnerability_Skills] [Vulnerability_Employment] [Disadvantage_Groups]**
46. Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector. **[Niche_Economic] [Niche_Technological]**
47. Large companies buying land and retaining workers will become a key strategy to de-risk digital transitions for small farms. **[Niche_Economic]**
48. By 2050, farmers older than 55 years old will lack the digital skills needed to comply with agricultural digitalization policies. **[Vulnerability_Skills] [Vulnerability_Employment] [Disadvantage_Groups] [Niche_Technological]**
49. Displaced rural workers will face long-term unemployment and exclusion from the digital economy. **[Vulnerability_Skills] [Vulnerability_Employment] [Disadvantage_Groups] [Niche_Technological]**
50. By 2050, the agriculture sector will experience a lack of generational renewal. **[Vulnerability_Employment] [Disadvantage_Groups]**
51. Smart farming technologies will become embedded in agricultural policy compliance frameworks, making tech adoption a condition for subsidies. **[Niche_Technological] [Niche_Economic] [Niche_Governance]**

6.2.2.2 Analysis of results

By 2050, fewer than 30% of small farms (below 10 hectares) will be able to independently adopt precision agriculture technologies without external support.		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	20%
Neutral	1	20%
Low	3	60%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?²	Consensus degree³
	No, remove	20%
Summary of qualitative comments	Respondents generally agreed that adoption among the smallest farms hinges on tailored training and supportive measures. One Italian commentator stressed that “specialized training” will be indispensable, while another noted the high cost of precision tools and slim margins make widespread uptake challenging. A third observed that “there are strategies and solutions for all types of farms,” suggesting that targeted programs could help bridge the gap. On impact, most viewed this scenario as transformative if achieved although one warned of “cultural risks for rural cohesion,” and another saw little impact without inclusive rollout.	

The twin transition will accelerate land concentration, reducing the number of family farms across Europe by 2050.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%

² Consensus is reached when the % over the total answers of combining “very high” and “high” in likelihood terms is >60%

³ Consensus degree is defined as: [(very high + high)/total number of answers]*100

Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Several respondents feared this would be a “great failure” if unchecked, pointing to market forces and compliance burdens that favour large operators. A second highlighted that both climate and digital mandates may unintentionally squeeze out smallholders, while a third argued that existing “strategies and solutions” could blunt these pressures. Impact-wise, contributors warned that many small farms in rural areas would be left behind, undermining local economies and community fabric. One respondent emphasized that family farms themselves are crucial, and their loss would have “deeply concerning” social consequences.	

Data collected by farmers will increasingly be monetized by third parties unless regulatory frameworks ensure data ownership and value sharing.		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	60%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	1	20%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Views were mixed but cautious: one respondent simply said “Let’s hope not,” while others stressed that data sharing and processing outside original farm contexts is already underway. A third noted that good governance could ensure farmers themselves capture value rather than intermediaries. When asked about impact, many pointed to persistent legal-literacy gaps at the farm level, meaning exploitation could continue under the radar. Others felt that better frameworks would ultimately enhance efficiency, “making agriculture more efficient” through responsible data use.	

Blockchain technology will significantly increase transparency and trust in agricultural supply chains by 2050.		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	3	60%
Low	2	40%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	1	20%
Neutral	4	80%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	0%
Summary of qualitative comments	Most participants were ambivalent or unfamiliar: responses ranged from “No idea” to acknowledging blockchain’s promise but noting uneven uptake across actors. Several were unable to gauge its likelihood, suggesting that sectoral diversity may slow consensus around immutable ledgers. Impact assessments were similarly cautious: while some foresaw meaningful benefits in short, local chains, others doubted systemic transformation absent broader cooperation. The sense was that blockchain could help in niche applications but won’t be a universal panacea.	

By 2050, rural depopulation will be a reality in the EU		
Likelihood	Number of answers	% over the total
Very high	3	60%
High	0	0%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%

Summary of qualitative comments	Respondents broadly agreed this trend is already in motion. One hoped for “alternative solutions,” another pointed out that automation and new agri-tech could even help repopulate villages if paired with supportive policies. Yet most saw continued exodus unless rural development strategies are significantly enhanced. On impact, participants emphasized that depopulation would “affect everything from services to work,” eroding Europe’s rural fabric. A common refrain was that failure to incentivize rural livelihoods would exacerbate social and economic divides.
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Technological investment in agriculture (AI, biotech) will primarily benefit large agribusinesses. (i.e. non-adapted to the local context)		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Many respondents agreed that ROI timelines and scale advantages favour major players. A minority cautioned that thoughtful transfer mechanisms could help distribute benefits more evenly. Impact assessments warned of stark structural inequalities: one highlighted risks of “rural disenfranchisement,” and another stressed that without corrective policies, gains will accrue to a shrinking number of large operators.	

Generalist digitalization policies (i.e. non-adapted to local context) in agriculture will be a paradigm by 2050.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%

Neutral	0	0%
Low	0	0%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Responses were generally skeptical or undecided: some admitted they couldn't fully assess, while others compared the risk to urban-centric policies in healthcare and education that overlooked rural needs. A third noted that policy design has always shaped adoption, for better or worse. On impact, most warned of severe mismatches between top-down mandates and on-the-ground realities, arguing that rigid frameworks could stifle innovation in diverse agricultural settings.	

Data-driven agriculture will be a reality in 2050, with efficient and interoperable data lakes and data hubs as key components of modern farming.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	0	0%
Neutral	3	60%
Low	0	0%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	1	20%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	20%
Summary of qualitative comments	While one respondent couldn't judge, others saw clear momentum: realizing "efficient and interoperable" systems depends on aligning incentives (especially for SMEs in smallholder regions) and on building shared infrastructures. A third affirmed that "that's the way" forward once frameworks mature. Impact forecasts were positive: data-driven systems could unlock better farm incomes, sustainability metrics, and evidence-based policymaking. Many stressed that transparent, farmer-centric governance will determine whether these gains materialize.	

Small farmers will increasingly shift to fine food, local production and proximity-based distribution to stay competitive.		
Likelihood	Number of answers	% over the total

Very high	0	0%
High	2	40%
Neutral	3	60%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Participants agreed this shift is already emerging: several pointed to reorganizing markets around quality, while others cautioned it will likely remain a niche opportunity for some producers rather than a universal model. A third saw it as one of several potential avenues for farm diversification. Impact perspectives highlighted that local and regional gains could be substantial, but success hinges on robust marketing, product recognition, and consumer engagement.	

Large investment funds will own more than 65% of the fields in Europe by 2050.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Most respondents were unsure or skeptical, noting that while institutional land acquisition is growing, economic conditions and regulatory checks will shape outcomes. One worried it could already be happening, another said it “depends on the economic context.”	

	Impact assessments were mixed: some felt consolidation could provide liquidity and professional management, while others warned it may render land access “accessible only to a small cohort.”
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Urban consumers’ demand for high-quality, traceable food will be a major driver of small farmers’ adoption of new business models		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	60%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Respondents generally believed city dwellers’ preferences could spur artisanal and traceable-food markets, though one doubted it would go mainstream without scalable logistics. Another simply said “should happen” if supply chains align. They judged impacts as potentially profound, opening premium value-chain niches, yet cautioned that without accessible channels, only a few may benefit.	

Artificial intelligence will be widely used in decision-making processes for crop and livestock management in Europe by 2050.		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	0	0%

Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments		
	Most respondents were intrigued but unsure: a couple said they “cannot answer,” while others envisioned generative AI advising on everything from planting regimes to herd health, provided systems are locally adapted. Impact projections emphasized that AI’s benefits can be empowering or extractive (driven by who controls and validates the tools) and that careful oversight will determine whether AI amplifies farmer agency or simply automates external prescriptions.	

Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems.		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	10%
Low	0	0%
Very low	0	0%
Impact		
Likelihood	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action		
	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments		
	Nearly everyone agreed compliance burdens are intensifying: one respondent called the effect “strong,” another noted sharper rises in the lead-up to 2050, and a third pointed out that larger estates face similar pressures too. Impact-wise, participants warned of systemic strains on the very rural base environmental rules aim to protect, potentially squeezing out the smallest operators unless support mechanisms are scaled.	

Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labor.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	3	60%
Neutral	1	10%
Low	0	0%

Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments	Respondents viewed this as a “dramatic scenario” for manual labor, particularly migrant workers, though one noted the effect varies by crop and region. Most judged the impact as a substantial rise in displacement, albeit with a counter-view that automation could make agriculture more sustainable overall, but not without deliberate workforce transition programs.	

Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	3	60%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	2	40%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	80%
Summary of qualitative comments	A majority were hopeful (one simply said “let’s hope so,” another pointed to institutional pilots, and a third saw clear bilateral benefits) while some remained cautious about misaligned incentives or resistance. Impact forecasts were positive: alliances could unlock fresh investment, shared R&D, and modular solutions that outpace siloed sector approaches.	

Large companies buying land and retaining workers will become a key strategy to de-risk digital transitions for small farms.

Likelihood	Number of answers	% over the total
Very high	1	20%
High	0	0%
Neutral	2	40%
Low	2	40%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	20%
Summary of qualitative comments	Opinions varied: some hoped family-run farms would retain autonomy, others observed that corporate land-holding can secure investment and skills retention. One noted that smallholders might still find room to operate through cooperative models. On impact, many felt such a trend could centralize decision-making and marginalize independent producers, though others argued it might stabilize rural employment if managed inclusively.	

By 2050, farmers older than 55 years old will lack the digital skills needed to comply with agricultural digitalization policies		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	0	0%
Neutral	3	60%
Low	0	0%
Very low	1	20%
Impact	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	3	60%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	20%
Summary of qualitative comments	Respondents noted that emerging generations are already digital natives and could bridge the gap if mentorship schemes are in place.	

	Impact evaluations warned that without dedicated training for older cohorts, technology adoption may stall, even as younger entrants refresh the sector.
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Displaced rural workers will face long-term unemployment and exclusion from the digital economy.		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	0	0%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%
Summary of qualitative comments	Nearly all agreed on the urgency of professional retraining to prevent permanent exclusion. One respondent cited agriculture as a case study of this risk, while another pointed out that new ag-sector roles could offset some losses. Impact concerns focused on societal fallout: deepened inequalities and regional disenfranchisement if automation-driven displacement is not countered by job-creation and upskilling.	

By 2050, the agriculture sector will experience a lack of generational renewal		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	2	40%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	60%
High	1	20%
Neutral	1	20%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	60%

Summary of qualitative comments	Several participants expressed regret and anticipated that digitization might both deter and entice young entrants depending on outreach. They judged the impact as severe: continued aging will strain knowledge transfer and farm succession, threatening sector vitality unless youth engagement is prioritized.
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Smart farming technologies will become embedded in agricultural policy compliance frameworks, making tech adoption a condition for subsidies		
Likelihood	Number of answers	% over the total
Very high	1	20%
High	1	20%
Neutral	2	40%
Low	1	20%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	40%
High	1	20%
Neutral	2	40%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	40%
Summary of qualitative comments	Responses ranged from uncertainty to affirmation that policy reforms (such as CAP modernization) are already embedding remote sensing and data checks. Others stressed that true progress requires coupling technology with on-farm support rather than mere enforcement. On impact, many cautioned that tech-heavy compliance could exclude resource-poor farmers, while others saw it as an efficiency gain that, if well-managed, “would contribute making agriculture more efficient.”	

6.2.3 Second round of interviews

6.2.3.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [21]	Interviewees identify new business dynamics emerging at both ends of the scale: on the one hand, small advisory firms and cooperative teams are stepping in to support small-scale farms; on the other hand, large investors and corporations are increasingly entering the sector, drawn by perceived profitability. This trend raises concerns around the	“If investment funds are entering agriculture, it’s because they see profitability. Maybe they can make it work.”(11)

	erosion of the “farm family” model and the risk of excessive corporatisation. Several participants highlight the need for a return on sustainability investments, income diversification and affordability of technology. While digital tools can enhance efficiency, many farmers face high capital expenditures (CAPEX), rising machinery costs and limited access to financial guarantees. New forms of cross-sector collaboration e.g., with energy or tourism offer promising economic opportunities, but remain underexploited. Meanwhile, farmers experience economic stress due to volatile income cycles, land concentration for large businesses and limited services in rural areas.	“Large companies will be the ones with money to buy land, not farmers.” (11) <i>“Corporatization goes against the principle of farming family” (12)</i> <i>“There’s a big CAPEX. In France you must be a farmer to buy land. In Ireland anyone can.” (13)</i> <i>“Farming is becoming very complex. You’re a manager now. If you’re good with digital tools, you’ll adjust better.” (13)</i>
Niche_Governance [35]	Interviewees highlight that agricultural legislation, while intended to foster sustainability (e.g., under the Green Deal and Farm2Fork), is often perceived as complex, distant or misaligned with the realities on the ground. There is widespread demand for policies that are better adapted to territorial diversity, levels of sustainability per farm, and sector-specific characteristics. Public policy is seen as both a driver and a burden: farmers face increasing requirements for emission reduction, traceability, and environmental certification, while struggling with bureaucratic overload and unclear compliance mechanisms. Interviewees stress the need for capacity-building measures and advisory support alongside any regulatory change. Cooperatives and farm advisors are key intermediaries, but also face structural limitations. In parallel, data governance is becoming a critical issue. Farmers are concerned about the collection and use of their data by public bodies without consent, and call for clearer frameworks on data ownership, sharing and benefit distribution. Some propose that digitalisation should not only serve efficiency but also simplify administrative procedures and empower decision-making at the farm level.	<i>“We should detect variability and adapt public policies to each field. In agriculture we talk about precision — policies should follow the same logic.” (11)</i> <i>“Many laws are based on detecting substances that should not be there... but this creates a mess. The administration should monitor how it is applied.”(11)</i> <i>“Nobody has told the farmers they are doing it wrong. They feel left behind.”(11)</i> <i>“Digitalising everything won’t necessarily make you a better farmer — but the data could at least help reduce bureaucracy.” (11)</i> <i>“The cooperative is strong; they can ask the price they want. The farmer doesn’t have much negotiation power.” (13)</i> <i>“We live in a democracy, and long-term policy is incompatible with short political cycles... If we want to succeed, we need to make unpopular decisions.”(14)</i>
Niche_Social [25]	Interviewees describe agriculture as not only an economic activity but a way of life for many family farmers. They highlight intergenerational, cultural and lifestyle dimensions that shape farming trajectories, with a wide variety of profiles, from part-time farmers with off-farm jobs to highly educated young entrants. Farming is seen as deeply rooted in tradition, but also undergoing profound identity transformations driven by technology, corporatisation, and sustainability debates.	<i>“Small family farms are often older and struggle with precision agriculture... but if there’s a young person with curiosity, that’s a different story.” (11)</i> <i>“We also need to train consumers. Not everything is ecological, but there is a sustainable way.” (11)</i> <i>“We need better services and</i>

	Some narratives emphasise the loss of connection with land among younger generations, the “displacement” of rural youth, and a growing anti-farmer sentiment in the name of sustainability. In contrast, others point to signs of generational renewal, especially when farming becomes more attractive or better connected to digital or lifestyle aspirations. Formal education and training are also increasingly accessible, helping new farmers enter the system. Consumer behaviour and social values play a critical role. Food consumption trends — such as deglobalisation or proximity-based choices — are slowly shifting. Yet the burden of climate responsibility is often perceived as falling disproportionately on farmers. Interviewees call for more inclusive transition strategies that address both consumers’ role and the lived experience of rural actors.	<i>communications in the sector.” (11)</i> <i>“The sector must be more attractive, a place where you can live.” (11)</i> <i>“Ireland is ahead: broadband in rural areas is better than in cities. Rural housing is improving.” (12)</i> <i>“The main problem is ‘displacement’, kids don’t want to live in rural areas. There’s even anti-farmer sentiment based on sustainability.” (12)</i> <i>“There will be generational renewal. Maybe technology will make people want to ‘touch grass’.” (12)</i> <i>“We must create links between existing and future farmers. Traditions must be preserved.” (12)</i> <i>“There’s a rise in education: certifications, higher levels, people working in agri-industry... The model is shifting, and farmers are increasingly well-educated.” (12)</i> <i>“There are new food trends, like deglobalisation.” (13)</i> <i>“Other countries favour proximity. France has a market for it.” (13)</i> <i>“You have to fight against big forces. But consumers in some countries are ready for price premiums.” (13)</i> <i>“Family farming involves complex intergenerational dynamics.” (13)</i> <i>“Farmers feel blamed for climate change.” (13)</i> <i>“The transition will only work if consumers say yes and accept the consequences: less travel, less meat... otherwise we have real problems.” (14)</i>
Niche_Technological [26]	Technological innovation is broadly seen by interviewees as a key enabler for a more efficient, sustainable and resilient agricultural model. Precision farming, AI-based tools, and digitalisation are among the most discussed trends. However, adoption remains uneven, and many tools are still designed with limited user involvement or insufficient usability. Small farms, older farmers and those with less access	<i>“Going digital doesn’t mean you farm better — but it could help reduce bureaucracy.” (11)</i> <i>“Farmers that adopt digital tools, even for backoffice, are more efficient.” (12)</i>

	to training may struggle to adopt or benefit from these innovations. At the same time, interviewees raised concerns about data overload, interoperability, and the need to ensure farmers retain control over the data they generate. Digitalisation should not only increase efficiency but also reduce bureaucratic burden, they argue and the risk of exclusion due to technological barriers remains a challenge.	“Semantic interoperability has changed... but data isolation is worth nothing.” (12) “Precision agriculture has existed since the 80s... but only recently it has become a serious option. It’s still mostly research-based. We need farmers to adopt it.” (11) New crops like soy are emerging as a result of climate change, cross-sectoral innovation is rising.” (14) “We’re not growing land, we’re losing it. We must make the most of it with technology. But it’s risky and costly.” (12) “Digitalisation is a key trend. Many new tools are becoming available.” (13) “In livestock, it’s harder to reduce emissions; innovation must come from additives, genetic improvements, etc.” (13)
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6.2.3.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political 4]	Interviewees emphasise that agricultural policy operates within complex and sometimes contradictory political frameworks. The Common Agricultural Policy (CAP) is cited as a central mechanism, providing a unified framework across EU Member States. However, its implementation varies significantly by country, resulting in unequal effectiveness and perceptions of unfairness. Strategic Plans at national level enable flexibility, but also fragment policy impact across territories. Some interviewees describe a “state of paralysis” among decision-makers, prioritising member interests over long-term transformation. The gap between international commitments (e.g., the Paris Agreement) and national action plans is also noted. While the EU leads in emissions reduction, there is scepticism about the global equity of climate action, with concerns that disproportionate effort is expected from European farmers while others lag behind.	“There’s a state of paralysis... politicians just look after their members.” (12) “The Common Agricultural Policy provides a framework for 27 countries, but each Member State develops its own Strategic Plan. Ireland has been able to adapt, others, not so much.” (12) “The CAP is supposed to be equal across Member States, but reforms now allow each country to manage its own money... Each CAP is different, and so is its effectiveness.” (13) “The Paris Agreement is often praised, but how far are we from real change? Why should European farmers bear all the burden when others aren’t doing the same?” (14)

Regime _ Economic [4]	Interviewees point to persistent structural challenges in the agricultural economic regime. The sector is divided between traditional and organic production, but there is a lack of recognition and incentives for sustainable practices that do not fall strictly under the "organic" label. Farmers navigating this "grey zone" face economic disadvantages despite adopting environmentally responsible methods. High capital expenditure and investment risks further constrain economic choices. Many farmers become locked into costly production models, unable to shift due to sunk costs or financial limitations. Interviewees also highlight the ongoing trend toward land concentration, which enables economies of scale but marginalises smaller farms. National differences in food systems, such as the shorter, more value-driven supply chains in Italy compared to Ireland, also illustrate the uneven structure of agricultural economies across Europe.	<i>"There's ecological farming and traditional farming. But what's in between? A total mess. You can farm sustainably without being certified as organic, but that's not recognised." (11)</i> <i>"Farmers are getting locked into certain models because the capital cost is too high to change." (12)</i> <i>"The business model in Italy is completely different, short supply chains, more respect for food. It's not the same in Ireland." (12)</i> <i>"Land concentration is already happening. It allows economies of scale, but it makes it harder for small farms to survive." (13)</i>
Regime _Social [7]	Interviewees highlight that social norms, cultural attitudes and lifestyle expectations deeply shape agricultural practices and trajectories. Farming is seen both as a business and a way of life, with strong attachment to land and community practices. In some regions like Ireland, this connection is being reshaped by housing restrictions and changing aspirations, while in others, such as Italy, food culture and short supply chains influence agricultural models. Farmers are portrayed as resilient and pragmatic actors, often innovating under pressure and operating through informal knowledge-sharing channels like WhatsApp, podcasts or YouTube. However, they are also exposed to societal criticism, with some feeling blamed for the environmental consequences of policies they were told to follow. There is a growing need to better formalise communication and restore the perceived social value of farming.	<i>"Farmers operate in communities of practice; they share tips through WhatsApp, Telegram, or even podcasts. YouTube is huge in this sector. We need to formalise how we spread knowledge." (12)</i> <i>"People became very polarised. Farmers felt really bad about their role... and were blamed in 2014 for following policy." (12)</i> <i>"Farmers are very innovative and resilient. They solve things." (12)</i> <i>"They have a deep attachment to land. Everyone wants to build a house next to their field, but municipalities often prohibit it. This love for land comes with age, though it's changing." (12)</i> <i>"It's a hard lifestyle. But people can accept it, if they're not also suffering from everything else. A decent income is the key." (13)</i>
Regime _Technological [2]	Interviewees describe a paradox in the technological dynamics of the agricultural regime. While precision agriculture has been under development since the 1980s and is now becoming a serious option, there is also a perception that mainstream agriculture has fallen behind in recent years. This contrast suggests	<i>"Precision agriculture has been in place since the 1980s. In recent years, it has become a serious option." (11)</i>

	that while the sector was once a pioneer in adopting innovation, it now faces structural and systemic barriers that slow down the large-scale uptake of emerging technologies.	<i>"Agriculture used to be quick in adopting technology... but now it's falling behind." (12)</i>
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6.2.3.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [7]	Interviewees highlight several geopolitical and political tensions that shape the broader landscape for agriculture in Europe. The war in Ukraine and dependency on Russian gas have triggered a renewed focus on energy security, indirectly promoting renewable energy. This energy shift also affects agricultural systems, particularly regarding input costs and infrastructure investments. At the same time, European cohesion is tested by divergent national responses to global pressures, including trade tariffs, AI regulation, and food security concerns. The Common Agricultural Policy (CAP) and broader EU initiatives must balance national interests with continental solidarity, especially as right-wing and populist parties grow in influence and increasingly oppose climate commitments. Broader concerns also emerge about public trust in information, media independence and the politicisation of climate and sustainability debates, especially in transatlantic contexts.	"The AI Act uses a risk-based model: it includes transport and energy as critical infrastructure, but food production was not included." (12) "We're living turbulent times... Each member state looks after its own interest, but the EU must stay united." (12) "There's a shift in America... We need to become more productive too." (12) "ZERO TRUST INFORMATION: We have to pause and rethink the role of public broadcasters... It's getting complicated." (12) "There's fear of right-wing populist parties, like in Italy. They say no to climate action, and once in power, they will dismantle it." (14)
Landscape _Economic [2]	Interviewees identify several macroeconomic pressures shaping the future of agriculture. On the one hand, geopolitical tensions and the growing focus on food security are reframing the value of shorter supply chains, previously seen as a resilience strategy, now also posing potential limitations. On the other hand, international market dynamics continue to drive strong price volatility, affecting both consumers and producers. Commodities are particularly exposed, as global fluctuations directly impact the income stability of farming operations.	"Food security is becoming crucial in geopolitical terms... but shorter supply chains might actually become a problem." (12) "Prices could rise at consumer and producer level; the real issue is volatility. International markets set commodity prices, and that creates uncertainty." (13)
Landscape _Environmental [4]	Interviewees underline the growing influence of environmental and climatic changes on agriculture, describing them as a major driver of stress, uncertainty and transformation. Volatile weather patterns, including droughts, floods and extreme heat are already affecting crop viability, productivity and farming strategies. These changes are perceived as structural rather than episodic, suggesting the need for deep adaptation.	"The weather has a big impact on the food... They have to make everything in a certain scale. It is not cost-effective." (12) "Also, weather volatility related to climate change. More and more variation, years are becoming more scarce. Farmers should be extremely stressed." (13)

	Climate change is seen as a systemic force that “will change everything”, reshaping both the physical environment and the economic conditions under which farmers operate. Respondents note that farmers are increasingly vulnerable, with unpredictable yields, shifting planting cycles, and a need to rethink feasibility at smaller scales. In this context, resilience planning and feasibility studies become essential tools to mitigate risks.	“Climate change (adaptation) will change everything.” (13) “As a farmer: climate changes... Floodings, drought... too hot temperatures.” (14)
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6.2.3.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [8]	Interviewees express concern about the growing mismatch between agricultural workforce needs and the current training, skillsets and organisational structures. As farming becomes more complex and digitised, traditional labour profiles are being displaced or require urgent upskilling. There is a strong demand for multidisciplinary teams combining agronomic knowledge with technological expertise, especially in cooperatives and advisory services, where technical staff often face bureaucratic overload and lack specialised support. The transition to a more digital and precision-based model implies a deep reconversion of agricultural jobs. Yet, many of these new roles are not fully defined, and education systems struggle to keep pace. Schools are beginning to showcase modern tools like GPS-enabled tractors or satellite imaging to make the sector more attractive to younger generations, but interviewees agree that more systemic efforts are needed to retain and prepare future workers. At the farm level, automation is reducing certain manual tasks, yet labour shortages persist, particularly where farmers must act as both managers and technicians. The employment vulnerability is thus double: risk of technological obsolescence for older profiles and difficulty in attracting young, skilled talent for the new demands.	<i>"All the digitalisation of agriculture (without losing focus on agronomy) implies a reconversion of agricultural jobs." (11)</i> <i>"In cooperatives, technicians deal with too much bureaucracy — we need someone dedicated just to that." (11)</i> <i>"The agronomy degree is very broad... but the person leading a field should be both an expert in agronomy and familiar with technological solutions. We need new professional profiles." (11)</i> <i>"Multidisciplinary teams must exist: an agronomic advisor and an engineering technician." (11)</i> <i>"Schools organise workshops to make agriculture attractive to youth: smart tractors, GPS, satellites..." (11)</i> <i>"Farming is becoming very complex; they are becoming managers, with lots of paperwork. If you are good with digital tools, you'll adapt better... but there is labour shortage in farms." (13)</i> <i>"Automatic milking machines mean one can manage 200 cows part-time, but that's it. You can't do more. Technology can help some types of farmers." (13)</i>
Vulnerability_Finance [2]	Financial insecurity is a major source of stress for farmers, compounded by price volatility, climatic uncertainty and the structural costs of	<i>"Farmers should be extremely stressed. No guarantees for the farmer — this is inherent</i>

	land access. Interviewees underline the fragility of farming incomes, which remain highly dependent on unpredictable factors such as market dynamics or weather conditions. Even one bad season can have disproportionate effects, with few financial guarantees to mitigate the impact. Although the EU has implemented some support mechanisms for young farmers, such as inheritance tax exemptions or infrastructure grants starting a farm remains capital-intensive. The high upfront investment needed, often estimated in the millions, acts as a structural barrier to entry, especially in a context where access to land is increasingly limited and subject to speculation. These financial barriers exacerbate generational gaps and reduce the resilience of the sector.	<i>to farming, but it's getting worse. You can get the hit for one year, even if you have one good year..." (13)</i> <i>"You have to start with 2M to generate a farm, subject to price and weather volatility — even if you have access to land." (13)</i>
Vulnerability_Skills [14]	Skills gaps are emerging as a critical vulnerability for the agricultural sector, especially in the context of digitalisation and the green transition. Interviewees underline the challenges that older farmers and small family farms face in adapting to precision agriculture, often due to a lack of digital literacy or familiarity with emerging technologies. This generational divide risks leaving some groups behind, particularly where formal training or advisory support is lacking. There is a growing demand for new agronomic profiles that combine technical, agronomic and digital skills. Farmers are increasingly expected to manage complex systems, understand data flows, and adopt new technologies; yet many lack the resources or support to do so effectively. Interviewees emphasise the importance of targeted training, school-based workshops, and accessible advisory services to support this shift. Moreover, the average age of farmers remains high, which further exacerbates the difficulty of adapting to new demands. While some younger farmers (often digital natives) may be more open to change, structural barriers persist, especially for those outside of formal educational pathways.	<i>"There must be training for farmers. The services they hire generate data, often they don't even realise it. They must know what is being produced and how to use it." (11)</i> <i>"The agronomy degree requires knowing a bit of everything, but that's also a challenge... The farm manager must be both an agronomist and a tech-savvy expert. We need new profiles." (11)</i> <i>"Multidisciplinary teams are needed: agronomy advisors, engineering technicians..." (11)</i> <i>"Changes must come with training. You can't just impose cuts as in the EU Green Deal — you need tools to help farmers adapt." (11)</i> <i>"There are school workshops to make agriculture attractive to young people; with smart tractors, GPS, satellites..." (11)</i> <i>"The average farmer is 57 years old. But now the digital natives are coming in..." (12)</i> <i>"Some farmers automate everything and do it part-time... Technology can be the answer for some." (13)</i> <i>"There is resistance to change among farmers, while</i>

		<i>providers are shifting from analogue to digital.” (13)</i>
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6.2.3.5 Disadvantaged groups (DGs)

Interviewees identified multiple disadvantaged groups in the agricultural sector who may face increased vulnerability due to digitalisation, climate change, or regulatory shifts. Older farmers and small family farms may lack the digital skills or financial capacity to adopt precision agriculture tools, despite their potential to improve efficiency and sustainability. There is concern over the potential obsolescence of certain jobs (e.g., seasonal workers), as automation and technological requirements increase. Additionally, the sector is marked by an ageing workforce, complex access to land for young entrants, and high upfront costs, creating barriers for generational renewal. Stress, lack of guarantees, and emotional toll are also recurring themes, reflecting the precarious nature of the profession. These vulnerabilities underscore the need for targeted support, retraining, and inclusive transition policies.

Disadvantaged groups (DGs) [13]	Summary of description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Elderly smallholder in precision agriculture	Often less familiar with digital technologies, may struggle to adopt precision agriculture practices.	Risk of exclusion from digital transition and sustainability support schemes	<i>“Small family farms are often older and struggle with precision agriculture...” (11)</i>
Seasonal farm workers affected by automation	Roles like fruit pickers are at risk of disappearing due to mechanisation.	Job losses without realistic retraining options.	<i>“The seasonal workers that used to come for fruit picking... you can’t just turn them into robot designers.” (11)</i>
Farmers with limited digital skills	Difficulty understanding or using the data generated by their own farm operations.	Missed opportunities to benefit from digitisation and automation.	<i>“There must be training, most don’t even know what data is being generated.” (11)</i>
Farmers with limited financial resources	Farming requires expensive machinery, land, and inputs.	Risk of exclusion from innovation due to unaffordable investment.	<i>“It is risky and costly. Machines may be more expensive...” (12)</i>
Young entrants lacking access to land	Young people face high capital requirements and land access barriers despite policy support.	Difficulty entering or sustaining farming activity.	<i>“You need 2M to start a farm, even if you have access to land.” (13)</i>
Farmers overwhelmed by complexity	Increasing bureaucracy and complexity turn farmers into managers.	Those without support or skills may leave the profession.	<i>“Farming is becoming very complex; they are becoming managers.” (13)</i>
Farmers under high stress	The volatility of prices and climate shocks exacerbates emotional and financial precarity.	Mental health risks and burnout.	<i>“Farmers should be extremely stressed. No guarantees... one bad year is enough.” (13)</i>

6.2.4 Second round of Delphi survey

6.2.4.1 Elaboration of statements

Based on the interviews carried out during the second round, additional statements were included to the Delphi survey to remove the not agreed ones (see criteria in subsection 4). As in the first round, the MLP elements have been included in brackets.

52. Cloud-based platforms will deliver end-to-end “farm-to-fork” traceability visible to consumers **[Niche_Technological]** **[Niche_Social]**
53. Irrigation networks will use remote sensing and AI to schedule watering autonomously. **[Niche_Technological]**
54. By 2050, most family farms will join cooperatively governed data hubs. **[Niche_Technological]** **[Niche_Governance]**
55. Small farms will earn regular income from digitally managed ecosystem services (carbon credits, pollination). **[Niche_Technological]** **[Niche_Economic]**
56. By 2050, edge-computing nodes (e.g. 5G base stations) will cover nearly all cultivated land, enabling real-time AI on remote farms.
57. By 2050, farm management systems will bundle weather-indexed insurance via smart contracts. **[Niche_Technological]** **[Niche_Economic]**
58. Most small farmers will join pan-European digital cooperatives to coordinate production and market access across Member States. **[Niche_Technological]**
59. By 2050, public agencies and private ag-tech firms will jointly fund regional centers that deliver continuous digital literacy and business training. **[Niche_Governance]** **[Vulnerability_Skills]**
60. By 2050, receipt of any Common Agriculture Policy (CAP) subsidy will require farmers to complete a certified digital-skills programme. **[Vulnerability_Skills]** **[Niche_Governance]**
61. No farm under 10 ha will be able to adopt precision technologies without joining a formal support network of service providers or cooperatives. **[Niche_Governance]** **[Niche_Technology]**
62. EU and national agricultural digitalization schemes will include mandatory regional customization clauses to reflect local climate, soil and crop differences. **[Landscape_Political]**
63. National agricultural advisory services will integrate peer-coaching networks—pairing digitally proficient farmers with those needing upskilling. **[Vulnerability_Skills]** **[Niche_Governance]**
64. All public funding for farm digitalization will flow through farmer-led cooperatives or community hubs to ensure equitable distribution and cost-sharing. **[Niche_Governance]** **[Niche_Technology]** **[Vulnerability_Finance]** **[Niche_Economic]**

6.2.4.2 Analysis of results

Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems		
Likelihood	Number of answers	% over the total
Very high	3	75%

High	1	25%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	50%
High	2	50%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Respondents widely note that this trend is already underway and likely to intensify by 2050, driven by EU policies such as the European Green Deal, the Farm to Fork Strategy, and the CAP 2023–2027. Meeting requirements for eco-schemes, carbon reporting, and traceability demands resources that small farms often lack, creating disadvantages compared to larger operations. Some believe this could accelerate consolidation and reduce diversity in rural systems, while others see opportunities for small farms to benefit if they can access premium markets that value sustainable certification.	

Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labor		
Likelihood	Number of answers	% over the total
Very high	2	50%
High	2	50%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	4	100%
High	0	0%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Many comments describe automation as already in progress, with robotic harvesters, autonomous tractors, and AI-driven sprayers in use. Investments in agri-robotics are expanding in several European countries, and respondents expect adoption to continue,	

	particularly in controlled environments like vertical farming. While some see benefits in efficiency and addressing labor shortages, others highlight risks such as reduced job opportunities for low-skilled workers and negative social impacts in regions dependent on seasonal labor.
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Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector		
Likelihood	Number of answers	% over the total
Very high	3	75%
High	0	0%
Neutral	0	0%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Several respondents believe collaboration between agriculture and clean energy has strong potential to generate mutual benefits, though some question the direct role of such alliances in digitalization. Technology companies are viewed as potentially more influential in this area. Positive impacts mentioned include increased efficiency, innovation, and competitiveness, with examples such as tech sector funding for agricultural climate projects.	

Cloud-based platforms will deliver end-to-end “farm-to-fork” traceability visible to consumers		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	1	25%
Neutral	1	25%
Low	1	25%
Very low	0	0%

Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Participants report that initiatives in this field are already advancing, often supported by EU strategies, and implemented through blockchain and QR code systems. While some see this as a pathway to greater transparency, consumer trust, and accountability, others question its overall impact, pointing out challenges in infrastructure, investment, and the assumption that local production automatically equates to better environmental or social outcomes.	

Irrigation networks will use remote sensing and AI to schedule watering autonomously		
Likelihood	Number of answers	% over the total
Very high	3	75%
High	1	25%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	4	100%
High	0	0%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	This is viewed as both technically feasible and already partially implemented in some regions. Respondents reference projects that integrate satellite imagery, IoT sensors, and weather forecasting to optimize water use. Expected benefits include improved efficiency and cost reduction in drought-prone areas. Many see it as an inevitable development, though adoption will depend on affordability and infrastructure.	

By 2050, most family farms will join cooperatively governed data hubs		
Likelihood	Number of answers	% over the total
Very high	1	25%
High	2	50%
Neutral	0	0%
Low	1	25%
Very low	0	0%

Impact	Number of answers	% over the total
Very high	2	50%
High	1	25%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Some respondents note that similar collaborative structures already exist in certain regions, and believe participation could offer benefits such as shared knowledge, benchmarking, and collective bargaining power. Potential advantages include equitable access to services like AI tools, carbon markets, and insurance products. However, comments also raise issues around privacy, and the need for trust and clear benefits to ensure farmer participation.	

Small farms will earn regular income from digitally managed ecosystem services (carbon credits, pollination)		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	1	25%
Low	0	0%
Very low	1	25%
Impact	Number of answers	% over the total
Very high	1	25%
High	2	50%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Opinions are mixed. Supporters see potential for new income streams that reward regenerative practices, especially if digital tools can reduce transaction costs and improve verification. Examples from pilot projects in Europe were mentioned, but barriers include high monitoring and certification costs and limited access for small farms. Some are skeptical this will become a reality, while others suggest it could help shift CAP payments toward environmental outcomes.	

By 2050, edge-computing nodes (e.g. 5G base stations) will cover nearly all cultivated land, enabling real-time AI on remote farms		
Likelihood	Number of answers	% over the total
Very high	2	50%
High	1	25%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	75%
High	1	25%
Neutral	0	0%
Low	0	0%
Very low	0	0
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Several comments express confidence that near-universal coverage will be achieved well before 2050, at least in certain countries. Respondents link this infrastructure to enabling robotics, autonomous vehicles, and AI-driven decision-making in farming. Anticipated benefits include higher productivity and reduced waste, though some note that adoption will also depend on farmers' readiness to use the technology.	

By 2050, farm management systems will bundle weather-indexed insurance via smart contracts		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	3	75%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Responses indicate uncertainty about both the technology and its adoption. Some foresee a shift in agricultural insurance toward more public provision due to climate-related risks, while others see	

	potential benefits if implemented fairly, such as improving resilience and accessibility. Concerns include the technical feasibility, equity for smallholders, and trust in digital contract systems.
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Most small farmers will join pan-European digital cooperatives to coordinate production and market access across Member States		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	0	0%
Low	2	50%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	50%
High	2	50%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	While cooperatives are well established at the national and local level, pan-European digital cooperatives are considered rare and challenging due to cultural and language differences. Proponents believe they could strengthen market power, improve access to infrastructure, and enhance competitiveness. Risks mentioned include dependency on digital systems and the need for significant capacity-building.	

By 2050, public agencies and private ag-tech firms will jointly fund regional centers that deliver continuous digital literacy and business training		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	2	50%
Low	1	25%
Very low	1	25%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%

Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments		
	Several respondents note existing initiatives in digital training but point out that large-scale, coordinated centers remain limited. Joint public–private investment is seen as potentially transformative, especially for smallholders and older farmers. However, some doubt the likelihood of collaboration between sectors. Benefits mentioned include accelerated digital adoption, reduced inequalities, and improved skills, while challenges include political will and aligning interests.	

By 2050, receipt of any Common Agriculture Policy (CAP) subsidy will require farmers to complete a certified digital-skills programme		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	2	50%
Low	0	0%
Very low	0	0%
Impact		
	Number of answers	% over the total
Very high	1	25%
High	0	0%
Neutral	3	75%
Low	0	0%
Very low	0	0%
Consensus and action		
	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments		
	Participants generally view this as ambitious. While newer generations may already possess digital skills, such a requirement could disadvantage older or less digitally literate farmers unless accompanied by strong support. If applied inclusively, it could accelerate the adoption of advanced agricultural technologies and improve compliance with digital systems.	

No farm under 10 ha will be able to adopt precision technologies without joining a formal support network of service providers or cooperatives		
Likelihood	Number of answers	% over the total
Very high	3	75%
High	0	0%
Neutral	0	0%
Low	1	25%
Very low	0	0%

Impact	Number of answers	% over the total
Very high	2	50%
High	1	25%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Many respondents agree that the cost and technical demands of precision agriculture are significant barriers for small farms, making support networks or cooperative arrangements essential. Without access, small farms risk losing competitiveness and falling behind in sustainability and productivity. Some anticipate that technology could become more accessible over time, but disparities between large and small farms are expected to persist.	

EU and/or national agricultural digitalization schemes will include mandatory regional customization clauses to reflect local climate, soil and crop differences		
Likelihood	Number of answers	% over the total
Very high	1	25%
High	2	50%
Neutral	0	0%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	25%
High	3	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	There is strong agreement that local adaptation would improve the relevance and effectiveness of digital tools. Respondents highlight that current subsidies and tools often fail to address micro-regional needs, and digitalization could help tailor solutions. Challenges include the complexity and cost of individualized assessments, and the need for better coordination and local expertise.	

National agricultural advisory services will integrate peer-coaching networks – pairing digitally proficient farmers with those needing upskilling		
Likelihood	Number of answers	% over the total

Very high	1	25%
High	1	25%
Neutral	1	25%
Low	0	0%
Very low	1	25%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	1	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Some countries are already experimenting with peer-to-peer approaches, and respondents generally see potential benefits in knowledge sharing and reducing isolation among farmers. Doubts remain about effectiveness, and universal adoption.	

All public funding for farm digitalization will flow through farmer-led cooperatives or community hubs to ensure equitable distribution and cost-sharing		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	50%
Neutral	1	25%
Low	1	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	75%
Neutral	0	0%
Low	1	25%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Views are divided. Supporters believe this could ensure fairer distribution of resources, shared costs, and increased efficiency. Others caution that some cooperatives have shifted toward private company models and might not prioritize equitable access. Concerns include potential exclusion of certain farmers and reduced direct access to funds.	

6.2.5 Third round of interviews

6.2.5.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [6]	In the agricultural sector, economic pressures are reshaping the viability of traditional activities, particularly livestock farming. A significant portion of farmers' income, especially in livestock, comes from subsidies, creating a dependency that limits profitability and discourages new entrants. Younger generations are often unwilling to accept the financial sacrifices involved, perceiving better quality of life and income opportunities in other sectors such as tourism. As the number of farmers declines, those remaining must work increasingly larger areas to sustain their income levels, further intensifying the economic strain on the sector.	<i>"In livestock farming, around 50% of a farmer's salary comes from subsidies, making them more like a public employee tending sheep" (22). "If you want to make money, you should not be a farmer. You need to invest a lot for a smaller profit" (22). "There are fewer and fewer farmers, and those who remain have to work more land to maintain their income level" (22). "You could make more money in the tourism sector" (22).</i>
Niche_Governance [10]	Governance in agriculture faces significant challenges rooted in mistrust, fragmented representation, and the exclusion of key stakeholders. Establishing effective cooperation between farmers, industrial actors, and public institutions is seen as crucial, but the process can be slow, as illustrated by multi-year efforts to draft shared statutes. Decision-making is often restricted to those with formal water rights, leaving environmental activists and other ecosystem guardians without a voice. Sectoral divisions, mutual blame, and lack of inclusive governance structures undermine collective responses to crises such as water scarcity. A more integrated approach that balances resource management, public representation, and ecological concerns is viewed as essential for long-term sustainability.	<i>"There is a lot of mistrust between actors, with blame between neighbours, between farmers depending on their products, and between sectors" (22). "When the problem has become very severe, without good and efficient governance, this is what happens" (22).</i>
Niche_Social [5]	Social dynamics in agriculture are shaped by generational shifts, economic dependency, and the need for greater cooperation. Many farmers rely heavily on subsidies, which can reduce motivation for profitability and discourage younger generations from entering the sector, particularly as they are less willing to accept the sacrifices required. The farming population is shrinking, with remaining farmers working larger areas to maintain income levels. Building a culture of trust and collaboration is identified as a key sociological challenge, especially to replace competition with cooperation. At the same time, the absence of mechanisms to integrate	<i>"The younger generation is not willing to make sacrifices; people believe they live better than past generations" (22). "We need to foster a climate of trust to cooperate instead of competing; this is a highly relevant sociological impact" (22). "We should find a mechanism to integrate environmental activists who have no water rights; there is no agent safeguarding the ecosystem" (22).</i>

	environmental advocates into decision-making leaves important ecological perspectives unrepresented.	
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6.2.5.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [4]	Political decision-making in water governance is marked by sectoral prioritization and institutional fragmentation. During droughts, urban supply to Barcelona is safeguarded, often at the expense of agricultural areas such as the Baix Ter, while sectors like tourism face no comparable restrictions. In Catalonia, the lack of agricultural regulation by the water authority (ACA) leaves responsibilities to the agriculture department, contributing to a low food self-sufficiency rate of 40–45%, far below FAO's recommended 70% threshold for food security. Outdated water management regulations still assume resource abundance, hindering adaptation to scarcity and contributing to environmental degradation.	<i>"In drought periods, guaranteeing supply to Barcelona means restricting agricultural water in Baix Ter" (22).</i> <i>"Tourism has not faced restrictions; is it fair to water the lawns of campsites?" (22).</i> <i>"Catalonia's food self-sufficiency is around 40–45%; below 70% is considered at risk of famine by FAO" (22).</i>
Regime _ Economic [9]	The economic sustainability of agriculture is constrained by limited capitalisation, low farm-gate prices set on international markets, and strict domestic regulations that reduce competitiveness against imports produced under less stringent standards. Farmers face rising costs for inputs like pest control and fertilisers, while returns per unit of land continue to decline, forcing expansion into larger areas. Structural issues, such as the abandonment of marginal lands and reduced water availability compared to past decades, further undermine productivity. Policies that prioritise water allocation for sectors with higher economic return per cubic metre, such as tourism, over agriculture exacerbate the sector's vulnerability.	<i>"The price paid to farmers is set internationally without tariffs, even though pest treatments and fertilisers are cheaper elsewhere" (22).</i> <i>"There are many marginal lands turning into forests, with rivers carrying half the water they did 50 years ago" (22).</i> <i>"In drought plans, less restriction is considered for those generating more economic return per cubic metre of water, such as filling a tourist's swimming pool over irrigating an apple tree" (22).</i> <i>"Economic return per unit of farmland is decreasing; farmers need more land to sustain income" (22).</i>
Regime _Social [4]	Social attitudes toward agriculture are shaped by changing consumption priorities and generational expectations. In the past, households devoted a much larger share of their income to food, but this has dropped significantly, enabling older generations to enjoy greater material comforts such as cars and holidays. Younger generations are reluctant to forgo these benefits, influencing their willingness to engage in farming.	<i>"People used to spend 40% of their income on food, now it is around 15%" (22).</i> <i>"Younger generations are unwilling to give up the comforts that older generations enjoy" (22).</i> <i>"We import products we cannot produce because of our own regulations" (22).</i>

	At the same time, self-imposed production standards lead to reliance on imports, sometimes resulting in lower product quality, but sustaining the perception of an improved lifestyle compared to previous generations.	
Regime_Environmental [3]	Environmental pressures on agriculture are intensifying due to land abandonment, declining water availability, and outdated regulatory frameworks. Marginal lands are reverting to forest, altering ecosystems and reducing water flow in rivers to half of what it was 50 years ago. Hydraulic management often operates under the outdated assumption of abundant water, leading to policies that fail to address current scarcity and contribute to the qualitative degradation of water resources. This mismatch between ecological reality and governance capacity hampers the sector's ability to adapt to climate and resource pressures.	<i>"Many marginal lands are becoming forests, with rivers carrying half the water they did 50 years ago" (22).</i> <i>"Water regulations are still based on the idea of an infinite resource, but water is no longer there" (22).</i> <i>"Management focuses on paper water, not real water, leading to problems and qualitative degradation of resources" (22).</i>

6.2.5.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [1]	Political stability is closely linked to food security, with global conflicts and geopolitical instability posing significant risks to agricultural supply chains in Europe. International disruptions can quickly translate into shortages, making domestic resilience and self-sufficiency key political priorities. The FAO has been warning for years that Europe could once again face the threat of famine if these vulnerabilities are not addressed.	<i>"The FAO has been warning for five years that Europe could once again face famine; any global instability, such as wars, can lead to supply shortages" (22).</i>
Landscape _Economic [2]	Global market instability and geopolitical shocks present significant risks for agricultural resilience and food security in Europe. FAO warnings highlight that countries with food self-sufficiency below 70% face heightened vulnerability to famine, a threshold that parts of Europe, including Catalonia, fail to meet. In a context where global disruptions such as wars can rapidly translate into supply shortages and inflation, ensuring greater domestic production capacity is becoming a strategic imperative	<i>"FAO says countries with less than 70% self-sufficiency risk famine; this is why inflation in Ukraine was so high" (22).</i> <i>"FAO has warned for the last five years that Europe could face famine again; any global instability, such as wars, can cause shortages" (22).</i>
Landscape _Environmental [1]	Water scarcity is emerging as one of the most pressing environmental challenges for agriculture. Without reliable access to water, food production is directly threatened, making the sector highly vulnerable to climate variability and drought. Stakeholders emphasise the urgent need for mechanisms that can maintain food output with reduced water availability, a goal that requires both technological innovation and systemic adaptation.	<i>"Without water there is no food; we must find mechanisms to produce the same amount of food with the same water" (22).</i>

6.2.5.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [2]	The agricultural workforce is shrinking, with fewer farmers remaining in the sector. Those who continue face the pressure of managing larger areas of land to sustain their income levels. This structural challenge not only increases workloads but also raises questions about fairness and the allocation of resources, particularly in times of scarcity, when water use for non-essential purposes such as maintaining campsite lawns is contested.	<i>"There are fewer and fewer farmers, and those who remain have to work more land to maintain their income level" (22). "Is it fair to water the lawns of campsites?" (22).</i>
Vulnerability_Finance [6]	Financial vulnerability in agriculture is marked by a strong dependency on subsidies, particularly in livestock, where they represent around half of farmers' income. This reliance discourages profitability and increases the risk for those with fewer resources. Water allocation policies during drought exacerbate these challenges, as urban supply is prioritised over agricultural use, often without compensation for farmers. Perceptions of unfairness intensify when higher-value urban or tourism-related uses are favoured, reinforcing economic insecurity in rural areas.	<i>"The most affected are those with fewer resources" (22). "In times of drought, guaranteeing supply to Barcelona means restricting agricultural water in Baix Ter. I receive less money, but I am not compensated" (22). "Drought plans talk about fewer restrictions for those who generate more economic return per cubic metre of water. Filling a tourist's pool brings a much greater return than watering an apple tree" (22).</i>

6.2.5.5 Disadvantaged groups (DGs)

Economic, environmental, and governance pressures in agriculture disproportionately affect certain vulnerable groups. Farmers with fewer resources face reduced profitability due to heavy reliance on subsidies, while water allocation policies during droughts often prioritise urban or tourism uses without compensating agricultural losses. This exacerbates inequalities, particularly in areas, where trust between stakeholders is already undermined by historical tensions. Broader socio-economic trends, including generational shifts and limited willingness among younger people to accept the sacrifices of farming, further threaten sector renewal.

Disadvantaged groups (DGs) [9]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Farmers with fewer resources	Dependence on subsidies and low profitability create structural economic vulnerability.	Reduced capacity to invest, risk of economic exit from farming.	<i>"The most affected are those with fewer resources" (22)</i>

Livestock farmers	Around half of income comes from subsidies, making the profession unsustainable without external support.	Limits profitability and deters long-term commitment to farming.	<i>"In livestock, 50% of a farmer's salary comes from subsidies. You need to invest a lot of money for a smaller benefit" (22)</i>
Farmers in drought-prioritised areas	Agricultural water use restricted in favour of urban supply, with no compensation.	Loss of income and production capacity, perceived unfairness.	<i>"In times of drought, guaranteeing supply to Barcelona means restricting agricultural water in Baix Ter... I receive less money, but I am not compensated" (22)</i>
Farmers competing with tourism and urban uses	Tourism and urban activities often enjoy fewer restrictions and greater economic valuation per water unit.	Reinforces perception of inequity and resource allocation bias.	<i>"Drought plans talk about fewer restrictions for those who generate more economic return per cubic metre of water. Filling a tourist's pool brings a much greater return than watering an apple tree" (22)</i>
Younger generations	Reluctance to make the sacrifices associated with farming compared to other sectors.	Reduced generational renewal in agriculture.	<i>"Younger generations are not willing to give up comforts; they think they live better than previous generations" (22)</i>

6.3 Energy

6.3.1 First round of interviews

6.3.1.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [6]	In the energy sector, economic niches are emerging around new decentralised business models. Interviewees discussed how energy consumers are becoming active players both generating and managing energy through roles like "prosumers" or participation in virtual energy plants. This reflects a broader democratisation of energy responsibility and the rise of entrepreneurial ecosystems (e.g., virtual energy plans, aggregators, citizen-managed platforms, etc.). The economic logic is shifting toward service-oriented models and shared ownership or control.	<i>"A democratisation of responsibility is happening, instead of large hidroelectric centrals, we find little batteries at home" (6)</i> <i>"I, as a citizen I'm generating and consuming, I'm a prosumer" (7)</i> <i>"Fossil fuels companies are instead of shifting to renewables, diversifying until it is more profitabe" (6)</i>

Niche_Governance [7]	Emerging governance niches in the energy sector go around decentralisation, citizen participation, and redefined regulatory frameworks. Interviewees noted a growing push toward models where individuals become both energy users and co-responsible actors within distributed systems. This shift is supported by a more advanced, though complex, regulatory landscape in Europe. At the governance level, prosumers, cooperatives and community-based organisations are beginning to reshape how authority and responsibility are distributed in the energy transition, challenging the centralised and state-dominated logics of the existing regime.	<i>"We are going towards energy self-management" (7)</i> <i>"The complexity of the energy system will continue to increase" (6)</i>
Niche_Social [8]	The social dimension of energy niches highlights how citizens are increasingly becoming active participants in the energy system. The transition towards decentralised and decarbonised energy involves new roles such as the <i>prosumer</i> (producer-consumer), which creates both opportunities and barriers. Experts point out the need for more vocational workers and specialised profiles, as well as the potential for social exclusion due to the complexity of the energy transition and conflicting views on community self-consumption versus centralised energy models.	<i>"A prosumer is a household consumer who interacts with the electric system (knowingly or not) by installing solar panels and batteries at home, and deciding when to inject energy into the grid. This adds flexibility to the system" (6)</i> <i>"We'll need much more specialisation across all areas. (6)"</i> <i>"There is an increasing shortage of skilled professionals for technical roles. (6)"</i>
Niche_Technological [17]	Technological innovation in the energy sector is closely tied to renewables expansion, digitalisation and system flexibility. The interviews highlight trends such as offshore wind power, second-life batteries (recycling), grid flexibility mechanisms and the emergence of virtual power plants. Experts point to the diversification of technological solutions: residential batteries, large-scale photovoltaics, biogas, and the integration of smart distribution systems. Decentralisation and electrification are at the core of these developments, with the <i>prosumer</i> becoming a key figure in this reconfiguration of energy flows.	<i>"There's a macrotrend from conventional thermal plants to photovoltaic and wind energy." (6)</i> <i>"The share of renewables in the energy mix will increase to meet set climate targets." (6)</i> <i>"Self-consumption and industry are shifting toward photovoltaics. Electrification extends beyond housing and industry to include mobility, with electric vehicles becoming mobile batteries to stabilise the grid." (6)</i> <i>"Biogas and biofuels will continue to grow." (6)</i> <i>"Hydroelectric plants are being upgraded with reversible systems and battery storage." (6)</i> <i>"Residential batteries will become widespread." (6)</i>

6.3.1.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime _ Economic [7]	Currently there is a mismatch between public incentives and administrative capacity, and volatility in business models. Experts underline that although prices of photovoltaic modules have plummeted due to overproduction, a lack of planning and slow permitting processes hinders investment. Public-private collaboration exists, but bureaucratic delays and inconsistent subsidies disrupt market confidence. Instruments like CAE (Energy Savings Certificates) in Spain or EU ETS (Emission Trading System) are perceived as complex, and while they offer financial incentives for decarbonisation, they often burden companies in an uneven way. Additionally, the diversity of business models around photovoltaics (renting, consulting, discount platforms) creates confusion among consumers and difficulty in scaling profitable models.	<i>"The price of photovoltaic modules is extremely low, there's more supply than demand." (6)</i> <i>"There's very little planning and a lot of inefficiency." (6)</i> <i>"There are many business models around solar panels renting, consultancy, discounts, but none have clearly succeeded yet." (6)</i>
Regime _ Technological [4]	The current technological regime of the energy system is marked by inertia and legacy infrastructures that are difficult to replace quickly. The fast expansion of renewables has led to inefficiencies and challenges in recycling outdated photovoltaic panels, which were less efficient and had shorter lifespans. While technological improvements are underway, existing infrastructures remain a limiting factor, both in operational terms and in the integration of decentralised or more efficient technologies. Experts highlight that despite progress, the system's rigidity and infrastructural dependence slow down the energy transition.	<i>"With the rapid growth of renewables, the value chain faces issues. especially with recycling. Older photovoltaic panels are inefficient and have a shorter commercial life, though newer ones are more efficient." (6)</i> <i>"The energy sector is highly inertial; you can't change infrastructures overnight." (6)</i> <i>"The problem lies in the infrastructure I have to use." (7)</i>

6.3.1.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _ Political [5]	Experts stress the need for Europe to reduce its dependency on countries like Russia and China, both for fossil fuels and for the supply of critical raw materials used in renewable energy technologies. The lack of European manufacturing infrastructure for key components such as solar panels or batteries, coupled with international tensions (e.g., tariffs on Chinese imports), poses significant barriers. Moreover, some actors are perceived to be intentionally slowing down the transition for political or commercial interests.	<i>"There's a clear trend towards recycling critical raw materials because we don't have them in Europe. We need a stable supply chain that doesn't depend on China." (6)</i> <i>"Trade war: tariffs on Chinese products. Spain has no manufacturing capacity for photovoltaic or hybrid panels, everything comes from China."</i>

		<i>Europe doesn't have much either."</i> (6)
Landscape _ Economic [3]	The macroeconomic landscape around energy is shaped by dependency on fossil fuels, geopolitical tensions, and the volatility of global supply chains. The European transition to renewables is motivated in part by economic vulnerability, specifically, dependence on cheap Russian oil, which created price instability during international crises. Another structural issue is the economic risk of investing in regions with conflict potential, like rare earth deposits in Ukraine. Political and commercial interests also influence energy markets globally, particularly in relation to fossil fuel trade.	<i>"Europe wants to shift to renewables because we don't have oil, we rely on Russia, and Russian oil used to be the cheapest. But when we had to buy it elsewhere, prices went up."</i> (6) <i>"Ukraine has rare earth deposits key for energy production, but they're close to the Russian border. No company wants to invest due to the war risk. That's why the US-Ukraine agreement includes protection of these deposits and transferring exploitation rights to the US."</i> (6) <i>"There are strong political and commercial interests behind oil trade."</i> (6)
Landscape _ Technological [2]	At the macro level, technological trends are exerting strong influence over the energy system. Digitalisation, the rise of data centers, and energy demand increases are shaping energy infrastructure choices. Experts highlight that high national energy demand can still be met through polluting sources like coal. At the same time, digitalisation and uneven infrastructure investments (e.g. natural gas networks) may create two-speed energy systems with unequal technological access or environmental outcomes.	<i>"If a country's demand increases significantly, the demand for energy must be met with available generation sources and this can still mean coal. That's not changing."</i> (6) <i>"Data centers (digitalisation) are creating incentives for natural gas... if you build certain gas infrastructures, you generate two-speed energy systems."</i> (6)

6.3.1.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [2]	Interviewees describe how the boom in photovoltaic energy (fueled by geopolitical shocks and public subsidies) led to an overdimensioned sector and subsequent job losses. Although the transition promises more and better jobs than those in extractive industries, mismatches in timing and skills create a precarious environment for certain workers, while in others the conditions are better.	<i>"There was a photovoltaic boom two years ago (due to the war in Ukraine and subsidies). The sector became oversized, and now many people have lost their jobs."</i> (6) <i>"The transition is creating more jobs than it's destroying. These are higher quality jobs than in mining or oil extraction... new</i>

		<i>professional roles are emerging.” (6)</i>
Vulnerability_Finance [2]	Financial vulnerability in the energy sector is tightly linked to energy poverty. Interviewees emphasise that low-income households face challenges such as choosing suboptimal contracts, not benefiting from the lowest energy rates, or lacking access to subsidies. Despite existing support measures (like energy audits or social tariffs) systemic barriers remain. The affordability of energy and thermal comfort remains a major concern, often overlapping with broader social inequalities.	<i>“Energy poverty is just poverty, in general.” (7)</i>
Vulnerability_Skills [4]	Vulnerability related to skills in the energy sector is tied to the increasing technological complexity of the transition. Interviewees report a lack of trained professionals for tasks like installing photovoltaic systems or managing bioenergy. This shortage slows deployment and creates a bottleneck in the transition. Moreover, current workers face the risk of skill obsolescence, as rapid innovation demands continuous upskilling. There is also uncertainty about the technical knowledge required for new energy roles, especially for disadvantaged groups.	<i>“There are many technological challenges: engineers, programmers, architects are needed. We lack trained workers in photovoltaic panels. Bioenergy clusters have created vocational training programs, but there’s still a shortage.” (6)</i> <i>“There’s a lot of specialization in energy. The problem is, if a new technology comes along, you’re obsolete. That’s what happened with the photovoltaic boom.” (6)</i> <i>“We need technified professionals in the energy sector” (7)</i> <i>“There used to be gas boiler installers, now we need electricians. We don’t have enough professionals.” (6)</i>

6.3.1.5 Disadvantaged groups (DGs)

According to the interviews, DGs in the energy sector include workers from fossil-fuel-related industries (e.g., gas boiler installers or oil extraction), highly specialized workers in outdated renewable energy technologies suffering shifts in trends, people with low energy literacy, households experiencing energy poverty, and elderly citizens unfamiliar with digital systems.

Experts emphasize that the rapid transformation of the energy landscape is generating both opportunities and structural vulnerabilities. Workers from the extractive and fossil sectors are especially affected by job losses due to sectoral decline or automation. Meanwhile, some specialized profiles risk obsolescence as technologies evolve faster than upskilling programs. At the user level, low energy literacy and a lack of technical knowledge limit access to emerging opportunities such as energy communities, self-consumption schemes or new tariff systems. Elderly individuals, in particular, may struggle to navigate increasingly complex systems. Finally, energy poverty remains a key issue, as vulnerable households face difficulties maintaining thermal comfort and accessing affordable contracts or subsidies.

Disadvantaged groups (DGs) [6]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Workers in outdated fossil sectors	Low-skilled workers from extractive industries (coal, oil, gas) or fossil fuel infrastructure (boiler installers, gas stations) may lose their jobs in the energy transition.	High risk of unemployment due to job obsolescence, with few reskilling pathways in place.	<i>"There has been a change in boilers, but reskilling?" (7)</i>
Highly specialized workers in RE obsolete technologies	Workers overly specialized in certain technologies (e.g., specific PV models) may become obsolete with rapid innovation.	Skills become outdated quickly, leading to vulnerability if reskilling is not available or accessible.	<i>"High specialization in energy. If a new technology arrives, you're obsolete. Example: solar panel boom." (7)</i>
People with low energy literacy	Citizens with low understanding of tariffs, subsidies or technical aspects of energy systems.	Risk of being excluded from benefits or exposed to scams and complexity in an increasingly digital and decentralized energy system.	<i>"Greater complexity in the energy sector, easier for others to take advantage of us. More options, more difficulty." (6)</i>
Low-income households (energy poverty)	Households unable to afford energy or access basic energy services.	Direct exposure to discomfort, illness or exclusion due to unaffordable bills or inability to invest in efficiency.	<i>"Energy poverty is just poverty, in general." (7)</i>
Elderly people	Elderly citizens may struggle more with adapting to the new energy model, particularly due to technological barriers.	Increased exclusion from digital energy services or self-consumption models due to generational digital divide.	<i>"But what about older people?" (7)</i>

6.3.2 First round of Delphi survey

6.3.2.1 Elaboration of statements

65. The transition towards wind and solar energy will continue accelerating. **[Niche_Technological]**
66. The main factors that will lead the energy transition will be the increase of prices, geopolitical aspects and environmental concerns. **[Landscape_Political] [Landscape_Environmental] [Landscape_Economic]**
67. By 2050, more than 50% of European households will participate as prosumers (producing all their energy for self-sustaining and selling the excess). **[Niche_Social] [Niche_Governance] [Niche_Technological]**
68. Regulatory and bureaucratic barriers (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050. **[Regime_Political] [Regime_Legal]**

69. The shortage of skilled labor (e.g., installers, technicians) will significantly hinder the energy transition. **[Niche_Social] [Vulnerability_Skills] [Disadvantaged_Groups]**
70. By 2050, energy bills will be more difficult to understand for the average consumer, increasing the risk of individuals not understanding and potentially paying extra charges. **[Niche_social] [Niche_Economic] [Niche_Technological]**
71. Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses. **[Niche_Social] [Niche_Economic] [Vulnerability_Skills] [Vulnerability_Employment] [Disadvantaged_Groups]**
72. Non-renewable energy producers and energy distributors will be key players in the renewable energy market, due to its diversifying strategy. **[Niche_Economic] [Niche_Governance]**
73. The future of the energy sector will be renewable 100% by 2050 **[Niche_Technological]**
74. The labor market will experience an increase in demand for highly skilled jobs, such as engineers. **[Vulnerability_Skills] [Niche_Social]**
75. The R&D&i on battery recycling will be a decisive factor in EU energy independence by 2050. **[Niche_Technological]**
76. Activism in the energy sector will create obstacles for large-scale renewable projects, particularly wind farms. **[Niche_Social]**
77. New financial models will arise due to the energy market incentives created by private companies and its technification. **[Niche_Economic] [Niche_Governance]**
78. Highly specialized workers in technologies with high substitution potential (e.g., certain solar panel models) will be affected by innovations in the energy market. **[Niche_Social] [Niche_Technological] [Vulnerability_Skills] [Vulnerability_Employment] [Disadvantaged_Groups]**
79. Fossil fuel phase-out policies will generate long-term regional unemployment risks in areas historically dependent on extractive industries. **[Vulnerability_Employment] [Disadvantaged_Groups] [Niche_Economic]**
80. By 2050, decentralized energy systems will dominate the European grid, reducing reliance on large-scale centralized infrastructure. **[Niche_Technological]**
81. Households without solar panels will face higher long-term energy costs, exacerbating inequality. **[Vulnerability_Finance] [Disadvantaged_Groups]**
82. Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumer rights. **[Niche_Technological]**
83. By 2050, energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs. **[Niche_Social] [Niche_Governance]**
84. EU energy dependency on third countries for critical materials (e.g., lithium, rare earths) will be a strategic vulnerability. **[Landscape_Political] [Landscape_Economic]**

6.3.2.2 Analysis of results

The transition towards wind and solar energy will continue accelerating.		
Likelihood	Number of answers	% over the total
Very high	4	44.44%
High	2	22.22%

Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	5	55.56%
High	2	22.22%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Consensus and action	Is consensus reached? ⁴	Consensus degree ⁵
	Yes, maintain	66.67%
Summary of qualitative comments	Respondents almost unanimously agreed that the global push for decarbonization, backed by ongoing investments in green technology and the rise of electrification, will keep driving wind and solar deployment forward. Many highlighted that solar panels will increasingly dot urban rooftops while wind farms expand in less populated regions, though this uneven rollout could strain local grids. A few cautionary voices pointed to shifting political priorities that might slow momentum or to maturing markets where installation rates could plateau. Overall, participants felt there's little alternative "rational way" forward and that this trajectory remains firmly on track. In terms of impact, the group was overwhelmingly optimistic: they foresaw major benefits for energy security and grid stability, new economic opportunities, and decisive progress toward climate goals. One respondent even described the positive effects as "huge" for competitiveness and livelihoods. While questions remain about balancing growth with policy and infrastructure constraints, the consensus was that accelerating wind and solar will be transformative.	

The main factors that will lead the energy transition will be the increase of prices, geopolitical aspects, and environmental concerns		
Likelihood	Number of answers	% over the total
Very high	7	77.78%
High	0	0%
Neutral	2	22.22%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	4	44.44%
High	3	33.33%

⁴ Consensus is reached when the % over the total answers of combining "very high" and "high" in likelihood terms is >60%

⁵ Consensus degree is defined as: [(very high + high)/total number of answers]*100

Neutral	2	22.22%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Respondents pointed to escalating energy prices, supply-chain geopolitics, and environmental pressures as the primary levers driving investment in renewables and efficiency. Many highlighted that volatile relations with critical-materials suppliers further amplify urgency, while a few underscored the role of competitiveness and policy incentives alongside pure economics. They judged these combined drivers to have profound system-wide impacts: reshaping industry structures, supply chains, and labor markets, and compelling both public and private actors to reorient around clean energy. A minority noted that regional differences in policy and infrastructure readiness will modulate outcomes, but consensus held that these factors will fundamentally transform the energy landscape.	

By 2050, more than 50% of European households will participate as prosumers (producing all their energy for self-sustaining and selling the excess).		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	22.22%
Neutral	5	55.56%
Low	0	0%
Very low	2	22.22%
Impact	Number of answers	% over the total
Very high	1	11.11%
High	3	33.33%
Neutral	4	44.44%
Low	1	11.11%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	22.22%
Summary of qualitative comments	Opinions on reaching a 50 percent prosumer rate by mid-century were more divided. Several respondents noted that technology and policy are moving us in that direction, but affordability, grid integration challenges and the need for supportive regulation could block mass self-generation, especially in dense urban areas. Others pointed out that prosumer numbers are already climbing, though geographic and infrastructure limits may cap full self-sufficiency. A	

	couple of participants simply weren't convinced it's feasible at scale given the complexity of retrofitting existing buildings and uneven market readiness. On the impact side, those who see the milestone being met believe it would be nothing short of transformative, enhancing energy independence, resilience and citizen engagement, and spawning new energy-market models supported by advances in storage and smart grids. A few were neutral or skeptical, suggesting that without major grid upgrades and regulatory overhaul, the broader system effects may fall short of the promise.
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Regulatory and bureaucratic barriers (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050.		
Likelihood	Number of answers	% over the total
Very high	3	33.33%
High	4	44.44%
Neutral	2	22.22
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	4	44.44%
High	4	44.44%
Neutral	0	0%
Low	1	11.11%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Respondents almost unanimously identified permit backlogs, inconsistent national regulations, and capacity constraints within authorities as persistent hurdles for new projects. A few optimists hoped that streamlining procedures and clarifying rules could ease the burden, but most felt that without major structural reforms, red tape will endure as a critical choke point. In terms of impact, participants rated these barriers as having very high negative effects: extended approval timelines threaten to stall installations, undermine investor confidence, and slow progress toward climate targets. While some argued that greater transparency and harmonized frameworks might mitigate delays marginally, consensus held that bureaucratic inertia remains the single biggest drag on deployment.	

The shortage of skilled labor (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050		
Likelihood	Number of answers	% over the total

Very high	1	11.11%
High	5	55.56%
Neutral	1	11.11%
Low	1	11.11%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	3	33.33%
High	2	22.22%
Neutral	2	22.22%
Low	1	11.11%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	66.67%
Summary of qualitative comments	Many respondents warned that training pipelines for technicians, electricians, and retrofit specialists will struggle to match surging demand, leading to recurring labor gaps. Although a minority noted that universities and vocational schools are ramping up programs, most anticipate persistent regional mismatches and capacity limits. On impact, the group judged this shortage to have very high consequences: without enough qualified workers, project rollouts could slow dramatically, costs may spike, and key decarbonization milestones risk being missed. Several emphasized that aggressive reskilling initiatives and apprenticeship schemes are essential, but even these measures may only partially close the gap.	

By 2050, energy bills will be more difficult to understand for the average consumer, increasing the risk of individuals not understanding and potentially paying extra charges.		
Likelihood	Number of answers	% over the total
Very high	1	11.11%
High	3	33.33%
Neutral	3	33.33%
Low	1	11.11%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	1	11.11%
High	3	33.33%
Neutral	4	44.44%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	No, remove	44.44%

Summary of qualitative comments	Several participants warned that as dynamic tariffs, tiered pricing, and new billing line-items proliferate, consumers may struggle with opaque and highly technical statements. Others countered that regulatory mandates and provider-led simplifications could preserve clarity, stressing the importance of clear communication and educational outreach. On impact, most rated consumer confusion as a serious risk: it could dampen engagement in energy-saving programs, erode trust, and slow adoption of smart-grid solutions. A few argued that intuitive billing formats and strengthened customer support services could meaningfully reduce these risks, but the prevailing view was that complexity must be actively managed.
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Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses.		
Likelihood	Number of answers	% over the total
Very high	4	44.44%
High	3	33.33%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	2	22.22%
High	3	33.33%
Neutral	2	22.22%
Low	1	11.11%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Respondents agreed that as fossil-fuel extraction declines, many low-skilled roles in mines, rigs, and processing plants will disappear, despite just-transition measures already underway. While some noted that reskilling programs exist, the majority felt these efforts may not scale quickly enough to prevent significant displacement. They assessed the social and economic fallout as substantial: without robust retraining, affected workers risk long-term unemployment, deepening regional inequalities and straining social safety nets. Advocates for stronger EU-wide policy frameworks stressed that only comprehensive, inclusive strategies can avert a social crisis.	

Non-renewable energy producers and energy distributors will be key players in the renewable energy market, due to its diversifying strategy.		
Likelihood	Number of answers	% over the total

Very high	1	11.11%
High	5	55.56%
Neutral	4	33.33%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	6	66.67%
Neutral	3	33.33%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	66.67%
Summary of qualitative comments	Most participants observed that incumbent utilities and distributors are actively diversifying into renewables, leveraging existing grids, capital, and market expertise to secure leadership in new projects. A few questioned whether legacy players could innovate fast enough, but the dominant view was that their resources and networks make them indispensable. They judged this involvement to have significant benefits—accelerating deployment through scale and smoothing market fluctuations—but warned that proper regulation will be necessary to prevent unfair concentration and to ensure competitive dynamics. Overall, incumbents' entry into renewables was seen as a catalyst for faster transition.	

The future of the energy sector will be renewable 100% by 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	33.33%
Neutral	3	33.33%
Low	1	11.11%
Very low	2	22.22%
Impact	Number of answers	% over the total
Very high	4	44.44%
High	3	33.33%
Neutral	2	22.22%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	No, remove	33.33%

Summary of qualitative comments	Opinions were divided: some felt strong policy commitments and technological progress make a fully renewable mix achievable, while others doubted that investment rates, grid upgrades, and political resolve will align in time. A few stressed that a broader suite of renewables beyond wind and solar will be essential for feasibility. Despite the split, most agreed that achieving 100% renewables would deliver massive climate benefits and energy security gains. Skeptics cautioned that overpromising without clear roadmaps could undermine public trust and policy credibility if intermediate targets are missed.
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The labor market will experience an increase in demand for highly skilled jobs, such as engineers.		
Likelihood	Number of answers	% over the total
Very high	3	33.33%
High	4	44.44%
Neutral	1	11.11%
Low	1	11.11%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	33.33%
High	4	44.44%
Neutral	1	11.11%
Low	1	11.11%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Nearly all contributors expected surging demand for technical specialists (engineers, software developers, data scientists, and digital-energy experts) as renewables, smart grids, and electrification create new complexity. While some felt current talent pools may suffice briefly, most foresaw a widening skills gap without targeted talent development. They anticipated this trend reshaping workforce composition and regional competitiveness: areas with strong educational pipelines will attract investment, and high-value roles will command premium wages. A handful noted that adjacent disciplines, like social sciences and economics, will also spawn specialized positions.	

The R&D&I on battery recycling will be a decisive factor in EU energy independence by 2050.		
Likelihood	Number of answers	% over the total
Very high	4	44.44%

High	3	33.33%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	3	33.33%
High	4	44.44%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Respondents emphasized that advances in battery recycling R&D&I are critical to closing the materials loop (recovering lithium, cobalt, and nickel) and reducing reliance on imports. While a few doubted recycling alone can meet future demand, most saw it as indispensable for long-term material security. They rated the impact of successful recycling innovations as very high: lowering costs, mitigating geopolitical risks, and strengthening the resilience of EV and grid-storage supply chains. A minority warned that without aligned policy incentives and investment, research may lag behind needs.	

Activism in the energy sector will create obstacles for large-scale renewable projects, particularly wind farms.		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	5	55.56%
Neutral	2	22.22%
Low	1	11.11%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	1	11.11%
High	4	44.44%
Neutral	2	22.22%
Low	1	11.11%
Very low	1	11.11%
Consensus	Is consensus reached	Consensus degree
	No, remove	55.56%
Summary of qualitative comments	Many participants noted that local activism, like environmental protests to NIMBY campaigns, already delays or blocks wind farms and other large installations. While some saw such activism as a	

	healthy check on developers, most expect it to remain a variable but persistent hurdle unless engagement strategies improve. On impact, they described regionally uneven but often substantial project slowdowns, cost overruns, and investor reluctance. Several suggested that early stakeholder inclusion, transparent planning, and benefit-sharing agreements could convert resistance into support over time.
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New financial models will arise due to the energy market incentives created by private companies and its technification.		
Likelihood	Number of answers	% over the total
Very high	4	44.44%
High	3	33.33%
Neutral	1	11.11%
Low	1	11.11%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	22.22%
High	4	44.44%
Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Survey respondents predicted that evolving incentives (green certificates, capacity markets, subscription services) will spur innovative financing structures, blending proprietary platforms with flexible consumer offerings. A minority questioned whether such models are truly novel, but most saw clear momentum toward service-oriented and risk-sharing arrangements. They judged the impact on investment and market liquidity to be significant: new models can broaden participation, reduce upfront costs, and realign stakeholder interests. A few remained skeptical about the pace of change, noting that core energy economics often resist rapid overhaul.	

Highly specialized workers in technologies with high substitution potential (e.g., certain solar panel models) will be affected by innovations in the energy market.		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	5	55.56%
Neutral	3	33.33%
Low	0	0%
Very low	1	11.11%

Impact	Number of answers	% over the total
Very high	0	0%
High	5	55.56%
Neutral	3	33.33%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached	Consensus degree
	No, remove	55.56%
Summary of qualitative comments	Most respondents observed that rapid innovation can render niche technical expertise obsolete as new solutions emerge, though they acknowledged that agile specialists can pivot through continuous learning. A few argued that strong foundational skills and adaptable training pathways will allow workers to transition rather than be left behind. They saw the impact of obsolescence as moderate to high: without proactive upskilling, talent shortages may arise in cutting-edge segments, hampering R&D and deployment. Several highlighted the need for certification programs and knowledge-transfer initiatives to maintain a robust talent pool.	

Fossil fuel phase-out policies will generate long-term regional unemployment risks in areas historically dependent on extractive industries.		
Likelihood	Number of answers	% over the total
Very high	2	22.22%
High	5	55.56%
Neutral	0	0%
Low	1	11.11%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	3	33.33%
High	3	33.33%
Neutral	0	0%
Low	2	22.22%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Participants largely agreed that communities built around coal, oil, and gas stand to lose significant employment as facilities close, despite nascent just-transition schemes. While retraining programs exist, most felt they need broader scope and scale to fully offset job losses.	

	They assessed the social and economic fallout as substantial: regional disparities could widen, public services may be strained, and political resistance could intensify. Some argued that timely economic diversification and investment in new industries can moderate these risks over a generation.
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By 2050, decentralized energy systems will dominate the European grid, reducing reliance on large-scale centralized infrastructure.		
Likelihood	Number of answers	% over the total
Very high	2	22.22%
High	5	55.56%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	3	33.33%
High	5	55.56%
Neutral	1	11.11%
Low	0	0%
Very low	0	0%
Consensus	Is consensus reached	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Respondents foresaw rapid expansion of rooftop solar, community microgrids, and peer-to-peer energy trading, driven by technology advances and policy support for energy democracy. A minority cautioned that technical integration and coordination hurdles may slow full dominance, but the prevailing view was that decentralization will play a leading role. They judged the impact positively: decentralized architectures can boost resilience, cut transmission losses, and empower consumers, though they underscored the need for robust cybersecurity and regulatory alignment. Overall, decentralization is expected to transform grid operations and market structures.	

Households without solar panels will face higher long-term energy costs, exacerbating inequality.		
Likelihood	Number of answers	% over the total
Very high	1	11.11%
High	5	55.56%
Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total

Very high	5	55.56%
High	1	11.11%
Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	66.67%
Summary of qualitative comments	Many contributors warned that as network tariffs and grid-service charges rise, non-solar homes will bear an increasing share of system costs, deepening energy affordability divides. A few suggested that collective or community solar schemes could partly alleviate this, but most stressed that inequality remains a critical risk. They rated the impact as severe: disproportionate cost burdens could worsen social disparities, fuel political backlash, and hinder the transition's public acceptance. Several advocated for targeted subsidies, sliding-scale tariffs, or shared generation models to ensure equitable access.	

Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumer rights		
Likelihood	Number of answers	% over the total
Very high	5	55.56%
High	1	11.11%
Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	5	55.56%
High	1	11.11%
Neutral	2	22.22%
Low	0	0%
Very low	1	11.11%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	66.67%
Summary of qualitative comments	Survey participants agreed that the proliferation of smart meters, IoT sensors, and cloud-based control systems will heighten the need for rigorous data governance to protect privacy and build trust. Some warned that existing regulations may lag behind technology, exposing consumers to potential misuse of sensitive data.	

	They unanimously rated governance failures as having very high negative consequences: privacy breaches, erosion of trust, and regulatory backlash could stall digital innovations. Conversely, robust governance frameworks were seen as catalysts for accelerating digitalization and consumer buy-in.
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By 2050, energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs.		
Likelihood	Number of answers	% over the total
Very high	6	66.67%
High	1	11.11%
Neutral	1	11.11%
Low	0	0%
Very low	1	11.11%
Impact	Number of answers	% over the total
Very high	4	44.44%
High	4	44.44%
Neutral	0	0%
Low	1	11.11%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	77.78%
Summary of qualitative comments	Most respondents felt that as dynamic tariffs and smart-grid incentives spread, understanding consumption patterns, pricing mechanics, and grid interactions will be indispensable for cost optimization and system flexibility. While a few doubted that broad consumer expertise is achievable, the consensus was that educational initiatives and intuitive tools are critical. They judged enhanced energy literacy to have very high positive impact: informed consumers can drive demand-side responsiveness, reduce overall system costs, and support grid stability. Conversely, knowledge gaps risk low participation rates and system inefficiencies.	

EU energy dependency on third countries for critical materials (e.g., lithium, rare earths) will be a strategic vulnerability.		
Likelihood	Number of answers	% over the total
Very high	4	44.44%
High	5	55.56%
Neutral	0	0%
Low	0	0%
Very low	0	0%

Impact	Number of answers	% over the total
Very high	5	55.56%
High	2	22.22%
Neutral	1	11.11%
Low	1	11.11%
Very low	0	0%
Consensus	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Respondents flagged the EU's reliance on external suppliers (particularly China) for battery and renewable-tech materials as a persistent strategic vulnerability. While research and recycling efforts could mitigate some risks, most agreed that alternative sourcing agreements and material-substitution innovations are necessary. They rated this dependency's impact as substantial: supply-chain disruptions can spike costs, delay projects, and compromise energy security. A handful noted that emerging circular-economy measures and substitute materials may soften the blow, but consensus held that the challenge demands concerted policy and investment action.	

6.3.3 Second round of interviews

6.3.3.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [11]	Interviewees describe the energy transition as unevenly shaped by affordability, regulation and labour gaps. Decentralised models such as prosumer energy or microgeneration are growing, but incentives remain weak where electricity is cheap, and administrative burdens remain high. Households are pragmatic: they adopt solar or retrofit only when it brings clear economic benefit. At the same time, structural challenges persist — fossil fuels remain economically attractive, heating upgrades are expensive, and grid infrastructure expansion is slowed by labour shortages. New niche markets are emerging around energy storage, backup services and industrial heat reuse, but require investment, market design and policy support. Interviewees also highlight the need to upskill energy workers and shift labour from declining sectors (e.g. automotive) to new domains. Without proactive reskilling and economic redistribution, transition benefits may remain inaccessible to many.	<i>"The centralised market is shrinking... costs fall on people who can't afford it."</i> (17) <i>"They need market for power-reserved, frequency response... converting waste streams."</i> (17) <i>"Households will install solar panels... as long as they save money."</i> (18) <i>"If we reduce fossil fuel, it would be to reduce dependency. But fossil fuels are still too cheap."</i> (18) <i>"As a prosumer you can use energy, but if you produce, you pay taxes and do paperwork."</i> (18) <i>"Transition costs a lot: heating, retrofitting... rich people can afford it."</i> (18)

		<i>"Reskilling and upskilling will be key." (18)</i> <i>"We don't have enough labor to install infrastructure to double the grid." (17)</i> <i>"The private sector is going there, but they're not serious about it yet." (19)</i>
Niche_Governance [16]	Permitting delays, bureaucratic inefficiencies and lack of coordination across levels of government are seen as persistent bottlenecks, especially for renewables. While there is growing interest in community engagement and cooperative financial models, these remain underdeveloped in practice. Decentralisation and prosumer models are often mentioned as desirable, but current regulations do not provide sufficient incentives or clarity. Some experts stress that justice must be a central goal of public subsidies; yet in practice, wealthier groups are often the main beneficiaries. Stronger, more targeted governance frameworks are needed to ensure that public support mechanisms are fair, transparent and effective.	<i>"Regulatory and bureaucratic barriers... will remain the main bottleneck for renewable energy deployment across EU countries in 2050." (17)</i> <i>"They say they help you to fill out the forms, but it does not happen." (17)</i> <i>"The planning and design process of wind farms has changed a lot; now they are engaging communities a lot." (17)</i> <i>"Being more transparent and fairer is needed. It is the rising trend." (17)</i> <i>"The regulation side is crucial." (18)</i> <i>"Technically it's possible to have prosumers... but there's no strong policy support." (19)</i> <i>"There are no serious policies towards the private sector; the in-place rules are not enough." (19)</i>
Niche_Social [11]	Interviewees emphasise the social dimension of the energy transition, especially the need for public engagement and community-based approaches. While there is growing awareness of climate challenges, many citizens still lack familiarity with energy systems or feel disconnected from policy decisions. Successful examples shared by experts, such as farmers supporting wind farms for community benefit, show that local engagement and shared values can foster acceptance. However, the transition risks polarisation if vulnerable groups feel excluded or targeted. Interviewees warn that people with limited resources or static mindsets may be drawn to populist narratives unless they clearly see the advantages for themselves. The pace of transition is seen as insufficient, with calls for faster, decentralised and socially-rooted action. Social trust, mindset change and inclusive communication are all seen as key to making the transition socially viable.	<i>"People are not used to 'buy energy'." (17)</i> <i>"Many citizens have reasons to be concerned... without authentic engagement and pre-planning, with care for biodiversity and land..." (17)</i> <i>"Households will make solar panels... as long as they save money." (18)</i> <i>"In the climate change context... it's difficult to argue for policies, we don't know what will happen." (18)</i> <i>"Poor people will be very vulnerable to populist speeches... if we don't take them with us, it's a very big problem." (18)</i> <i>"The energy transition should finally be done by consumers. Public opinion is quite</i>

		sensitive.” (19) “They are bombarding people with lots of information... we need more community-based, decentralised measures.” (19) “We need more speed.” (19)
Niche_Technological [23]	Interviewees foresee a strong push towards decentralised and diversified energy systems, with solar power, batteries and automation playing key roles. Solar is expected to dominate due to its cost-efficiency and low material intensity, while smart technologies to manage demand (e.g. automated washing or EV charging) are gaining relevance. Yet, data openness remains limited, and the energy sector still lags behind others in digitalisation. Several experts stress that no single technology will suffice. A balanced mix (solar, wind, biogas, thermal) is needed to stabilise the grid. Short-term reliance on fossil fuels is still seen as necessary to ensure flexibility, especially in contexts like Germany where the nuclear phase-out has created new vulnerabilities. Storage remains a major challenge, with green hydrogen facing limitations due to water availability and low efficiency. Beyond technical issues, geopolitics and resource control shape the outlook. Globalisation is seen as receding, with growing concerns about dependence on foreign raw materials. Strategic diversification and stronger EU coordination are considered urgent. While industry shows ambition, political reluctance to fund stabilising technologies (like flexible gas plants) may delay implementation.	“Decentralised energy systems will dominate the European grid... it will continue.” (17) “Batteries are a recent trend... automation of demand control needs to improve.” (17) “Turning data into relevant inputs... the energy sector always has to catch up.” (17) “We need a lot of water for green hydrogen... sea water is not efficient.” (18) “We need technologies that stabilise the grid... but they cost money and politics don’t want to spend it.” (18) “There is not a single technology... it’s like a puzzle.” (18) “Without gas we can’t stabilise the system... the nuclear phase-out was too fast.” (18) “We are making the same mistakes, still importing fossil fuels.” (18) “It’s difficult to separate the energy sector from the others... it generates all of them.” (19) “We are not going towards an energy transition, but an energy addition.” (19) “Solar will be the main actor; then energy efficiency and electric vehicles.” (19) “This can’t be reversed. It’s too hard to go back to fossil fuels.” (19) “Long-term we are pushing R&D and recycling to avoid dependency.” (19)

6.3.3.2 Regimes

Regimes *Number of citations per category in brackets [X]	Summary of description	Literal citations *Number of interview in parenthesis (), see section 3.2
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Regime_Political [9]	The energy transition is deeply influenced by political dynamics, electoral cycles and geopolitical pressures. Experts underline the misalignment between long-term transition goals and short-term political incentives, especially within democratic systems. Making the transition work would require unpopular decisions, yet these are often postponed or avoided due to political risks. Shifts to the right across Europe have influenced the energy mix, with climate-sceptic parties threatening to reverse progress. Energy security crises, like the war in Ukraine, have paradoxically accelerated renewables in some cases, especially where dependency on fossil imports was exposed. However, disparities between EU member states in ambition and capacity remain strong, undermining collective action.	<i>"In the last 3–4 years the energy mix is changing and is reflecting the politics in Europe (change to the right)." (17)</i> <i>"The countries that have more power are the ones that have non-renewable energy." (17)</i> <i>"Disparities between EU member states in terms of intention... it's all about the money." (18)</i> <i>"We live in a democracy... political cycles are not compatible with long-term goals." (18)</i> <i>"If we want to succeed, we need to make unpopular decisions, otherwise nothing changes." (18)</i> <i>"Why should Europeans do all this effort if others are not doing it?" (18)</i> <i>"Energy security helps to promote renewables... Germany suffered from gas dependency." (18)</i>
Regime _ Economic [4]	The current energy regime is marked by financial asymmetries and short-term disincentives that undermine long-term transition goals. While renewable energy is expanding, its lower profit margins reduce its attractiveness for large-scale investors. Those with insider knowledge of energy markets are better positioned to benefit, often securing short-term gains such as reduced bills. Experts also highlight deep disparities across EU member states in terms of investment capacity and political will, reinforcing the role of economic inequalities in shaping transition trajectories. Despite the long-term vision of decarbonisation, decisions remain driven by immediate financial concerns.	<i>"Profit margin in renewable energy is far much less." (17)</i> <i>"The people who understand the market are getting financial benefits." (17)</i> <i>"Short-term decrease in their energy bill." (17)</i> <i>"Energy transition is a long-term thing, but many short-term obstacles... it's all about the money." (18)</i>
Regime _ Social [3]	Social acceptance of the energy transition is shaped by deeply rooted values and political narratives. Community consultations reveal that people prioritise wellbeing, health, clean air and the visual landscape, more than abstract goals like emissions reduction. This reflects a need for transition policies to connect with lived experiences and local values. At the same time, public discourse around climate action remains vulnerable to political polarisation. Right-wing populist narratives can erode trust and stall progress, especially when the tangible social benefits of the transition are not clearly communicated.	<i>"What came up again and again was health and wellbeing, the value of community, clean air, landscape, visual aesthetic." (17)</i> <i>"Paris Agreement — everybody says 'the best thing we made,' but... not a big change." (18)</i> <i>"Afraid of right-wing parties... as soon as they're in power, they will cancel it." (18)</i>

Regime_Technological [4]	The current technological regime is struggling to keep pace with the decentralisation and complexity brought by the energy transition. While solar PV adoption is rising (often supported by power providers) the broader grid infrastructure may not yet be ready to accommodate it. Experts cite cases like blackouts in Spain as signals of systemic instability. A stable and responsive smart grid is seen as essential, yet underdeveloped. Centralised energy sources continue to dominate global power dynamics, reflecting both geopolitical inertia and technological lag in adapting infrastructure to new energy models.	<i>"The blackout in Spain was related to solar energy." (17)</i> <i>"The countries that have more power are the ones that have non-renewable energy." (17)</i> <i>"There are power providers who are working with people to help to install solar PV." (17)</i> <i>"The smart grid needs to be stable; this may not be happening, for example in Spain." (18)</i>
Regime_Environmental [5]	Interviewees express skepticism about the dominant environmental assumptions within the current energy regime. While the transition to renewables is broadly supported, experts warn that it often overlooks hidden environmental costs; especially those outsourced beyond EU borders. The case of lithium extraction for batteries illustrates this contradiction: labelled "green," yet highly damaging. There is concern that emissions reductions are not pursued with sufficient seriousness, and that the environmental narrative of the transition lacks a holistic, global perspective. Experts call for stronger accountability on the externalities of both fossil fuels and low-carbon technologies.	<i>"The reduction of emissions is not as needed... we don't do too much about it." (18)</i> <i>"Just because it is renewable doesn't mean it's good for the environment." (18)</i> <i>"The big damage we make to harvest the lithium is incredible!" (18)</i> <i>"Renewable energy is not always good." (18)</i> <i>"We import environmental damage... we have to think of all the transition globally." (18)</i>

6.3.3.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape_Political [6]	The global political landscape is seen as a major influence on the pace and direction of the energy transition. Interviewees point to the concentration of critical materials (especially for batteries) in China, which not only dominates mining operations but also adds value through processing and manufacturing. This creates dependencies that raise geopolitical concerns for Europe and the US. Countries with fossil fuel reserves continue to hold disproportionate geopolitical power, slowing down incentives for transformation. Experts also highlight that poorer countries face different priorities, and political focus in regions like the EU or US may shift depending on external pressures such as taxation or strategic alliances. The political asymmetries between actors complicate coordinated global action on climate and energy.	<i>"Batteries are key — geopolitical consequences (supply chain...)." (17)</i> <i>"The countries that have more power are the ones that have non-renewable energy." (17)</i> <i>"China holds 90% of the critical material for batteries... they're building value added. China is very ahead of EU and US." (17)</i> <i>"A lot of dependency to taxes to USA... changes of focus." (18)</i> <i>"It's not so easy for poor countries to solve plastic problems... we have other problems." (18)</i> <i>"Any disruption affecting China</i>

		<i>is going to increase the challenges of transition.” (19)</i>
Landscape _ Economic [2]	At the macro-economic level, interviewees highlight structural risks that could hinder the energy transition. The international nature of business puts strong pressure on energy costs, making renewables appear less competitive in the short term. This perception limits investment in sustainable alternatives, especially in sectors sensitive to global price dynamics. Concerns are also raised about the geopolitical concentration of critical materials, particularly rare earths. With China dominating their production and supply chains, any disruption (whether political or economic) could mirror past energy crises and significantly delay the transition. Global dependencies and market volatility are thus seen as major landscape-level constraints.	<i>“The business is international... energy cost is too much. If they changed to more renewables, it would be more expensive.” (18)</i> <i>“There are concerns on disruptions on critical materials... most are produced in China. Any disruption affecting China is going to increase the challenges of transition.” (19)</i>

6.3.3.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Finance [6]	Interviewees highlight that lower-income households and renters face limited access to decentralised energy systems, even when subsidies are available. Upfront costs for heating upgrades or solar installations are prohibitive, and the structure of current support mechanisms often benefits wealthier individuals who can afford to invest. Even prosumer models, often portrayed as democratizing energy, require capital and financial literacy. Experts note that the benefits of subsidies and market participation frequently bypass those most in need. A shift from investment-based to operational-cost-based support is suggested to address these inequalities. Additionally, structural housing factors (such as lack of balcony space for renters) further limit access.	<i>“Transition costs a lot of money: heating costs. Rich people can afford comodidad.” (18)</i> <i>“A lot of consumers are renters, how many people have balconies suitable for this?” (18)</i> <i>“There will be problems in their access of finance, accessing health and financial benefits of being a prosumer.” (17)</i> <i>“Financial structures are more likely to be owned by cooperatives and are more likely to be back to the community.” (17)</i> <i>“We see that benefiteres from subsidies are richer people... they should pay on operational cost.” (19)</i> <i>“A distributed solar system — you’d have cheaper solar panels.” (17)</i>
Vulnerability_Skills [5]	Interviewees underline the growing risk of skill-based exclusion in the energy transition. Low-skilled workers in extractive sectors such as coal, oil and gas are expected to face displacement, with few transferable skills	<i>“Low-skilled workers in extractive industries... will face job losses.” (17)</i> <i>“There are no transferable skills, but individuals in the</i>

	available to support their integration into new energy industries. Even where technical skills exist, the pride and identity tied to traditional sectors can hinder reskilling efforts. The shortage of qualified labour (particularly energy engineers) is already slowing down infrastructure deployment. Experts emphasise that reskilling and upskilling will be crucial, but not all individuals have the same capacity or mindset to adapt. Resistance to change, combined with educational or cognitive barriers, may deepen skill-based inequalities unless proactive training and inclusion strategies are deployed.	<i>sector have pride... 'I'm a coal mining worker'." (17)</i> <i>"We don't have enough labor to install infrastructure to double the capacity of grid." (17)</i> <i>"Of course there will be changes; reskilling/upskilling will be key." (18)</i> <i>"People with a static mindset... consume a lot of shit (TV, social media)... they do not reflect what they see." (18)</i>
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6.3.3.5 Disadvantaged groups (DGs)

Disadvantaged groups identified in the energy sector interviews include low-skilled workers in fossil fuel industries, renters, low-income households, and individuals with limited adaptive capacity or technological access. These groups are at risk of being left behind in the transition due to structural, financial and cognitive barriers. Workers in traditional sectors face job losses with few transferable skills, while low-income consumers struggle to benefit from subsidies or access decentralised technologies like solar panels. Renters often lack the autonomy or infrastructure to adopt these systems. Interviewees also emphasise the psychological dimension: individuals with more rigid mindsets may be less likely to adapt to changing energy realities.

Disadvantaged groups (DGs) [9]	Summary of description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Low-skilled fossil fuel workers	Facing job losses due to the decline of coal, oil and gas sectors; limited skill transferability	Risk of unemployment and long-term exclusion from green labour markets	<i>"Low-skilled workers in extractive industries... will face job losses." (17)</i>
Engineers in nuclear sector	Nuclear expertise may become obsolete as focus shifts to renewables	Loss of technical capital and professional identity	<i>"All the engineers closer to know about nuclear energy will be lost." (17)</i>
Renters	Often lack rooftops or balconies, limiting access to solar systems or decentralised energy solutions	Excluded from prosumer models and energy savings	<i>"A lot of consumers are renters, how many people have balconies suitable for this?" (18)</i>
Low-income households	Cannot afford solar panels, retrofitting or electric vehicles, even with subsidies	Excluded from transition benefits; vulnerable to energy poverty	<i>"Transition costs a lot of money... rich people can afford it" (18)</i> <i>Even with grants, they can't buy electric vehicles, or solar panels, or constructions." (19)</i>

Individuals with low adaptive capacity	People with rigid mindsets, low media literacy or limited critical thinking	Lower capacity to engage with or support energy transition efforts	<i>"People who have a more static mindset... they do not reflect what they see." (18)</i>
Politically vulnerable low-income groups	At risk of being swayed by populist narratives if left behind in transition	Potential resistance to climate policies and social instability	<i>"Poor people will be very vulnerable to populist speeches... we have to take them with us." (18)</i>

6.3.4 Second round of Delphi survey

6.3.4.1 Elaboration of statements

85. By 2050, energy security will consistently be prioritized over sustainability in EU energy policy. **[Niche_Governance] [Niche_Technological] [Landscape_Political]**
86. The acceleration of renewable energy deployment will have continued, and grid resilience will face new challenges as illustrated by the solar-related blackout in Spain. **[Niche_Technological]**
87. The sharing of energy-sector knowledge and technology will have ceased between the EU and the rest of the world. **[Niche_Technological] [Niche_Governance] [Landscape_Political]**

6.3.4.2 Analysis of results

The transition towards wind and solar energy will continue accelerating		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	5	62.5%
Neutral	1	12.5%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	6	75%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	87.5%
Summary of qualitative comments	Solar deployment is seen as certain, while wind expansion may face restrictions, making offshore wind necessary. Eventually, a plateau could be reached where storage systems and power electronics must be installed to operate a grid with near-full renewable generation. Cost reductions and policy mandates are expected to sustain ongoing growth, with current main drivers being the expansion of renewable generation. However, a high share of renewables could lead to a grid with low inertia, and if not planned	

	correctly, this might increase blackouts and degrade power quality through voltage instability and frequency oscillations. Accelerated deployment will reshape the energy mix and drive rapid emissions reductions.
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Regulatory and bureaucratic barriers (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	5	62.5%
Neutral	2	25%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	4	50%
Neutral	1	12.5%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	62.5%
Summary of qualitative comments	Participants identify bureaucratic processes as a long-standing and ongoing obstacle to energy deployment. They highlight permitting delays and complex regulations as issues that will persist into 2050, potentially slowing capacity expansion. Some mention that these barriers may combine with technical constraints, forcing developers to adopt costly compliance workarounds and reducing the overall speed of the transition.	

The shortage of skilled labor (e.g., installers, technicians) will significantly hinder the energy transition		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	1	12.5%
Neutral	3	37.5%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	2	25%
Neutral	4	50%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree

	No, remove	50%
Summary of qualitative comments	Some believe it will create new jobs and that the labor market will adapt, but the shortage is already evident and worsening due to demographic trends such as an aging workforce, limited generational replacement, and slow progress in specialized training. The rapid growth of the renewable energy sector has driven demand beyond the current capacity of vocational and technical education, making it highly likely that a skills gap will hinder project execution in the coming years. If this persists, there will be significant delays in planning, installing, and commissioning projects, threatening decarbonization timelines. The shortfall will likely raise costs through reliance on expensive international contractors or intensive training programs, reducing returns on investment and creating bottlenecks in grid connection and maintenance.	

The main factors that will lead the energy transition will be the increase of prices, geopolitical aspects and environmental concerns		
Likelihood	Number of answers	% over the total
Very high	5	62.5%
High	3	37.5%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	7	87.5%
Neutral	1	12.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Unless breakthrough technologies such as nuclear fusion emerge, these remain the primary drivers of change. Price surges, geopolitical instability, and heightened environmental awareness will strongly influence the transition, accelerating policy reforms and investment in clean energy.	

Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	3	37.5%
Neutral	3	37.5%
Low	1	12.5%

Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	37.5%
Neutral	5	62.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	This is expected, particularly in Europe and wealthier countries, though the extent will depend on regional and national policies. Automation and plant closures will almost certainly reduce these roles. The impact on job markets will be significant, highlighting the need for skill enhancement across the workforce. Without rapid reskilling and just-transition programs, job losses will be widespread.	

Non-renewable energy producers and energy distributors will be key players in the renewable energy market, due to their diversifying strategy.		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	4	50%
Neutral	2	25%
Low	0	0%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	1	12.5%
High	4	50%
Neutral	3	37.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	62.5%
Summary of qualitative comments	Some believe they are not interested due to their core business being in a different sector, with large, fixed infrastructures to maintain. However, their financial and infrastructural resources, along with lobbying power, position them to play a decisive role. Many major oil, gas, and utility companies are already redirecting capital toward wind, solar, and storage. If they engage fully, the effect on market dynamics will be substantial. Prosumers will also be important actors. Their balance sheets, infrastructure, and customer bases could accelerate deployment and reshape market structures.	

The energy labor market will experience an increase in demand for highly skilled jobs, such as engineers		
Likelihood	Number of answers	% over the total
Very high	4	50%
High	3	37.5%
Neutral	1	12.5%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	50%
Neutral	4	50%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	87.5%
Summary of qualitative comments	The shift to clean energy will generate new technical careers, with high demand for engineers and possibly even more for technicians. Electrification, grid modernization, and R&D needs make this demand almost certain. Without adequate training, skill shortages could slow innovation and project delivery.	

The R&D&I on battery recycling will be a decisive factor in EU energy independence by 2050		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	6	75%
Neutral	1	12.5%
Low	0	0%
Very low	0	0
Impact	Number of answers	% over the total
Very high	1	12.5%
High	5	62.5%
Neutral	2	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	87.5%
Summary of qualitative comments	Advances in recycling and circularity, including standardized and repairable batteries, are considered critical to integrating renewables and reducing dependence on imported raw materials. Scaling up recycling technologies is already a major policy priority.	

	Breakthroughs in this area would significantly strengthen energy autonomy.
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New financial models will arise due to the energy market incentives created by private companies and their technification		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	6	75%
Neutral	1	12.5%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	37.5%
Neutral	5	62.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	While some see innovation already underway through corporate PPAs, green bonds, and energy-as-a-service platforms, others believe policy and societal buy-in will drive change more than finance. New models may optimize funding options and complement existing structures, but are unlikely to replace traditional capital flows entirely.	

Fossil fuel phase-out policies will generate long-term regional unemployment risk in areas historically dependent on extractive industries		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	2	25%
Neutral	3	37.5%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	37.5%
Neutral	5	62.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	37.5%

Summary of qualitative comments	Closures of mines and plants in fossil-fuel-dependent regions are expected as investment declines. Without robust reallocation and retraining measures, these areas risk sustained unemployment and economic decline, although resource extraction in other sectors may offer partial mitigation.
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Decentralized energy systems will dominate the European grid, reducing reliance on large-scale centralized infrastructure		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	50%
Neutral	2	25%
Low	1	12.5%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	2	25%
Neutral	2	25%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	While decentralization could enhance energy democracy and resilience, some doubt governments' willingness to adopt it due to operational challenges. Growth in prosumers, microgrids, and storage is already notable, but improper management could heighten blackout risks. A shift toward local control would require major changes in network design, regulation, and investment.	

Households without solar panels will face higher long-term energy costs, exacerbating inequality		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	4	50%
Neutral	1	12.5%
Low	2	25%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	37.5%
Neutral	3	37.5%
Low	1	12.5%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree

	No, remove	50%
Summary of qualitative comments	Some respondents consider this unlikely, citing subsidies, community solar, and tariff structures that can protect non-solar households. While cost disparities could emerge, they are expected to be small and mitigated by support measures, limiting the potential for increased inequality.	

Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumers rights		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	3	37.5%
Neutral	2	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	5	62.5%
Neutral	3	37.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Digitalization is advancing rapidly, with widespread deployment of smart meters, IoT devices, and cloud-based management tools. This trend is seen as inevitable, but it raises privacy and governance concerns. Strong regulatory frameworks will be needed, though their impact will be more on compliance than on the pace of the transition.	

Energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	4	50%
Neutral	2	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	2	25%
Neutral	3	32.5%
Low	1	12.5%

Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	There is skepticism about broad public uptake, but awareness and understanding of energy systems are considered important for effective participation. Time-of-use tariffs and prosumer engagement are likely to increase literacy demands. While high literacy could influence market dynamics, low literacy would limit demand response and grid optimization potential.	

EU energy dependency on third countries for critical materials (e.g., lithium, rare earths) will be a strategic vulnerability		
Likelihood	Number of answers	% over the total
Very high	4	50%
High	2	25%
Neutral	1	12.5%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	1	12.5%
Neutral	3	37.5%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Dependency on imports, particularly from Asia, is expected to persist given Europe's limited reserves and manufacturing capabilities. High reliance makes the system vulnerable to supply disruptions, which could delay renewable technology deployment and threaten energy security.	

By 2050, energy security will consistently be prioritized over sustainability in EU energy policy		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	4	50%
Neutral	0	0%
Low	3	37.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	4	50%

Neutral	2	25%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	62.5%
Summary of qualitative comments	Some see this as already happening, while others believe sustainability will remain central, with security concerns spiking only during crises. Policy frameworks like the Green Deal are seen as embedding sustainability long-term, even if short-term shifts occur.	

The acceleration of renewable energy deployment will have continued, and grid resilience will face new challenges as illustrated by the solar-related blackout in Spain		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	5	62.5%
Neutral	0	0%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	5	62.5%
Neutral	0	0%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	100%
Summary of qualitative comments	Renewables growth is expected to continue, bringing with it stability challenges. While some see resilience issues as manageable, others warn that blackouts and curtailments could require significant infrastructure upgrades.	

The sharing of energy-sector knowledge and technology will have ceased between the EU and the rest of the world		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	1	12.5%
Low	4	50%
Very low	3	37.5%
Impact	Number of answers	% over the total
Very high	3	37.5%

High	0	0%
Neutral	1	12.5%
Low	4	50%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	0%
Summary of qualitative comments	A complete halt is seen as unlikely due to the scale of global collaborations, but if it occurred, the impact would be severe, hindering technological progress. Even partial restrictions would slow innovation, though not stop it entirely.	

6.3.5 Third round of interviews

6.3.5.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [6]	Interviewees highlight the emergence of new economic models such as prosumerism, energy communities, and battery-based energy sharing. While the vision of citizen-led production and exchange has been circulating for years, its implementation remains limited, with prosumer participation still seen as marginal. However, financial innovation is gaining ground, with expectations that decentralised energy schemes and differentiated tariff structures will expand, especially in countries like Austria. Despite this, fossil fuels remain highly profitable and continue to benefit from unaccounted externalities. For new models to thrive, interviewees suggest that regulation must adapt; for example, by setting fair contributions to grid infrastructure or enabling community-level revenue sharing.	<i>"The prosumer idea is in 'baby shoes'... maybe only 2% of people are thinking like this." (22)</i> <i>"Trend of more financial schemes; how to share energy directly in communities." (21)</i> <i>"Energy communities could be more affordable at some point... with different segmented fees." (21)</i> <i>"Fossil fuels are still very profitable, and they are not paying the price for that." (21)</i> <i>"They are building a big project around the creation of a battery... they have a business model." (21)</i>
Niche_Governance [22]	Interviewees describe a growing tension between regulatory complexity and grassroots innovation. While decentralised energy initiatives like plug-and-play solar and local cooperatives are gaining traction, bureaucratic and political barriers remain strong. Shifting regulations, often driven by electoral cycles, undermine continuity and disproportionately affect small cooperatives with limited capital. National governments, particularly under conservative leadership, are seen as reverting to fossil fuel-friendly policies. Concerns also emerge about governance blind spots: biodiversity is often excluded from energy plans, technical professions remain siloed from environmental concerns, and gender divides persist in cooperative leadership. Meanwhile, energy literacy is	<i>"In Germany, you have very low barriers to install [balcony solar]... only need to complete a file." (22)</i> <i>"We are just making the shortest-term decision." (22)</i> <i>"We need strict regulations." (22)</i> <i>"Political changes are very impactful... now the conservative party talks negatively about photovoltaic." (22)</i> <i>"Growing opposition of big distribution companies... networks not working with all</i>

	expected to grow at the local level, with municipalities developing plans and battery projects expanding. Still, energy community engagement remains uneven, with time constraints and marginalisation limiting participation for many. A more inclusive and stable governance approach is needed: one that bridges technical and environmental domains, simplifies processes, and ensures energy transition policies support all scales of initiative, not just the largest actors.	<i>the volume of solar panels.” (22)</i> <i>“The increase of complexity will create unaware citizens... 100 paragraphs in the renewable energy law of Germany.” (22)</i> <i>“Energy communities could be more affordable... with segmented fees.” (21)</i> <i>“It will be regulated how much money you put into the energy network.” (21)</i> <i>“Bureaucracy will be one of the bottlenecks, but it is not the only one.” (21)</i> <i>“Some regulations affect smaller cooperatives... politicians don’t take this into account.” (21)</i>
Niche_Social [30]	The social dimension of energy transition is marked by contrasting dynamics of empowerment, resistance, and inequality. On one hand, interviewees highlight positive trends: growing energy literacy, greater citizen interest (partly triggered by the Ukraine war), and increasing participation of women and migrant communities in the renewable energy workforce. Training initiatives, both technical and organisational, are expanding, with pathways from amateur to expert, and even potential for school-based energy education. However, structural and cultural barriers remain. Populist movements and right-wing parties are fuelling resistance to renewable energy, particularly wind and photovoltaic installations, often framed as threats to landscapes or national identity. The complexity of regulation also deters citizen understanding, especially among marginalised communities with less time or capacity to engage. Gender gaps persist within energy cooperatives, with men dominating technical tasks while women are relegated to communication roles. Interviewees call for broader questions around responsibility, sustainability, and social standards, including concerns about job quality, AI’s role in replacing workers, and safety conditions in a warming climate.	<i>“The prosumer idea is in ‘baby shoes’... maybe only 2% of people are thinking like this.” (22)</i> <i>“We need to think more globally. What would be the level everyone could reach on Earth?” (22)</i> <i>“What are our social standards for low-skilled workers? If scale-up doesn’t go with quality, we don’t want it.” (22)</i> <i>“There will be 2–3 months where employees won’t be able to work on roofs due to climate change.” (22)</i> <i>“Extreme right-wing parties are a threat to the transition.” (22)</i> <i>“The increase of complexity will create unaware citizens... 100 paragraphs in the renewable energy law.” (22)</i> <i>“Many people do not have time for energy communities... different types of engagement are needed for marginal communities.” (21)</i> <i>“People will hate photovoltaic farms... populist movement.” (21)</i> <i>“Movements against renewable energy, especially wind, can be bad for the transition.” (21)</i>

		<i>"Ukraine war has woke up the interest of people in the energy system." (21)</i> <i>"Women will also be more present in the energy sector." (21)</i> <i>"New employment opportunities in renewables can help integrate migrant people." (21)</i> <i>"Perhaps some citizens will learn about energy systems at school." (21)</i>
Niche_Technological [23]	Technological innovation in the energy sector is advancing on multiple fronts, from decentralised systems and batteries to plug-and-play solar panels. Interviewees highlight the growing relevance of flexible, community-scale infrastructure, such as balcony solar or small heating networks, as more accessible alternatives to large, unpopular installations like photovoltaic farms. The sector is also experiencing a strong push for digitalisation, although opinions vary. While some see clear opportunities in software tools and automation, others remain sceptical of AI's role and stress the importance of high-quality, not just high-volume, data. The pace of change, and the technical complexity it entails, is a key challenge with growing demands on skills to manage interconnected networks and cross-sectoral applications, including agricultural photovoltaics. Nuclear and hydrogen remain contested: seen by some as outdated or unfit for large-scale deployment in places like Germany. The speed of transition and the pressure to scale up without sacrificing quality or social trust remain central concerns.	<i>"Plug-and-play solar panels on balconies are growing... it's a small thing, but it's something." (21)</i> <i>"Decentralised energy systems will happen... heating networks can be used by smaller and larger communities." (21)</i> <i>"The investment in batteries will be a way to secure energy and avoid blackouts." (21)</i> <i>"We are going towards a scenario of complexity... decentralisation requires other skills." (22)</i> <i>"I'm not really convinced on how AI helps... we don't need big data, we need good data." (22)</i> <i>"The scale-up needs quality." (22)</i>

6.3.5.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [2]	The political landscape around energy is shaped by both regulatory facilitation and shifts in strategic priorities. In Germany, streamlined procedures for small-scale solar installations reflect a political will to lower entry barriers for citizens. At the same time, the prominence of energy security has grown, particularly in the context of geopolitical instability, influencing national energy strategies and reinforcing the perceived urgency of domestic renewable deployment.	<i>"In Germany, you have very low barriers to install plug-in solar panels... only need to complete a file." (22)</i> <i>"Energy security has become more important." (21)</i>
Regime _ Economic [1]	Within the current economic regime, energy cooperatives represent a pragmatic and value-	<i>"Every cooperative is really dedicated to energy"</i>

	driven model. They combine affordability with flexibility by focusing on low-entry options like rooftop solar, which require less upfront capital. This allows them to diversify progressively into wind or heating, while maintaining their role as part of a broader transition movement. Their success reflects an alternative approach to energy investment that is more accessible and decentralised.	<i>transition... they use photovoltaic in roofs, and sometimes buy some wind energy or heat to diversify.” (21)</i>
Regime _Social [5]	The energy regime continues to reflect uneven levels of public engagement and awareness. While some efforts aim to increase inclusivity, such as cooperatives training women in photovoltaic skills, much of the population remains disconnected from the energy sector. Low energy literacy, limited time, and unequal access to knowledge reinforce social gaps, as energy communities often emerge among well-resourced individuals. In parallel, resistance to renewables also stems from local concerns, particularly landscape preservation. The tension between renewable deployment and environmental protection reveals the need to align social acceptance with ecological goals.	<i>“People have very low literacy... they have a contract with the provider and that is all.” (21)</i> <i>“Energy communities are born by people who are settled... they have time and money.” (21)</i> <i>“She has been working with a cooperative to train women for photovoltaic, but there is still a lot to do.” (22)</i> <i>“People are against wind energy because they want to preserve forests.” (21)</i>
Regime _Technological [4]	Within the current technological regime, some core systems such as wind and solar energy are expected to remain dominant due to their proven efficiency. However, there is also frustration with the pace of deeper innovation. While technical efficiency has improved incrementally, foundational principles in turbine design and other systems are perceived as stagnant. This stagnation contrasts with the urgency of the energy transition, suggesting a disconnect between technological continuity and the scale of change required.	<i>“Photovoltaic on buildings will stay for sure.” (21)</i> <i>“Wind energy will continue as it is very efficient.” (21)</i> <i>“Efficiency has evolved immensely, but it is still the same principles.” (22)</i> <i>“In the last 50 years, the efficiency of turbines has been the same.” (22)</i>
Regime _Environmental [4]	Environmental considerations remain under-addressed in the current energy regime. While technological efficiency has improved, its foundational principles have not changed. Interviewees express concern that the dominant focus on renewables often overlooks broader ecological impacts, particularly biodiversity. Batteries are cited as a key example, with lithium extraction carrying heavy environmental costs. Without integrating biodiversity and ecosystem concerns into mainstream planning, the transition risks reproducing extractive patterns under a green label.	<i>“Batteries... a lot of environmental impact to get the lithium.” (22)</i> <i>“Energy transition is not everything, if we do not consider the biodiversity.” (21)</i> <i>“People in the energy sector are so concentrated in renewables, they do not think of biodiversity loss.” (21)</i>

6.3.5.3 Landscapes

Landscapes *Number of citations per category in brackets [X]	Summary of description	Literal citations *Number of interview in parenthesis (), see section 3.2
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Landscape _Political [9]	The political context surrounding energy transition is marked by geopolitical dependency, colonial legacies and domestic polarisation. Interviewees criticise reliance on imported energy vectors like hydrogen from North Africa, calling it a continuation of extractive practices. The dependency on China for solar panel manufacturing is also seen as a strategic vulnerability, now belatedly acknowledged. Domestically, shifting political majorities directly affect regulatory stability and public trust. Green party proposals often face media backlash, while conservative parties are perceived to slow down or reverse progress on renewables. Broader geopolitical events, such as the war in Ukraine, have reshaped public perceptions but have not fully translated into long-term political commitment.	<i>"We need renewables with our EU resources, not importing... importing hydrogen from North Africa is going back to colonialism."</i> (22) <i>"China is producing all our panels. This is not new, but now everybody is crying because we ignored our dependence."</i> (22) <i>"When the green parties propose ideas, they are backlashed... they were destroyed from a media campaign."</i> (22) <i>"In Germany they just got a conservative party... so they are going a bit backwards."</i> (21)
Landscape _Economic [2]	Economic uncertainty in the energy transition is heightened by the volatility of political landscapes. Interviewees point out that prosumers, citizens who invest in renewable energy production, face significant risks linked to long-term investment decisions. Shifts in political leadership, especially toward more conservative or right-wing positions, could lead to abrupt changes in energy policies, undermining financial predictability and trust in the system.	<i>"If a right-wing party wins, the energy system changes... that's a lot of risk for long-term investments."</i> (21)
Landscape_Social [5]	Social dynamics at the landscape level present growing challenges to the energy transition. The spread of misinformation and distrust is intensifying, with interviewees expressing concern over the sheer volume of unverified content and the rise of populist rhetoric. Extreme right-wing parties and organised movements opposing renewable energy are seen as significant threats, often mobilising public sentiment against projects like photovoltaic farms. This backlash is not only ideological but also linked to perceived disruptions to local environments and ways of life.	<i>"The amount of unverified information is so big... many journalists work to debunk fake news, but it's exponential."</i> (22) <i>"Extreme right-wing parties are a threat for the transition to renewable energy."</i> (22) <i>"People will hate the photovoltaic farms as they are destroying the landscape."</i> (21) <i>"Movements against renewable energy."</i> (21)
Landscape _Environmental [1]	Environmental concerns at the landscape level are tied to broader debates on responsibility, consumption and sustainability. Interviewees argue that energy security must not be addressed in isolation from ecological limits and conflict risks. Western consumption patterns are described as excessive, and calls are made for a shift toward self-reliance using domestic resources in a more sustainable and restrained way, in line with environmental standards.	<i>"We can't think of energy security without thinking of wars... it is a matter of responsibility to use our own resources sustainably."</i> (22)

6.3.5.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [11]	Labour market impacts from the energy transition present both risks and opportunities. Interviewees stress the emergence of new employment paths in renewables, particularly for underrepresented groups such as women and migrants. However, these opportunities often collide with entrenched social norms, care responsibilities, or lack of training, especially among women. Heatwaves and outdoor work conditions add further challenges, particularly in low-income or manual roles. Concerns also emerge about low-skilled jobs being displaced or replaced by automation, leading to a potential decline in work quality and job stability.	<i>"In Germany in 2025 many women are not working, as childcare is expensive and the tradition is that women stay at home. But the green parties are looking for employees... how do we engage these women?" (22)</i> <i>"New employment opportunities for renewable energy are great to integrate migrant people." (21)</i> <i>"How far will AI go to substitute people in the factories?" (22)</i> <i>"If the scale-up does not go with quality, then we do not want it." (22)</i>
Vulnerability_Finance [6]	Financial barriers remain a key obstacle for the energy transition, particularly for disadvantaged groups. Women, who are more often renters and tend to earn lower wages, face greater challenges in accessing solar panels or becoming proactive prosumers. Energy communities are typically initiated by individuals with time and financial stability, which can exclude others. Additionally, some regulations disproportionately affect smaller energy cooperatives with limited capital, favouring larger players. There are also concerns about whether the sector can attract the required engineering talent given current salary expectations.	<i>"Renters... people who can't afford solar panels or heating systems. Women are more often renters than homeowners, as they earn less money." (21)</i> <i>"Many energy communities are born by people who are settled, they have time and money to dedicate to it." (21)</i> <i>"DGs: women... without skills to save money becoming a proactive prosumer." (22)</i>
Vulnerability_Skills [9]	The energy transition demands a wide range of new technical skills. While companies are starting to offer tailored training programmes, both basic and advanced skills are needed, from installing solar panels to managing complex electrical systems. However, low-skilled workers face significant challenges in keeping up with quality standards, and a mismatch persists between available training and sector demands. Women, in particular, are underrepresented and often lack the skills or confidence to participate. Although there are reskilling opportunities (e.g. from the automotive to energy sector), concerns remain about how to engage those with low digital literacy, especially when educational efforts focus only on already interested individuals.	<i>"You need both easy competences to build a powerplant and more advanced skills like electrical systems." (22)</i> <i>"People have very low literacy, they have a contract with the provider and that is all." (21)</i> <i>"Women... without skills to save money becoming a proactive prosumer." (22)</i>

6.3.5.5 Disadvantaged groups (DGs)

The energy transition presents uneven impacts across different social groups. Women, low-income renters, migrants and low-skilled workers face structural barriers in participating equally as prosumers or workers. While many initiatives aim to increase diversity, digitalisation, technical complexity and affordability gaps continue to exclude marginalised populations. Women are often underrepresented in technical roles and more likely to rent, limiting access to home-based renewable solutions. Refugees and migrants are being integrated through job programmes, but capacity gaps and stereotypes persist. Smaller cooperatives with limited capital may also be disproportionately affected by regulatory barriers.

Disadvantaged groups (DGs) [22]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Women	Underrepresented in technical energy roles, often relegated to communication tasks or excluded due to gender norms and care responsibilities	Exclusion from decision-making, limited technical participation, dependency on others for retrofitting and maintenance	<i>"Women without skills to save money becoming a proactive prosumer." (22) "The stereotype and roles of men are so strong that women do not even get interested." (22) "Many women not working, as the childcare is still very expensive... but how do we engage these women?" (22)</i>
Renters and low-income households	Often unable to afford solar panels or heating upgrades; renters cannot install systems themselves	Inability to participate as prosumers, left out of subsidy benefits	<i>"People who can't afford solar panels. Women are more renters than homeowners, as they earn less." (21) "Renters." (21)</i>
Migrants and refugees	Targeted for inclusion in green job programmes	Potential integration in renewable energy workforce	<i>"New employment opportunities for renewable energy is great to integrate migrant people." (21) "They are working with marginalised people (refugees), women." (21)</i>
Low-skilled and ageing workforce	Often excluded due to increasing technical complexity of systems	Need for reskilling, risk of job loss or marginalisation	<i>"What are our social standards for low-skilled workers?" (22) "Reskilling people from the automotive sector to the energy sector." (22)</i>
Small cooperatives	Struggle to comply with changing regulations due to limited capital and administrative capacity	Risk of being crowded out by larger players	<i>"Some regulations affect smaller energy cooperatives that do not have a lot of capital." (21)</i>

6.4 Mobility and transport

6.4.1 First round of interviews

6.4.1.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [10]	The interviews highlight a major transformation in the economic model of the mobility sector, especially the automotive industry. Traditional product-based business models (e.g., car ownership) are being replaced by service-oriented models such as subscription or shared mobility (e.g., mobility-as-a-service). Experts emphasize that this shift questions the long-term sustainability of automotive manufacturing, especially given the industry's significant contribution to national and regional GDP in some EU countries. The economic weight of private companies contrasts sharply with the limited role of public administrations and third sectors. This imbalance raises concerns about unequal influence and accountability in shaping the future of mobility. While the transition to electric vehicles and battery recycling creates new economic opportunities, it also accelerates the decline of traditional automotive structures and introduces uncertainties for consumers, especially with the rising costs of vehicle ownership. Innovation is seen as a double-edged sword: while autonomous vehicles and battery recycling could generate new markets, the transition is likely to result in a leaner and more concentrated industrial base.	<i>"The electric vehicle will dominate in developed countries. It will still be more expensive than combustion ones" (8)</i> <i>"The total cost of ownership is no longer so advantageous; it creates doubts for consumers" (10)</i> <i>"If autonomous vehicles scale up, it will be a game changer. I wouldn't need to own a car anymore. I could just book a self-driving one via app. This would require massive regulation" (10)</i> <i>"When transport is considered a right, business models should not take priority" (9)</i> <i>"We are heading towards a smaller industrial base" (8)</i> <i>"The automobile sector is in decline" (9)</i>
Niche_Governance [8]	The governance challenges highlighted in the mobility sector by the experts go around multi-level coordination, regulation of emerging technologies, and the role of public institutions in guaranteeing equitable access. Interviewees underscore the fragmented nature of public administration in many EU countries, which creates difficulties for the creation of coherent and inclusive mobility policies. Experts call for public operators to share data and align values-based strategies to manage the transition effectively. The transition to new models like mobility-as-a-service (MaaS) requires deep behavioral and institutional change, including policies that integrate private vehicles into shared systems. Regulation becomes essential to oversee the electrification process, the emergence of autonomous vehicles, and the expansion of public transport. The European	<i>"We are on the right track with vehicle electrification. The European Commission is leading that push, aiming for nearly full electrification by 2035" (10)</i> <i>"Electric public transport is coming. Technology is ready, and regulation is catching up. Nearly 100% of vehicles could soon be electric" (10)</i> <i>"The electrification debate is becoming more political and less about technology. This makes decisions harder. We are giving free parking and subsidies to the rich, which raises fairness issues" (10)</i>

	Commission is seen as playing a key role in promoting electrification targets, but implementation remains complex due to political tensions and the growing influence of the private sector (e.g., Tesla, VW, Chinese companies). Governance frameworks are also under pressure to ensure fairness: policy incentives (e.g., free parking, subsidies) may disproportionately benefit wealthier groups, raising concerns about equity in the transition.	<i>“Public operators need to reach agreements. We must share the needs of different actors through tools like data lakes. Multi-level governance is complex—e.g., Catalonia versus Spain—and public administration is fragmented. We need policies grounded in values” (9)</i> <i>“People have a right to mobility, including access to health and cultural services. Transport should remain public. The administration is trying to privatize this. Taxi drivers should be part of the transition” (9)</i>
Niche_Social [10]	The interviews underscore that the mobility transition has deep social implications, particularly regarding equity, rights, and behavioral change. A key concern is the framing of transport as a social right rather than a commodity. When access to mobility is seen as essential for health, education, or employment, purely profit-driven models are questioned. Experts argue that this reframing should inform public policy and infrastructure planning. There is also concern about the unequal distribution of impacts and benefits. The transition toward new mobility models (e.g., subscription services, mobility-as-a-service) must be designed inclusively, accounting for the needs of vulnerable groups like low-income users or those dependent on traditional modes like taxis. Social pressure is also pushing for the redefinition of KPIs in transport companies to measure societal rather than purely economic outcomes. Furthermore, changes in consumer behavior, like reduced private car usage, are already happening and must be addressed through inclusive design and regulation to ensure no one is left behind in the shift toward shared and digital mobility ecosystems.	<i>“There is pressure on the taxi-driver monopoly, this comes from social groups demanding more inclusive services” (9)</i> <i>“Mobility-as-a-service is changing our behaviors. For instance, sharing your private car with public transport services” (9)</i> <i>“Owning a vehicle is no longer very advantageous. Consumers are starting to question whether it’s worth it” (10)</i>
Niche_Technological [17]	The interviews reveal that the mobility transition is strongly shaped by a diversification of technologies adapted to different geographies, use cases and vehicle types. While electric vehicles (EVs) are perceived as optimal for short distances and urban transport, experts suggest that hydrogen may play a more significant role in long-distance logistics, public transport, and freight. Each technology is viewed as complementary, not exclusive, with specific advantages depending on range, charging needs, and infrastructural availability.	<i>“In short-distance routes, EVs are a good solution. For long-distance routes, it’s more complex” (9)</i> <i>“Hydrogen may be better for logistics or long-distance vehicles like trains and trucks. But it depends on political willingness and cost reduction” (9)</i> <i>“We’re going toward more autonomy, ecodesign and</i>

	Battery innovation is also a central theme. Interviewees mention the shift toward lighter, longer-lasting and more ethically sourced components (e.g., LFP batteries instead of cobalt-based ones), and emphasize the growth of battery recycling as a strategic industrial priority. The role of ecodesign, autonomy extension, and the emergence of circular economies around battery reuse were particularly stressed. Experts foresee the expansion of technologies such as autonomous driving, drone delivery for last-mile logistics and the electrification of buses, all of which require stronger public-private partnerships and major infrastructure upgrades. The development of solid-state batteries, smaller and more powerful grid boxes, and extended vehicle lifespans are key for enabling a zero-emission fleet by 2035. Maturity of technology is not just a matter of innovation but of political willingness and social regulation.	<i>lighter batteries. From 400km today to 800km by 2050" (8)</i> <i>"Battery innovation is accelerating: solid-state batteries, LFP instead of cobalt, better density, and more ethical sourcing" (10)</i> <i>"EV cost is still higher than combustion vehicles, especially in developing countries, but adoption is accelerating" (8)</i> <i>"Drones could be used for last-mile delivery in urban areas" (8)</i> <i>"Battery recycling is still underdeveloped; right now, they mostly go to electric landfills. We must create a recycling industry" (8)</i> <i>"Every technology covers a different mobility need. There won't be a single solution" (9)</i>
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6.4.1.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [3]	While electrification is perceived as inevitable, political actors are seen as potential bottlenecks that may delay rather than block the transition. Experts suggest that instead of preserving the status quo or protecting obsolete industrial models, political leaders should actively promote industrial reconversion. Public administration is described as a key actor in the governance of transport, especially in segments like rail. Its role is not merely operational but also strategic, as public operators are expected to drive the social dimension of the transition. However, experts highlight that this role must be fully internalized moving beyond declarations to effective public leadership in decarbonization. There is a consensus that the political environment influences the speed and scope of transformation. While political resistance may delay changes, structural shifts in mobility such as electrification and public transport reform are ultimately seen as irreversible.	<i>"Politicians should focus on how to reconvert the industry, not on maintaining it at all costs" (9)</i> <i>"The social role of transport is being carried out by public operators, especially rail. The main actor here is public administration, but this needs to be fully internalized" (9)</i> <i>"Electrification will happen, no matter what. Politics can only slow it down, not stop it" (8)</i>
Regime _ Economic [4]	The interviews expose several economic barriers within the dominant regime that slow down the transition toward sustainable mobility. Key among these is the influence of powerful industrial and	<i>"The automotive industry is on steroids. What happens when the doping ends? The entire value chain depends on it" (8)</i>

	energy lobbies, which actively resist change and seek to protect their interests. Private operators in transport and automotive manufacturing (especially tiered suppliers) are described as reluctant to engage in transformation processes. Experts underscore that resistance is not only technological but deeply rooted in economic self-preservation. There is also criticism of energy companies' superficial adaptations, such as rebranding or investing in alternative sources without genuine transformation. These actions are perceived as greenwashing, aimed at maintaining market share while evading structural shifts. Another major barrier is the industry's economic dependency on global supply chains. For instance, the high cost and strategic importance of batteries (up to 70% of a car's value) pose a vulnerability: many European manufacturers lack domestic battery production capacity and rely heavily on imports, which reduces their profitability margins. Interviewees warn that current profitability may be artificially inflated due to subsidies or market protection. As these supports phase out, the real resilience and competitiveness of the industry will be tested.	<i>"70% of a car's value is the battery, but European manufacturers can't produce them locally. They import them, which reduces profit margins. EVs generate less economic margin than traditional vehicles" (10)</i> <i>"There are strong industrial lobbies, especially in transport and automotive, that are blocking change. Tier 1, 2, 3 suppliers just don't want to evolve" (9)</i> <i>"Energy companies resist renewable transitions. CEPSA is rebranding, REPSOL is exploring new sources... but these are greenwashing tactics" (9)</i>
Regime _Social [2]	From a social perspective, interviewees highlighted the role of European values in shaping the future of mobility. These include commitments to equality, environmental protection and social welfare, which are expected to remain core priorities across all EU strategies, particularly in alignment with the Sustainable Development Goals (SDGs). These values serve as a normative anchor for the just transition and help frame the expected role of mobility as a basic right. At the same time, experts pointed to growing tensions within the automotive industry's social fabric. The sector is currently undergoing a restructuring process that includes widespread job losses and declining labor security across the value chain. There is an urgent need to anticipate these changes through large-scale reskilling strategies, especially for those working in traditional mechanic roles. These social dynamics reflect both structural pressures and normative continuities: while the system is under strain, the foundational values of the welfare state and labor protection still frame expectations and demands from both industry and policymakers.	<i>"EU values such as equality, environmental protection and welfare will remain in place. These are core elements of European strategy, like the SDGs and similar commitments" (9)</i> <i>"The automotive industry in Europe is struggling. A lot of people are being laid off; the entire value chain is under pressure. We need to invest in retraining, especially for mechanical jobs" (10)</i>
Regime _Technological [2]	At the regime level, experts pointed to existing technological strategies that structure the European mobility transition. A combined focus on hydrogen and electrification has emerged as the dominant	<i>"EU strategies are combining hydrogen and electrification as the main pillars of the transition" (9)</i>

	framework, shaping investments and innovation pathways. This dual-track strategy, led by EU institutions, provides technological directionality and signals long-term support for low-emission solutions. However, significant technological uncertainties persist particularly in the logistics segment. Emissions from trucks and tractors are difficult to mitigate, and experts noted that Europe has yet to find a reliable pathway for these segments. International experimentation, such as battery-swapping pilots in China, highlights the complexity and variability of solutions across different regions. These reflections underscore that while the technological regime is becoming more defined around electrification and hydrogen, sector-specific challenges (like freight transport) remain unsolved, requiring further innovation and adaptation.	<i>“Logistics are a big source of emissions. Trucks, tractors... we’re facing real difficulties here—it’s hard to predict the outcome. In China they are testing battery swapping as an alternative” (10)</i>
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6.4.1.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [1]	Political uncertainty, particularly linked to ideological shifts, may undermine the recognition of mobility as a public right.	<i>“Political willingness and uncertainty are critical.”(9)</i>
Landscape _Economic [2]	Experts highlight how macroeconomic conditions may slow the transition. Cuts in innovation and digital infrastructure investment due to financial uncertainty could hinder progress. Additionally, high energy prices and rising vehicle costs risk making electric mobility less competitive, especially in more vulnerable markets.	<i>“We are seeing cuts in innovation investment—possibly linked to the economic downturn. We need to invest in other technologies too, such as data lakes and green energy... But the financing is being reduced” (9)</i> <i>“The transition could be slowed down due to high energy prices, which make electric vehicles less competitive, and also due to the rising cost of cars overall” (10)</i>
Landscape _Social [1]	Experts warn that right-wing positions rejecting this principle could dismantle the foundational values of current mobility policy frameworks.	<i>“Right-wing parties saying things like ‘transport is not a right’, this would break all values.”</i>

6.4.1.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>

Vulnerability_Employment [4]	Workers across the automotive value chain face increasing job insecurity due to electrification, automation and industrial restructuring. Factory employees are at risk of early retirement schemes or layoffs due to higher automation levels. In parallel, older mechanics may struggle to adapt to electric vehicle repairs, exposing a generational gap in technical preparedness. At the same time, electrification may create new job opportunities for women in roles historically less accessible within the automotive sector. Experts highlight the urgent need for upskilling and reskilling measures to prevent further exclusion.	<i>“Can older mechanics really handle an electric vehicle breakdown? This is a disadvantaged group, an ageing mechanic.” (8)</i> <i>“One benefit: more women can access the sector.” (8)</i> <i>“The factory worker is the disadvantaged group here: factories will be far more automated. It’s all about recycling and early retirement plans.” (8)</i> <i>“Right now, the automotive industry in Europe is suffering. Lots of people are being laid off. The entire value chain is struggling. There has to be reskilling, especially in mechanics.” (10)</i>
Vulnerability_Finance [4]	Economic vulnerability is perceived as less prominent in the mobility transition, yet several concerns persist. Rural populations with lower purchasing power and poor households face affordability barriers, especially with the higher cost of electric vehicles. Experts warn that subsidies and incentives often disproportionately benefit wealthier urban users, reinforcing inequality. The politicisation of electrification decisions adds further complexity, potentially sidelining those most financially vulnerable.	<i>“Disadvantaged groups: rural areas with lower purchasing power.” (8)</i> <i>“Poor people are disadvantaged: electric vehicles are just more expensive than normal ones. There is an issue of affordability.” (10)</i> <i>“You end up offering free parking and grants to people who are already relatively rich.” (10)</i>
Vulnerability_Skills [2]	The shift to electric and digitalised mobility systems poses a significant skills gap, particularly among older workers and traditional mechanics. Experts stress the urgent need for reskilling, especially in electric vehicle maintenance and battery technologies. The entire automotive value chain is under pressure, and without targeted upskilling programmes, many workers may be left behind.	<i>“Can older mechanics really handle an electric vehicle breakdown?” (8)</i> <i>“There must be reskilling in mechanics.” (10)</i>

6.4.1.5 Disadvantaged groups (DGs)

According to the interviews, DGs in the mobility sector include employees from private transport platforms facing regulatory exclusion, rural populations with fewer transport options and lower purchasing power, workers from the traditional automotive industry impacted by automation and older mechanics lacking

skills to adapt to electric vehicles. Additionally, affordability barriers and poorly targeted incentives create financial exclusion for low-income urban residents.

Experts note that the transition toward electrification and new mobility models (e.g., shared, autonomous, or subscription-based transport) is disrupting existing employment structures. Workers in private ride-hailing or delivery services may not be absorbed into public systems. In parallel, the shift to electric vehicles is accelerating automation in automotive factories, threatening job stability for traditional workers. Reskilling is urgently needed, especially for older or low-skilled profiles such as combustion mechanics.

On the user side, rural and low-income households face the dual challenge of reduced service availability and unaffordable clean mobility solutions. Experts also warn about urban inequalities, where incentives like free parking benefit wealthier individuals, while those lacking space to charge EVs or with lower digital access are left behind.

Disadvantaged groups (DGs) [9]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Drivers in private platform companies	Workers in private mobility services (e.g., Cabify, Uber) may be excluded if public sector reclaims control.	High risk of job loss or exclusion if these services are not integrated into future public mobility systems.	<i>"Private transport companies are meant to disappear... What happens to their employees if they are not integrated into the public sector?" (9)</i>
Automotive factory workers (blue-collars) (e.g., assembly roles)	Workers across the automotive manufacturing sector may be affected by automation, downsizing and industrial restructuring.	Job losses and forced early retirements due to increased automation and plant optimization. Low chances of reskilling.	<i>"Factory workers will be hit: plants are becoming highly automated. Some will be recycled or forced into early retirement." (8)</i>
Mechanics with outdated skills (combustion vehicles)	Older mechanics with traditional skill sets face difficulties adapting to electric vehicle maintenance.	Skills mismatch and exclusion from the new EV maintenance market.	<i>"Can older mechanics deal with an EV failure? They're a DG." (7)</i>
Low-income individuals unable to afford EVs	People with limited capital are unable to afford the high upfront costs of electric vehicles, even with subsidies.	Exclusion from electric mobility; persistence of social inequality in access.	<i>"EVs are more expensive than combustion ones. Poor people are left out." (8)</i>
Urban residents without access to EV charging	Citizens living in dense areas without private garages or parking spots cannot easily charge EVs.	Structural exclusion from EV use despite living in areas where car use may still be necessary.	<i>"Urban citizens without parking spots are the ones who have a problem." (8)</i>
Rural low-income populations	Rural residents with low purchasing power face both income and territorial vulnerability (longer	Transport poverty and lack of access to affordable mobility solutions.	<i>"Disadvantaged group: rural zones with lower purchasing power." (10)</i>

	distances, fewer services).		
Low-income households	People with limited income struggle to access electric mobility due to high upfront costs.	Reduced access to EVs and risk of exclusion from sustainable mobility models.	<i>"Poorer groups are disadvantaged: electric vehicles are simply more expensive than traditional cars." (10)</i>
Urban low-income users without private parking	City dwellers without dedicated parking struggle to adopt EVs and benefit from charging infrastructure.	Excluded from EV incentives and infrastructure-based benefits.	<i>"In urban areas, people without a parking spot are the ones truly disadvantaged." (10)</i>
Citizens dependent on cars but without sustainable alternatives	Urban or peri-urban citizens who must use cars (e.g., for commuting), but lack affordable EVs or alternatives.	Caught in a mobility trap; burdened by fuel costs, regulation, or exclusion.	<i>"You live in the city but have to go to a place with no public transport. You need a car but can't charge it." (10)</i>
Automotive value chain employees (blue-collar and white-collar)	Workers throughout the automotive value chain are exposed to layoffs due to sectoral transition.	Risk of unemployment without adequate reskilling pathways.	<i>"The entire automotive value chain in Europe is suffering; reskilling in mechanics is urgently needed." (8)</i>
Citizens affected by inequitable subsidy design and/or unfamiliar with bureaucratic procedures	Subsidies like free parking or grants often benefit wealthier citizens, excluding those most in need. Moreover, citizens with low bureaucratic literacy or access barriers (e.g., migrants, elderly) may not benefit from EV subsidies or incentives.	Unequal distribution of public resources, deepening social gaps and exclusion from public support schemes due to lack of access, knowledge or digital skills.	<i>"You end up giving free parking and grants to wealthier people, this is politically problematic." (10)</i>

6.4.2 First round of Delphi survey

6.4.2.1 Elaboration of statements

88. By 2050, electric vehicles (EVs) will account for at least 90% of new private vehicles purchases in Europe. **[Niche_Economic] [Niche_Technological]**
89. Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption. **[Vulnerability_Finance] [Niche_Governance] [Disadvantaged_groups]**
90. Fiscal incentives for EVs will be redesigned to better target low-income households. **[Niche_Governance] [Vulnerability_Finance] [Disadvantaged_groups]**
91. Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050. **[Niche_Economic] [Niche_Technological]**

92. Urban areas citizens will face more challenges than rural areas citizens in the transition to EV. **[Disadvantaged_groups]**
93. Mechanics without proper knowledge to repair electric engines will be negatively impacted by the transition towards electric mobility sector. **[Disadvantaged_groups] [Vulnerability_Employment] [Vulnerability_Skills]**
94. Limited knowledge of bureaucratic processes and financial literacy constitutes a significant vulnerability related to EV acquisition. **[Vulnerability_Skills] [Disadvantaged_groups]**
95. The automatisisation of the traditional automotive manufacturing jobs will impact blue-collar workers. **[Vulnerability_Employment] [Niche_Technological]**
96. Autonomous vehicle technology will remain marginal in urban transport until after 2050. **[Niche_Technological]**
97. New business models such as Mobility-as-a-Service (MaaS) models will be a reality in 2050. **[Niche_Governance] [Niche_Economic]**
98. The challenges of battery recycling will hinder the growth of the EV market in 2050. **[Niche_Technological] [Niche_Economic]**
99. Multi-level governance will pose challenges for coherent policy implementation in the electrified mobility sector. **[Regime_Political]**
100. Fiscal incentives and subsidies will boost consumer adoption of EV by 2050. **[Niche_Economic] [Niche_Governance] [Niche_Social]**
101. The logistic sector will experience problems adopting EV on the road. **[Niche_Technological]**
102. Regarding the batteries market, cobalt-based batteries will be left out **[Niche_Technological] [Niche_Economic]**
103. The lack of infrastructure to charge the EV vehicles will be a challenge for the transition. **[Niche_Technological]**
104. The whole value chain of EU automotive factories will be disrupted by 2050, due to changes in consumption patterns and the transition to EV. **[Niche_Economic] [Niche_Social] [Landscape_Political]**
105. Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs. **[Disadvantaged_groups] [Vulnerability_Finance]**
106. Battery reuse and recycling will enable circular business models in the automotive sector. **[Niche_Economic] [Niche_Technological]**

6.4.2.2 Analysis of results

By 2050, electric vehicles (EVs) will account for at least 90% of new private vehicle purchases in Europe		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	2	25%
Neutral	2	25%
Low	1	12.5%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	4	50%

High	2	25%
Neutral	2	25
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Respondents were split. Several argued that once OEMs lock in their EV production plans after 2030—and with supportive regulation, cost reductions, and growing urban-mobility policies—EVs are almost certain to dominate sales. Others doubted that the EU can scale charging infrastructure, incentives and consumer uptake fast enough, or questioned whether battery-electric is the final “eco-vehicle” solution. Despite this divergence, most agreed that if 90% penetration is reached, the environmental and energy-sector impacts will be very large.	

Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	4	50%
Neutral	0	0
Low	1	12.5%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	2	25%
High	4	50%
Neutral	1	12.5%
Low	0	0%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	All who spoke to this saw a clear risk of spatial and socioeconomic exclusion where charging points are unevenly distributed. Case studies from China and Ireland were invoked to show how rural drivers, flat-dwellers and lower-income groups could face persistent barriers to EV ownership and use. While a couple of respondents had no strong view, the prevailing sense was that lack of charging access would have a high negative social impact unless proactively addressed.	

Fiscal incentives for EVs will be redesigned to better target low-income households		
Likelihood	Number of answers	% over the total

Very high	0	0%
High	2	25%
Neutral	4	50%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	12.5%
High	2	25%
Neutral	3	37.5%
Low	2	25%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	Most participants believed that as equity concerns rise, subsidy schemes will evolve from flat purchase grants to means-tested rebates, scrappage bonuses and support for the used-EV market. However, they cautioned that political will, budget constraints and administrative complexity could slow reform. If redesigned effectively, the impact on broadening adoption and reducing mobility inequality would be high.	

Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050		
Likelihood	Number of answers	% over the total
Very high	4	50%
High	2	25%
Neutral	2	25%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	3	37.5%
Neutral	1	12.5%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	There was near-unanimous agreement that closing the materials loop (recovering lithium, cobalt and nickel) and building local gigafactories are essential to shield the EU from geopolitical supply shocks. Respondents also stressed social-justice must accompany technical advances, ensuring mining-region communities share in the circular-economy gains.	

Urban areas citizens will face more challenges than rural areas citizens in the transition to EV		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	2	25%
Low	2	25%
Very low	4	50%
Impact	Number of answers	% over the total
Very high	0	0%
High	0	0%
Neutral	4	50%
Low	2	25%
Very low	2	25%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	0%
Summary of qualitative comments	Opinions diverged: some noted urban pain points around parking-and-charging access in dense housing, low-emission zones and higher ownership costs, while others pointed to rural “charging deserts,” longer trip distances and poor network density. Most concluded that both environments have distinct barriers ,and that rural areas may actually be slightly worse off, so balanced infrastructure policy is needed.	

Mechanics without proper knowledge to repair electric engines will be negatively impacted by the transition towards electric mobility sector		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	1	12.5%
Neutral	4	50%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	12.5%
High	0	0%
Neutral	3	37.5%
Low	4	50%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	37.5%
Summary of qualitative comments	Nearly everyone agreed that garages and technicians lacking high-voltage diagnostics skills face significant disruption. Yet most also	

	saw a pathway through vocational retraining, certification programmes and OEM-supported upskilling, which can absorb many into the new maintenance ecosystem.
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Limited knowledge of bureaucratic processes and financial literacy constitutes a significant vulnerability related to EV acquisition		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	25%
Neutral	4	50%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	25%
Neutral	3	37.5%
Low	2	25%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	Respondents highlighted that navigating complex subsidy applications, financing options and technical jargon can deter less literate consumers. While a few believed one-stop digital platforms could ease this friction, the majority judged the procedural barrier's impact to be moderate-to-high without simplification and outreach.	

The automatization of the traditional automotive manufacturing jobs will impact blue-collar workers		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	4	50%
Neutral	0	0%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	5	62.5%
Neutral	1	12.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	87.5%

Summary of qualitative comments	All agreed that robotics, AI and lean-production techniques will eliminate a large share of manual assembly roles, some estimates cited up to 40% fewer positions. Although new “Industry 4.0” jobs will emerge, respondents warned that without coordinated reskilling and social-protection policies, many blue-collar workers will face serious hardship.
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Autonomous vehicle technology will remain marginal in urban transport until after 2050		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	0	0%
Neutral	2	25%
Low	4	50%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	2	25%
High	0	0%
Neutral	2	25%
Low	4	50%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	12.5%
Summary of qualitative comments	Respondents pointed to persistent technical hurdles (safety validation, sensor reliability), regulatory lag, public trust issues and necessary infrastructure upgrades as factors keeping AV penetration at niche levels (perhaps around 10%) well beyond mid-century. A minority held out hope for faster breakthroughs, but the dominant view was one of marginal deployment.	

New business models such as Mobility-as-a-Service (MaaS) models will be a reality in 2050		
Likelihood	Number of answers	% over the total
Very high	4	50%
High	2	25%
Neutral	1	12.5%
Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	0	0%
Neutral	2	25%
Low	4	50%

Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Everyone agreed that integrated, on-demand platforms are already emerging in major cities and will continue growing. While private car ownership won't vanish, MaaS offerings are seen as poised to claim a significant slice of urban trips, reshape revenue models and enable more data-driven, seamless travel experiences.	

The challenges of battery recycling will hinder the growth of the EV market in 2050		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	0	0%
Neutral	2	25%
Low	3	37.5%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	2	25%
High	2	25%
Neutral	1	12.5%
Low	2	25%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	25%
Summary of qualitative comments	Views were mixed. Some pointed to technical and economic bottlenecks in recycling supply chains, while others noted robust EU quotas (e.g. Regulation 2023/1542) and industrial R&D working in tandem. The consensus was that recycling complexity poses hurdles, but will evolve in parallel with market growth rather than block it outright.	

Multi-level governance will pose challenges for coherent policy implementation in the electrified mobility sector		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	3	37.5%
Neutral	3	37.5%
Low	0	0%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	2	25%
High	2	25%

Neutral	3	37.5%
Low	0	0%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	Respondents split between warnings of fragmented rules across local, national and EU tiers, and observations that well-coordinated industry lobbying often aligns these layers behind common objectives. Both sides agreed that governance design and stakeholder coordination will materially shape rollout speed.	

Fiscal incentives and subsidies will boost consumer adoption of EV by 2050		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	3	37.5%
Neutral	1	12.5%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	1	12.5%
High	1	12.5%
Neutral	4	50%
Low	2	25%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	62.5%
Summary of qualitative comments	Nearly all saw purchase grants, tax breaks and related levers remaining a core part of the policy toolkit, especially for price-sensitive segments, even as schemes are refined to avoid market distortion and ensure fairness. Ongoing calibration will be needed to sustain momentum without creating long-term dependencies.	

The logistic sector will experience problems adopting EV on the road		
Likelihood	Number of answers	% over the total
Very high	2	25%
High	1	12.5%
Neutral	5	62.5%
Low	0	0%
Very low	0	0
Impact	Number of answers	% over the total
Very high	2	25%

High	1	12.5%
Neutral	4	50%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	37.5%
Summary of qualitative comments	Respondents pointed to range constraints, heavy-vehicle battery costs, sparse charging corridors for trucks and operational inflexibility as key hurdles. Yet several noted promising pilot projects in fast-charging and battery-swap systems, suggesting that targeted infrastructure and business-model innovations can mitigate the worst impacts.	

Regarding the batteries market, cobalt-based batteries will be left out		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	1	12.5%
Neutral	6	75%
Low	0	0%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	0	0%
High	2	25%
Neutral	5	62.5%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	12.5%
Summary of qualitative comments	Most doubted a complete phase-out of cobalt chemistries by 2050, given cobalt's performance advantages in energy density and stability. They saw cobalt's share shrinking as alternatives mature, but predicted that niche, high-performance applications will still call for it.	

The lack of infrastructure to charge the EV vehicles will be a challenge for the transition		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	2	25%
Neutral	1	12.5%
Low	1	12.5%

Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	1	12.5%
Neutral	1	12.5%
Low	2	25%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	62.5%
Summary of qualitative comments	Every respondent flagged gaps in public and home charging, especially in rural areas and among renters, as one of the sector's top risks. Without rapid, equitable network expansion, range anxiety and accessibility inequities will persist, slowing adoption.	

The whole value chain of EU automotive factories will be disrupted by 2050, due to changes in consumption patterns and the transition to EV		
Likelihood	Number of answers	% over the total
Very high	1	12.5%
High	3	37.5%
Neutral	2	25%
Low	1	12.5%
Very low	1	12.5%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	3	37.5%
Neutral	1	12.5%
Low	0	0%
Very low	1	12.5%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	50%
Summary of qualitative comments	All agreed that electrification, digitalization and circular-economy pressures will force OEMs and suppliers to retool factories, reorganize supply chains and retrain workforces. Early movers stand to gain competitive advantage, while laggards risk obsolescence.	

Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	4	50%
Neutral	0	0%

Low	1	12.5%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	25%
High	3	37.5%
Neutral	2	25%
Low	1	12.5%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	87.5%
Summary of qualitative comments	Respondents uniformly stressed that lacking reliable home-or-street charging, whether due to flat living or rural location, will impose higher costs, reduced convenience and mobility exclusion. High-speed DC cluster roll-outs help corridors but leave many areas underserved.	

Battery reuse and recycling will enable circular business models in the automotive sector		
Likelihood	Number of answers	% over the total
Very high	3	37.5%
High	3	37.5%
Neutral	0	0%
Low	2	25%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	37.5%
High	3	37.5%
Neutral	1	12.5%
Low	0	0%
Very low	1	12.5
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	75%
Summary of qualitative comments	Most saw second-life applications (grid or community storage) and closed-loop recycling creating new commercial services, reducing raw-materials dependency and opening fresh revenue streams. They viewed circular business models as a cornerstone of sustainable mobility strategies.	

6.4.3 Second round of interviews

6.4.3.1 Niches

Niches code	Summary of description	Literal citations
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<i>*Number of citations per category in brackets [X]</i>		<i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [9]	Interviewees highlight how economic models within the mobility sector are undergoing a profound transformation, especially as the traditional ownership paradigm shifts. The automotive industry is expected to remain profitable not through increased unit sales, but through emerging business models such as “car-as-a-service,” including leasing, renting, and multiple-use sales per unit. At the same time, developing countries with rising middle classes represent new markets for conventional combustion vehicles, potentially creating economic asymmetries in the global transition. The economic viability of the transition is also shaped by critical material dependencies and the cost of transitioning fleets. European mobility still heavily depends on private vehicles and on industries in car-centric countries such as Germany, making structural change both economically sensitive and politically entangled. Alternative measures — such as subsidizing retrofitting instead of new purchases — are being proposed as more efficient and equitable ways to support the transition. However, reorganising mobility systems at scale will require not just technological innovation but deep economic reconfiguration across supply chains and consumer habits.	<i>“Even if the automotive industry sells fewer units, they won’t become less profitable. The model is evolving: renting, leasing, they’ll sell the same car many times.” (15)</i> <i>“Developing countries with a rising middle class are new targets. Internal combustion cars could be sold there, keeping that niche alive.” (15)</i> <i>“All the money should go to retrofitting vehicles, not buying new ones. France is already doing this.” (15)</i> <i>“We still rely a lot on private vehicles. But Europe also relies on countries like Germany, which are car-based economies.” (16)</i> <i>“The cost of the transition matters. The challenge is to reorganise mobility.” (16)</i> <i>“We depend on the adaptation of industry and availability of critical materials.” (16)</i> <i>“Combustion vehicles might become a niche, and production may move to third countries.” (16)</i>
Niche_Governance [5]	Interviewees argue that the current governance approach to mobility is constrained by outdated paradigms. The transition is largely being pursued within a private vehicle ownership logic, which is increasingly seen as incompatible with long-term sustainability goals. Experts stress the need for a fundamental shift towards integrated mobility systems and public transport-oriented planning, going beyond technological fixes. Experts call for a rethinking of what “mobility” means, both at regulatory and normative levels. Existing policies risk excluding those who still depend on private cars, especially in areas with limited alternatives. They advocate for more inclusive planning frameworks that recognise different needs and facilitate a just transition. Regulatory pressure is already pushing toward retrofitting existing vehicles rather than simply promoting new electric car purchases, but broader governance reforms are still needed.	<i>“Right now, the transition is being done in the present paradigm; this does not work. There should be integrated mobility.” (15)</i> <i>“This is going to happen by 2050 — there’s going to be more retrofitting. There’s regulation pushing towards this.” (15)</i> <i>“Europe would need to rethink completely what mobility means.” (16)</i> <i>“We really need to understand who is more on the need to use private cars, so we don’t exclude them from the transition. Current policies are excluding these people.” (16)</i> <i>“He wants to change the mobility paradigm towards public transportation.” (16)</i>
Niche_Social [6]	Interviewees highlight that mobility in Europe is still deeply structured around the use of private vehicles,	<i>“What we need first of all is to reduce the number of</i>

	especially in car-based countries like Germany. However, this dominant model is being questioned by emerging narratives that call for a reduction in individual car use and a deeper rethinking of what mobility actually means. The social dimensions of the transition are particularly sensitive: experts warn that rural populations, migrants and middle classes risk being excluded from the benefits of the twin transition due to their dependence on private cars or lack of representation. There is also a cultural and political undercurrent: the rural-urban divide is reflected in electoral outcomes, where feelings of exclusion may reinforce opposition to green policies. Respondents argue that mobility policies must be more inclusive and attentive to territorial contexts, avoiding strategies that implicitly penalise those outside metropolitan areas. Without careful attention to social realities, the transition may exacerbate existing inequalities.	<i>individual vehicles.” (15)</i> <i>“Europe would need to rethink completely what mobility means.” (16)</i> <i>“European mobility relies a lot on private vehicles.” (16)</i> <i>“We can see that in electoral results; rural areas tend to vote for right-wing. They are expressing that they are being excluded from certain dynamics of twin transition that are more relevant for urban population.” (16)</i> <i>“We really need to understand the rural context in mobility. This transition can exclude people in the rural areas.” (16)</i> <i>“Migrants can’t vote. So there’s the need to safeguard middle classes. These are policies done towards safeguarding middle classes.” (16)</i>
Niche_Technological [9]	Technological transformation is widely seen as inevitable in the mobility sector, particularly in the shift to electric vehicles (EVs). Experts emphasise that around 90% of private vehicles are expected to become electric, and that electrification will also shape public transport (trains, trams, metros). However, this shift is not without constraints. The current battery technology still lacks sufficient range for long-use vehicles, such as airplanes or trucks, pushing innovation towards synthetic fuels and biofuels for specific niches. Interviewees also note that the sector’s transformation is deeply dependent on access to critical materials and minerals, many of which are not produced in Europe. Without breakthroughs in material science or large-scale battery recycling systems, this dependency may delay the transition. The connection between energy and mobility is also underlined as central to planning the transition effectively.	<i>“90% of private vehicles are going to be electric vehicles.” (15)</i> <i>“Long-range, long-usage vehicles, airplanes... the present battery technology does not allow that. They are moving towards synthetic vehicles or even bio-fuels.” (15)</i> <i>“Big connection between mobility and energy.” (15)</i> <i>“Right now, the transition is being done in the present paradigm; this does not work. There should be integrated mobility.” (15)</i> <i>“This is going to happen by 2050, there is going to be more retrofitting.” (15)</i> <i>“Certainly recycling batteries is a way to go; but he can’t say if there’s going to be a serious industry on that.” (16)</i> <i>“In 2050 we can see almost completely electric public transportation (especially through trains, trams, metros).” (16)</i>

6.4.3.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Political [3]	Political dynamics are playing a central role in shaping the mobility transition. Experts point out that subsidies and incentives often favour those who already have purchasing power, widening social inequalities. Meanwhile, growing discontent is visible in electoral behaviour, particularly in rural areas where populations feel excluded from the green transition. The erosion of middle-class stability further fuels these tensions, making the transition a politically delicate process that requires more equitable policy design.	<i>"We are subsidizing people that have purchasing power." (16)</i> <i>"We can see that in electoral results; rural areas tend to vote for right-wing." (16)</i> <i>"It's a political matter; middle classes are losing ground." (16)</i>
Regime _ Economic [1]	The mobility transition is limited by the strong economic dependence, particularly in countries whose industrial fabric depends on internal combustion engines. Experts underline that for these nations, the speed of the transition will be conditioned by industrial interests. While some countries support the green agenda, they do so cautiously, balancing climate goals with the risk of economic disruption. This tension shapes both national policies and the pace at which decarbonisation moves forward across Europe.	<i>"Some countries are interested in the transition but not in a fast pace." (16)</i>
Regime _Social [1]	Social dimensions of the current mobility regime show significant tensions between urban and rural populations. Existing transition strategies tend to prioritise urban contexts, where the benefits of electrification, shared mobility and infrastructure upgrades are more readily implemented, while rural communities risk being left behind. Perceptions of exclusion and unfairness are growing, especially among those who feel disconnected from the goals of the twin green and digital transition. These dynamics are contributing to political polarisation and raise important challenges for the design of inclusive transition pathways.	<i>"They are expressing that they are being excluded from certain dynamics of twin transition that are more relevant for urban population." (16)</i>
Regime_Technological [2]	Battery technologies remain insufficient for long-range or heavy-duty vehicles, leading to uncertainty in specific sub-sectors such as aviation or freight. Moreover, the structure of national industries plays a major role: countries like Italy, with a strong legacy in combustion engine manufacturing, are hesitant to accelerate the transition, as this would entail profound industrial restructuring. This underlines how technological regimes are embedded in broader economic and industrial systems that resist abrupt change.	<i>"Long-range, long-usage vehicles, airplanes... the present battery technology does not allow that." (15)</i> <i>"Some countries are interested in the transition but not in a fast pace (like Italy). Many industries are based in combustion engines." (16)</i>
Regime_Environmental [1]	Although electrification is often portrayed as a clean solution, experts highlight that the extraction and processing of critical minerals required for electric vehicles remain highly unsustainable. There is growing concern that the environmental burden is	<i>"The process of obtaining minerals necessary for electrification of vehicles is not sustainable. The whole</i>

	simply being displaced along the supply chain. This calls into question the very foundations of current transition pathways, and suggests a need to rethink what sustainability means in the context of mobility.	<i>procedure is not sustainable.” (15)</i>
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6.4.3.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape _Political [2]	The political landscape surrounding the mobility transition is increasingly influenced by the global availability of critical raw materials. Geopolitical tensions and supply chain dependencies make access to essential minerals a strategic concern, especially for Europe. The continent’s limited domestic supply and strong reliance on third countries pose significant vulnerabilities. This external pressure intersects with internal political decisions about how fast and how broadly to implement electrification, potentially delaying or reshaping transition strategies. Technological innovation (e.g., in mineral substitution) could ease some of these constraints, but is not guaranteed.	<i>“Availability of critical materials; in this geopolitical situation they become even more critical.” (16)</i> <i>“Unless there’s a technological breakthrough in minerals (we can use different minerals that are relatively available in Europe). We are much behind and dependent on third countries.” (16)</i>

6.4.3.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Finance [1]	Financial constraints emerge as a significant barrier in the transition towards sustainable mobility. According to interviewees, the cost of vehicle transformation (particularly electrification) may place an uneven burden on individuals or groups who are unable to afford new or retrofitted vehicles. This creates a risk of exclusion and delays in adoption. Addressing this vulnerability requires systemic thinking: not only reducing costs but also rethinking the structure and accessibility of mobility systems.	<i>“It will depend also on the cost of the transition of the vehicles; the issue is to reorganize mobility.” (16)</i>

6.4.3.5 Disadvantaged groups (DGs)

Experts highlighted several disadvantaged groups that risk being left behind in the mobility transition. Among them are communities located near mineral extraction zones, which bear environmental burdens without direct benefits. Another key group includes users who are structurally dependent on private vehicles (such as workers or caregivers) but lack alternatives. Rural residents are especially affected, due to longer distances and insufficient public transport infrastructure.

Disadvantaged groups (DGs) [4]	Summary of description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>

Car-dependent citizens	People who need to use private vehicles daily, including due to job or family obligations	Risk of exclusion if public policy only supports collective modes or bans combustion vehicles without alternatives	<i>“We really need to understand who is more on the need to use private cars to not exclude them from the transition.” (16)</i>
Rural population	Inhabitants of rural areas face long distances and poor access to public transport	Higher vulnerability to mobility transition measures that ignore spatial inequality and infrastructure gaps	<i>“People that live in rural areas are way more damaged; less public transportation and distances.” (16)</i>
Residents near battery components extraction sites	Communities living around areas where critical minerals are extracted for the mobility transition	These groups may suffer environmental degradation and social neglect while receiving few benefits from the transition itself	<i>“Disadvantaged groups: those who live around the places where the minerals are extracted.” (15)</i>
Users without space for EV infrastructure	Individuals already lacking infrastructure for electric vehicles (e.g., charging stations)	These groups risk being left behind or further marginalised in a system geared towards electrification	<i>“Having the infrastructure that allows electric cars so they do not fail is key. But they are already being excluded.” (16)</i>

6.4.4 Second round of Delphi survey

6.4.4.1 Elaboration of statements

107. By 2050, retrofitting combustion vehicles into electric powertrains will be the dominant strategy for decarbonizing private transport across Europe. **[Niche_Governance]**
[Niche_Technological]
108. European public transport fleets—urban and rural—will be fully electric. **[Niche_Technological]**
109. By 2050, Europe will rely primarily on diversified battery chemistries based on domestically available materials, significantly reducing its dependency on imported critical minerals **[Niche_Technological]** **[Niche_Governance]** **[Landscape_Political]**
[Landscape_Economic]
110. Integrated rural mobility hubs combining shared electric vehicles, electric buses, and on-demand microtransit services will transform mobility in non-urban areas and reduce transport

poverty. [Niche_Social] [Niche_Governance] [Niche_Technological] [Vulnerability_Finance] [Disadvantaged_Groups]

111. Pan-European reskilling programs will have retrained the combustion-engine workforce for roles in EV maintenance, smart-mobility services, and battery recycling] [Vulnerability_Skills] [Vulnerability_Employment] [Disadvantaged_Groups] [Niche_Technological]
112. By 2050, subscription-based “car as a service” models will be the default, instead of private cars. [Niche_Governance] [Niche_Economic] [Niche_Social]
113. By 2050, Europe will have established a circular battery economy in which end-of-life batteries are systematically reclaimed and processed through urban-mining and recycling facilities. [Niche_Technological]
114. By 2050, the long-haul road freight and aviation sectors will rely primarily on synthetic fuels and bio-derived alternatives, as battery technologies will remain unsuitable for those use cases. [Niche_Technological]

6.4.4.2 Analysis of results

Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	5	55.6%
Neutral	3	33.3%
Low	0	0%
Very low	1	11.1%
Impact		
Impact	Number of answers	% over the total
Very high	3	33.3%
High	3	33.3%
Neutral	1	11.1%
Low	2	22.2%
Very low	0	0%
Consensus and action		
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	55.6%
Summary of qualitative comments		
Participants agree this risk exists, particularly in low-density areas where extending charging networks is challenging. Although most EU regions have widespread electricity access, the lack of convenient charging points could deter adoption. Public and workplace charging, curbside facilities, and competitive suppliers could help, but inequities may remain. Those without infrastructure access might still rely on shared electric vehicles or public transport.		

Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050		
Likelihood	Number of answers	% over the total

Very high	4	44.4%
High	3	33.3%
Neutral	1	11.1%
Low	1	11.1%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	4	44.4%
High	3	33.3%
Neutral	2	22.2%
Low	0	0%
Very low	0	0
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	Respondents see this as inevitable, driven by EU policy and the need for strategic raw-material security, though scaling and permitting pose challenges. Some stress that mobility autonomy, through energy supply and autonomous vehicles, may be more important than domestic manufacturing. Battery recycling is expected to reduce emissions and improve material circularity, while domestic production could strengthen industrial relevance, especially against competition from Asian producers. Political commitment is viewed as decisive.	

The automatization of the traditional automotive manufacturing jobs will impact blue-collar workers		
Likelihood	Number of answers	% over the total
Very high	5	55.5%
High	2	22.2%
Neutral	1	11.1%
Low	1	11.1%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	3	33.3%
High	3	33.3%
Neutral	2	22.2%
Low	1	11.1%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	Automation in the automotive sector is already advanced, but further innovations and EV powertrain simplification will likely reduce some manual roles. While certain supervisory and specialist positions will remain, many routine tasks may be eliminated,	

	potentially affecting thousands of jobs. Reskilling, job redesign, and possibly social measures such as basic income are seen as essential to mitigate these effects.
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New business models such as Mobilit-as-a-Service (MaaS) models will be a reality in 2050		
Likelihood	Number of answers	% over the total
Very high	1	11.1%
High	5	55.5%
Neutral	2	22.2%
Low	1	11.1%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	4	44.4%
Neutral	5	55.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	66.7%
Summary of qualitative comments	Many see MaaS as already emerging in cities, offering flexible, app-based mobility solutions that simplify transport and reduce the need for private vehicles. Adoption will likely be strongest in urban environments with high service density. While these models could streamline mobility and reduce congestion, they will not fully replace ownership, which is expected to persist in less dense areas.	

Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs		
Likelihood	Number of answers	% over the total
Very high	3	33.3%
High	3	33.3%
Neutral	2	22.2%
Low	1	11.1%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	22.2%
High	3	33.3%
Neutral	3	33.3%
Low	1	11.1%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	66.7%

Summary of qualitative comments	This is considered a real barrier, especially for apartment dwellers and those without reserved parking. Home charging offers cost and convenience benefits, while public charging can be less predictable and more expensive. Although some could rely on workplace or public chargers, the gap in convenience may slow adoption among these groups.
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Battery reuse and recycling will enable circular business models in the automotive sector		
Likelihood	Number of answers	% over the total
Very high	1	11.1%
High	6	66.7%
Neutral	2	22.2%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	0	0%
High	7	77.8%
Neutral	2	22.2%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	While pilots have shown potential, scalability is questioned due to low population density, decentralised demand, and insurance or operational costs. Success depends on achieving financial sustainability and tailoring solutions to diverse European rural contexts.	

By 2050, retrofitting combustion vehicles into electric powertrains will be the dominant strategy for decarbonizing private transport across Europe		
Likelihood	Number of answers	% over the total
Very high	1	11.1%
High	2	22.2%
Neutral	1	11.1%
Low	2	22.2%
Very low	3	33.3%
Impact	Number of answers	% over the total
Very high	1	11.1%
High	3	33.3%
Neutral	3	33.3%
Low	1	11.1%
Very low	1	11.1%
Consensus and action	Is consensus reached?	Consensus degree

	No, remove	33.3%
Summary of qualitative comments	Participants see retrofitting as a nice alternative but only for niches; EVs, which will become more affordable, are going to be the main option and dominant path. This kind of retrofitting is not cost-effective. The only scenario where that would happen was if the final price was competitive enough to make people willing to pay	

European public transport fleets (urban and rural) will be fully electric		
Likelihood	Number of answers	% over the total
Very high	3	33.3%
High	3	33.3%
Neutral	2	22.2%
Low	0	0%
Very low	1	11.1%
Impact	Number of answers	% over the total
Very high	1	11.1%
High	5	55.5%
Neutral	2	22.2%
Low	0	0%
Very low	1	11.1%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	66.7%
Summary of qualitative comments	Full electrification is seen as most achievable in urban fleets, where technology and infrastructure are already advancing quickly. Rural and long-distance coach services face greater challenges due to range limitations and charging availability, although fuel-cell electric vehicles could help address these duty cycles. Governments are considered the key actors with the capacity and political will to drive full fleet electrification, supported by regulatory measures such as the 2035 ban on selling CO ₂ -emitting vehicles. Even without full replacement, substantial electrification could meet emissions targets. While this shift is important for the energy transition, several note that private transport remains a far larger contributor to CO ₂ emissions	

By 2050, Europe will rely primarily on diversified battery chemistries based on domestically available materials, significantly reducing its dependency on imported critical materials		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	3	33.3%
Neutral	3	33.3%
Low	3	33.3%
Very low	0	0%

Impact	Number of answers	% over the total
Very high	2	22.2%
High	3	33.3%
Neutral	2	22.2%
Low	1	11.1%
Very low	1	11.1%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	33.3%
Summary of qualitative comments	While battery technology is advancing rapidly, particularly in chemistries like lithium iron phosphate (LFP), sodium-ion, sulfur, and through improved recycling, achieving a supply that is “primarily domestic” is seen as highly ambitious. Constraints include limited mineral availability within Europe, as well as challenges in mining, permitting, and scaling production. Some believe imports will remain necessary, although diversification could substantially cut dependency and reduce transport emissions. Success would require a more unified EU strategy alongside significant investment in domestic research and innovation	

Integrated rural mobility hubs combining shared electric vehicles, electric buses, and on-demand microtransit services will transform mobility in non-urban areas and reduce transport poverty		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	5	55.5%
Neutral	2	22.2%
Low	2	22.2%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	
High	3	
Neutral	3	
Low	1	
Very low	0	
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	55.6%
Summary of qualitative comments	While pilots have shown potential, scalability is questioned due to low population density, decentralised demand, and insurance or operational costs. Success depends on achieving financial sustainability and tailoring solutions to diverse European rural contexts.	

Pan-European reskilling programs will have retrained the combustion-engine workforce for roles in EV maintenance, smart mobility services, and battery recycling		
Likelihood	Number of answers	% over the total

Very high	1	11.1%
High	6	66.6%
Neutral	1	11.1%
Low	1	11.1%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	22.2%
High	6	66.6%
Neutral	1	11.1%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	Ongoing shifts toward EVs and hydrogen mobility are already driving workforce changes. Large-scale reskilling, supported by policy funding and industry needs, is expected to be essential for competitiveness. Such programs could also facilitate acceptance of new vehicle technologies and help align skills with emerging manufacturing and service demands.	

By 2050, subscription-based “car as a service” models will be the default, instead of private cars		
Likelihood	Number of answers	% over the total
Very high	0	0%
High	2	22.2%
Neutral	2	22.2%
Low	4	44.4%
Very low	1	11.1%
Impact	Number of answers	% over the total
Very high	0	0%
High	3	33.3%
Neutral	5	55.5%
Low	0	0%
Very low	1	11.1%
Consensus and action	Is consensus reached?	Consensus degree
	No, remove	22.2%
Summary of qualitative comments	Most respondents doubt it will completely replace ownership, given cultural attachment to car possession and rural mobility needs. Growth is expected in urban MaaS and subscription models, which could reduce ownership rates, but private fleets will likely remain significant.	

By 2050, Europe will have established a circular battery economy in which end-of-life batteries are systematically reclaimed and processed through urban-mining and recycling facilities		
Likelihood	Number of answers	% over the total
Very high	1	11.1%
High	6	66.6%
Neutral	2	22.2%
Low	0	0%
Very low	0	0%
Impact	Number of answers	% over the total
Very high	2	22.2%
High	6	66.6%
Neutral	1	11.1%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	Recycling mandates, second-life uses, and urban mining are expected to make circular battery systems mainstream. Proper management of used batteries would reduce environmental impacts and raw-material demand. Political and industrial commitment is considered critical to realising these benefits.	

By 2050, the long-haul road freight and aviation sectors will rely primarily on synthetic fuels and bio-derived alternatives, as battery technologies will remain unsuitable for those use cases		
Likelihood	Number of answers	% over the total
Very high	3	33.3%
High	4	44.4%
Neutral	1	11.1%
Low	0	0%
Very low	1	11.1%
Impact	Number of answers	% over the total
Very high	2	22.2%
High	2	22.2%
Neutral	5	55.5%
Low	0	0%
Very low	0	0%
Consensus and action	Is consensus reached?	Consensus degree
	Yes, maintain	77.8%
Summary of qualitative comments	E-fuels and biofuels are seen as more viable than batteries for aviation, while long-haul freight may adopt both electric and fuel-based solutions. Technological limits, especially in battery capacity, make sustainable fuels the most realistic path. Industry players are	

	expected to adapt their fuel production, though decarbonisation will depend on wider sectoral changes.
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6.4.5 Third round of interviews

6.4.5.1 Niches

Niches code <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Niche_Economic [3]	Economic perspectives in mobility point toward a gradual shift from traditional private vehicle ownership to more service-oriented and technology-driven models. Electric vehicles are seen as a competitive option thanks to mature technology and the availability of electricity distribution networks, though charging infrastructure still requires expansion. Future mobility could increasingly be shaped by concepts like “mobility as a service” and automated transport, both of which have the potential to reshape cost structures, investment priorities, and market opportunities.	“Electric vehicles, on the other hand, benefit from existing electricity distribution networks and more mature technologies, even if charging infrastructure still needs development” (24). “He foresees a potential shift toward mobility as a service” (24). “Automated transport such as robotaxis” (24).
Niche_Governance [4]	Governance in mobility involves both opportunities and constraints tied to infrastructure, regulation, and equity. Electric vehicles benefit from existing electricity distribution networks and mature technologies, though charging infrastructure still requires significant expansion. Urban residents may face specific challenges such as limited space for private chargers, pressure on public infrastructure, and higher parking and charging costs, yet they often have greater access to public transport and shared mobility options. Small logistics operators, like van drivers, risk being disproportionately affected by stricter emissions regulations. While public networks are improving, they must grow substantially to meet future demand. Furthermore, current government subsidy schemes tend to favor wealthier individuals able to pay upfront costs and navigate complex applications, creating a regressive impact that excludes lower-income groups.	“Public networks are improving, but must grow significantly to support widespread adoption” (24). “Electric vehicles, on the other hand, benefit from existing electricity distribution networks and more mature technologies, even if charging infrastructure still needs development” (24). “Public networks are improving, but must grow significantly to support widespread adoption” (24). “Government subsidy schemes often favor wealthier individuals who can afford upfront payments and navigate complex application procedures. This results in a regressive effect, excluding lower-income groups from benefits” (24).
Niche_Social [2]	Social dynamics in mobility are shaped by evolving transportation options and the potential disruption of traditional ownership models. Urban residents often have greater access to alternatives such as public transportation or car-sharing, reducing reliance on private cars. If autonomous vehicle technology develops at scale, it could further	“Urban dwellers may have more alternatives to car ownership, such as public transportation or car-sharing” (24). “If autonomous vehicle technology scales effectively, traditional car ownership

	accelerate this shift, especially in cities, although such a transformation remains uncertain.	<i>could decline, especially in urban areas. However, this outcome is still speculative” (24).</i>
Niche_Technological [12]	Technological pathways in mobility are defined by a strong momentum toward electrification, which is expected to be largely irreversible by 2050, though not implying full substitution of conventional vehicles. Electric vehicles are viewed as the most viable option for passenger transport, benefitting from existing electricity distribution networks and higher energy efficiency compared to hydrogen for light vehicles. Hydrogen fuel cells are considered more promising for heavy-duty applications such as trucks and buses, while skepticism persists about their viability for passenger cars due to lower efficiency, greater infrastructure demands, and challenges in scaling green hydrogen production. Current infrastructure remains insufficient for mass EV adoption, with notable gaps in charging availability, especially in private residences. Emerging technologies such as Mobility as a Service (MaaS) and autonomous transport could further disrupt ownership models, although their large-scale adoption remains uncertain. Additionally, long-term supply chain dependencies for critical raw materials are expected, and technological advances in aviation and maritime decarbonisation remain limited.	<i>“The shift toward vehicle electrification as largely irreversible by 2050, even if it may not mean full substitution” (24).</i> <i>“Hydrogen fuel cells are also being explored, particularly for heavy-duty vehicles like trucks and buses” (24).</i> <i>“I’m skeptic about hydrogen’s viability for light vehicles, due to lower energy efficiency, greater infrastructure requirements, and difficulty producing green hydrogen at scale” (24).</i> <i>“Electric vehicles benefit from existing electricity distribution networks and more mature technologies, even if charging infrastructure still needs development” (24).</i> <i>“There is consensus that current infrastructure is insufficient for a large-scale EV transition” (24).</i>

6.4.5.2 Regimes

Regimes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Regime_Technological [1]	Technological change within the current mobility regime is marked by steady progress in electrification across multiple transport modes. Rail transport is already largely electrified, while significant advancements are also being made in buses and trucks, indicating a gradual but consistent integration of low-emission technologies into established transport systems.	<i>“Electrification is already prevalent in rail transport and progressing in buses and trucks” (24).</i>

6.4.5.3 Landscapes

Landscapes <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Landscape_Political [2]	Political factors in mobility are closely tied to the geopolitics of supply chains. The sector’s reliance on critical raw materials such as lithium creates strategic vulnerabilities, as Europe is unlikely to achieve full self-sufficiency. This dependency	<i>“Geopolitical aspects may shift supply chains” (24).</i>

	heightens exposure to geopolitical tensions and trade disruptions, which could shape policy priorities and influence the pace and cost of the mobility transition.	
Landscape _ Economic [2]	Economic dynamics in the mobility sector are increasingly shaped by the geopolitics of supply chains. The transition to electric vehicles (EVs) will continue to rely heavily on raw materials such as lithium, for which Europe lacks self-sufficiency. This structural dependency is expected to persist, leaving the sector exposed to global market fluctuations and potential geopolitical disruptions that could impact availability and cost.	<i>“Europe is unlikely to become fully self-sufficient in raw materials for EVs (like lithium), predicting continued global supply chain dependencies unless major geopolitical disruptions occur” (24).</i>

6.4.5.4 Vulnerability dimensions

Vulnerability dimensions <i>*Number of citations per category in brackets [X]</i>	Summary of description	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Vulnerability_Employment [2]	Employment vulnerabilities in mobility center on two main groups. Automotive industry workers in Europe face uncertainty from both the structural shift toward electrification and the geopolitical volatility affecting supply chains for critical materials. At the same time, small logistics operators, such as van drivers, may find it difficult to meet tightening emissions regulations, risking job losses or costly fleet upgrades during the transition phase.	<i>“Automotive industry workers, particularly in Europe may be affected due to geopolitical shifts in supply chains” (24). “Small logistics operators, like van drivers, who may struggle to comply with emissions regulations during the transition phase” (24).</i>
Vulnerability_Finance [4]	Financial vulnerabilities in mobility stem from both infrastructure limitations and inequitable policy design. Urban residents often face higher costs for parking and charging, compounded by limited access to private charging stations. Government subsidy schemes tend to favor those with higher incomes, as they can afford upfront costs and navigate complex application processes, leaving lower-income groups excluded from the benefits of the transition. As a result, some individuals risk being priced out of car ownership entirely.	<i>“Limited space for private charging stations” (24). “Higher costs associated with urban parking and charging” (24). “Low-income individuals, who may be priced out of car ownership” (24).</i>
Vulnerability_Skills [2]	Skills-related vulnerabilities in mobility are linked to the technological shift toward electric vehicles and stricter emissions regulations. Traditional mechanics face declining demand for their services, as EVs require less maintenance and fewer repairs. Additionally, small logistics operators may lack the skills or resources to adapt to new compliance requirements, creating further barriers to participation in the transition.	<i>“Mechanics, as EVs require fewer repairs and maintenance” (24). “Small logistics operators, like van drivers, who may struggle to comply with emissions regulations during the transition phase” (24).</i>

6.4.5.5 Disadvantaged groups (DGs)

Disadvantaged groups in mobility reflect the uneven social and economic impacts of the sector's transition toward electrification and new mobility models. Urban residents may face physical and financial constraints in adapting to electric vehicle use, while low-income groups are disproportionately excluded from incentives and ownership due to structural barriers. Workers in traditional automotive and logistics sectors risk job loss or reduced income without targeted reskilling measures, and small operators may struggle to meet tightening emissions regulations. Without inclusive policy design and equitable infrastructure development, these groups are at risk of being left behind in the mobility transition.

Disadvantaged groups (DGs) [6]	Description	Impact	Literal citations <i>*Number of interview in parenthesis (), see section 3.2</i>
Urban residents	Limited access to private charging and higher costs in cities	Increased difficulty adopting EVs due to infrastructure and financial constraints	<i>"Urban residents may face greater challenges: Limited space for private charging stations, increased pressure on public infrastructure, higher costs associated with urban parking and charging" (24)</i>
Low-income individuals	Excluded from subsidies and priced out of ownership	Reduced access to benefits of transition, risk of transport exclusion	<i>"He also critiques government subsidy schemes, which often favor wealthier individuals who can afford upfront payments and navigate complex application procedures. This results in a regressive effect, excluding lower-income groups from benefits" (24)</i>
Automotive industry workers	Vulnerable to supply chain shifts	Job insecurity from geopolitical changes and restructuring of industry	<i>"Automotive industry workers, particularly in Europe, due to geopolitical shifts in supply chains" (24)</i>
Mechanics	Reduced maintenance needs for EVs	Potential job losses and need for reskilling	<i>"Mechanics, as EVs require fewer repairs and maintenance" (24)</i>
Small logistics operators	Struggle to comply with new emissions rules	Financial strain and potential market exit	<i>"Small logistics operators, like van drivers, who may struggle to comply with emissions regulations during the transition phase" (24)</i>

7 Scenario building results

As outlined in subsection 5, this section corresponds to step 6 of the analytical procedure, which focuses on the conceptualisation of three alternative futures for 2050. For each sector, the Delphi survey statements are first classified according to whether they represent positive or negative dynamics in relation to the twin transition. These classifications provide the basis for the elaboration of three internally coherent scenarios: optimistic, neutral and pessimistic.

The optimistic scenario describes a future in which the twin transition has been successfully managed, resulting in an inclusive and sustainable transformation where negative impacts have been mitigated. The neutral scenario depicts a partial achievement of transition goals, where progress coexists with persistent structural challenges, the emergence of disadvantaged groups, and unintended consequences. The pessimistic scenario portrays a context in which exogenous shocks have disrupted the regime, leading to regressive developments and severe negative impacts on disadvantaged groups.

Within the Multi-Level Perspective (MLP) framework, the optimistic scenario corresponds to a transformation pathway, in which deep structural changes occur across the niche, regime and landscape levels. The neutral scenario aligns more closely with an optimisation pathway, where existing systems adapt and improve without undergoing a fundamental transformation. The pessimistic scenario reflects a reconfiguration pathway, where certain elements are redesigned in response to pressures, but the underlying structures remain misaligned with the objectives of a fair and inclusive twin transition.

7.1 Housing and built environment

Classification of statements

Positive statements (supporting inclusive and sustainable twin transition)	Negative statements (opposing an inclusive and sustainable twin transition)
By 2050, the construction sector will be dominated by assembly and automation roles	By 2050, concrete will remain the dominant construction material
Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies	Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050
	Energy prices will significantly influence incentives for adopting sustainable housing solutions

Table 1: Housing and built environment Scenarios 2050 – Classification of Delphi survey #2 statements

Optimistic scenario – Twin transition takes place in the housing and built environment sector by 2050

By 2050, Europe's built environment is a model of climate resilience and health. Cities and towns have been reshaped through decades of investment in deep renovation and human-centered urban planning based on sustainable design. The housing stock is unrecognisable from the early 21st century; inefficient

buildings have been replaced or retrofitted into spaces that are comfortable year-round, regardless of the climate hazards.

All homes meet or exceed the highest performance standards for insulation and ventilation, Natural light is maximised through intelligent design, and passive heating and cooling systems maintain comfort without mechanical strain. Materials are locally sourced, and chosen for their circular life cycle, so that components can be reused or recycled without waste. In apartment complexes, green roofs and living walls are the norm, improving biodiversity and urban air quality while regulating building temperatures. Rainwater harvesting and greywater recycling are integrated into every building, reducing demand on municipal supplies and improving resilience.

Urban form has shifted toward mixed-use, walkable districts that integrate housing, services, and leisure spaces. Streets are shaded with trees and permeable pavements, cooling cities and preventing flooding during heavy rains. Public squares double as community gathering places and stormwater retention areas, designed to be both beautiful and functional. Parks and green corridors connect neighbourhoods, creating habitats for wildlife and safe spaces for people to meet and walk (or cycle).

Accessibility is universal. Homes and public buildings are barrier free, with wide corridors and adaptable interiors that can be easily modified to suit changing mobility needs. Wayfinding systems use clear signage to assist people with visual or cognitive impairments. For older adults and people with disabilities, assistive technologies are integrated into living spaces, from voice-controlled lighting and ventilation to fall-detection systems and responsive home layouts.

Affordability has been protected through long-term housing policy. Renovation subsidies and Inclusionary Zoning (IZ) ensure that low- and middle-income households benefit from transformation, avoiding the displacement that once accompanied urban renewal. In rural areas, buildings that were not used have been repurposed into modern housing clusters that support community life and provide access to essential services without long travel distances. Formerly neglected districts have been revitalised with high-quality, climate-adapted housing alongside cultural and commercial spaces that sustain local economies.

Neutral scenario – Partial achievement of the twin transition, coexistence of the new and the old (2050)

By 2050, Europe's built environment has seen substantial improvements, yet its transformation remains uneven. Many cities have well-renovated housing and climate-resilient public spaces, but the depth and quality of these changes vary widely between regions. In leading areas, older building stock has been retrofitted to high performance standards (insulated façades, airtight windows, advanced ventilation systems...) which create comfortable interiors with reduced environmental impact. In other parts of the continent, however, renovations have been partial or delayed, leaving some homes still vulnerable to extreme weather and high maintenance costs. Material choices have been central to the divergence. In forward-moving regions, procurement policies have favoured local sourcing and recycling content, creating circular supply chains that keep carbon low. In other areas, budget constraints have led to retrofits with mixed-quality materials, some of which have limited life spans or higher environmental impacts, which will be translated into future maintenance costs-

Urban areas have embraced a mixed pattern of redevelopment. City centers often feature attractive, walkable districts with integrated green infrastructure. Yet in less well-funded municipalities, urban heat islands persist, and public realm improvements remain modest. Access to high-quality housing is also inconsistent: in thriving regions, affordability policies and inclusive zoning have kept costs stable, but elsewhere, gentrification and limited supply have pushed low- and middle-income residents to peripheral areas with fewer services.

Rural settlements have benefitted in pockets from adaptive reuse of old buildings and the creation of compact housing clusters close to essential services. In some areas, underused administrative buildings have been converted into modern community spaces, revitalising small towns. In other rural zones, however, depopulation has left housing stock deteriorating, with insufficient investment to reverse decline.

Accessibility has advanced, but progress is uneven. In well-resourced districts, barrier-free design is standard: step-free entrances, adaptable interiors, tactile signage, and smart navigation aids serve people of all ages and abilities. Elsewhere, retrofitting for accessibility has been slow, particularly in older buildings with structural constraints and funding gaps have delayed upgrades. Public spaces show similar disparities. Many metropolitan areas have created shaded pedestrian corridors and neighborhood squares that host markets and events, strengthening social cohesion. But in some towns and city outskirts, public spaces remain fragmented, lacking safe walkways or spaces for community gathering.

Pessimistic scenario – Negative trends and landscape shocks (2050)

By 2050, Europe's housing and built environment reflect decades of fragmented policy and insufficient investment, accompanied by escalating climate pressures. While some projects stand as symbol of ambition, they are exceptions in a place where much of the building stock remains outdated and inefficient. The renovation wave of the early 21st century stalled midway, leaving millions of homes with poor insulation and obsolete ventilation systems. Extreme heat in summer and cold snaps in winter make these homes uncomfortable and expensive to maintain, putting pressure on household finances and public health systems.

Material choices have been driven more by short-term cost-cutting than by sustainability or resilience. In many areas, low-grade imports have replaced durable and low-carbon options, leading to rapid deterioration and a cycle of constant repair. The absence of coherent circular-economy policies means demolition waste continues to pile up, while valuable materials are discarded instead of reused. Some urban districts have become patchworks of crumbling façades and mismatched retrofits, undermining both visual cohesion and community pride.

Urban planning has failed to keep pace with climate risks. Many city centers still rely on impermeable pavements and poorly shaded streets, making them hotspots for both flooding and heat stress. In wealthier areas, residents have invested privately in innovations such as cooling technologies and rooftop gardens; elsewhere, the lack of adaptation has left entire neighbourhoods prone to repeated climate-related damage. Public spaces in underfunded municipalities are scarce or degraded, with broken pavements or poorly maintained parks.

Accessibility remains inconsistent. While some newer developments include barrier-free entrances and adaptable interiors, many older housing remain inaccessible to people with reduced mobility. Lack of lifts and narrow staircases, as well as absence of visual aids, limit participation in community life for older adults, disabled people and parents with small children. Rural areas face additional challenges, as depopulation has left housing stock to decay, and limited resources mean essential accessibility upgrades are rare.

Affordability has worsened in many urban centers. Housing shortages, combined with speculative investment in prime locations, have driven up rents and prices, pushing low- and middle-income residents to peripheral areas with limited infrastructure and poor-quality housing. Informal settlements have emerged in areas around cities in some regions, where residents occupy unfinished or abandoned buildings with minimal services.

7.2 Agriculture and food

Classification of statements

Positive statements (supporting inclusive and sustainable twin transition)	Negative statements (opposing an inclusive and sustainable twin transition)
Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector	Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems
Cloud-based platforms will deliver end-to-end “farm-to-fork” traceability visible to consumers	Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labour
Irrigation networks will use remote sensing and AI to schedule watering autonomously	
By 2050, most family farms will join cooperatively governed data hubs	
By 2050, edge-computing nodes (e.g. 5G base stations) will cover nearly all cultivated land, enabling real-time AI on remote farms	
No farm under 10 ha will be able to adopt precision technologies without joining a formal support network of service providers or cooperatives	
EU and/or national agricultural digitalization schemes will include mandatory regional customization clauses to reflect local climate, soil and crop differences	

Table 1: Agriculture 2050 – Classification of Delphi survey #2 statements

Optimistic scenario – Twin transition takes place in the agriculture and food sector by 2050

By 2050, Europe's agricultural sector embraced both sustainability and digitalisation, creating a system that is productive, resilient, and fair. Farms of all sizes use advanced tools to manage resources efficiently and meet high environmental standards. Cross-sector alliances between agriculture, clean energy, and technology have become common, pooling expertise and funding to deliver solutions that were once out of reach for smaller producers. These partnerships have driven the spread of innovations such as bioenergy-powered greenhouses, solar-powered irrigation, and digital tools for monitoring soil health.

Cloud-based traceability platforms now connect farms directly to consumers, making it possible for shoppers to see where their food comes from, how it was produced, and its environmental footprint. This transparency has built trust and opened new markets for farmers who meet high sustainability criteria, including premium prices for certified products. Smaller farms benefit by differentiating their produce and reaching customers willing to pay for quality and accountability.

Water use has become far more efficient thanks to irrigation networks guided by remote sensing and AI. Satellites, IoT sensors, and weather forecasts feed into autonomous systems that deliver water only when and where it is needed. This has cut costs, reduced waste, and helped farming adapt to more frequent droughts. Even in traditionally dry areas, crop yields are stable, and farmers are less dependent on expensive emergency water supplies.

Family farms thrive within cooperatively governed data hubs, which act as shared knowledge centers and service providers. These hubs give members access to AI-driven crop models, carbon market participation, and affordable insurance. The cooperative structure ensures that decisions are made locally and benefits are shared fairly. Small farms, once at risk of falling behind, have equal access to the same cutting-edge tools as larger operations.

Edge-computing coverage now extends to almost every field, enabling real-time AI for farm management. Autonomous tractors, drones, and robotics operate with precision, reducing input costs and improving productivity. Even the smallest farms can use precision technologies by joining formal support networks or cooperatives, which provide shared access to machinery, training, and maintenance. This arrangement has kept technology adoption high while preventing smaller producers from being left behind.

Policies now require agricultural digitalisation schemes to be tailored to local conditions, recognising that a vineyard in southern Spain has different needs from a dairy farm in northern Germany. This regional customisation has made digital tools more relevant, increasing their adoption and impact. Advisory services are staffed with local experts who understand the specific climate, soil, and crop challenges of each region.

For disadvantaged subgroups, the changes have been tangible. Low-income rural households have stable incomes through cooperative membership and access to premium markets. Migrants are employed in skilled roles operating and maintaining advanced agricultural systems, supported by training in both technology and language. Older farmers have remained active by working within cooperatives that handle the most complex digital tasks, while providing them with tools they can use confidently. People with disabilities find more accessible employment opportunities in roles involving monitoring, data analysis, and cooperative management. Rural communities once left behind now enjoy better infrastructure, stronger local economies, and improved food security.

Agriculture in 2050 is no longer divided between the winners and losers of change. Instead, it is a connected, adaptive system where sustainability and technology work together, and where even those in remote or small-scale settings share in the benefits of the twin transition.

By 2050, Europe's agricultural landscape shows progress toward sustainability and digitalisation, but the picture is uneven. Many farms have adopted advanced tools and more sustainable practices, while others continue with older methods, creating a patchwork of progress. Cross-sector alliances between agriculture, clean energy, and technology exist but are concentrated in certain regions and sectors. In areas with strong investment and political support, these partnerships have brought more efficient production and innovative practices. In other regions, limited funding and fragmented networks have slowed change.

Cloud-based traceability systems are widely used by large and export-oriented farms, helping them reach premium markets with detailed information on product origin and sustainability. However, smaller farms without the resources or capacity to meet certification standards often operate outside these platforms, selling into local markets where traceability is less valued. The gap between farmers able to benefit from consumer demand for transparency and those left in traditional supply chains has not been fully closed.

Irrigation networks that use AI and remote sensing operate in many regions, especially those prone to drought, but not everywhere. Some farmers still rely on conventional systems, either because of cost barriers or limited access to the infrastructure needed. As a result, water efficiency has improved overall, but unevenly, and some areas still face stress during prolonged dry spells.

Family farms participate in cooperatively governed data hubs in several Member States, gaining access to shared tools and services. Yet membership is far from universal, and many small farms remain outside these structures, either due to mistrust, lack of awareness, or reluctance to share data. Where hubs function well, they improve competitiveness and sustainability. Where they do not exist or are poorly managed, farms struggle to keep pace with technological change.

Edge-computing coverage enables real-time AI in many agricultural areas, but there are still gaps in remote and mountainous regions. Farmers in well-connected areas use autonomous equipment and data-driven decision-making, while those in poorly served zones rely on less precise methods, creating disparities in productivity and efficiency.

Policies for regional customisation of digitalisation schemes are in place, but their implementation varies. In some countries, adaptation to local climate and soil conditions have made tools more relevant and effective. In others, generic approaches have reduced their usefulness, leading to low adoption.

The impact on disadvantaged subgroups is mixed. Low-income rural households in regions with strong cooperative structures benefit from stable income and better market access, while those in less connected areas see fewer gains. Migrants continue to work in agriculture, but many remain in seasonal manual roles because training for higher-skilled positions is uneven. Older farmers adapt better in areas where cooperatives provide technical support, but in less organised regions they face challenges in meeting new requirements. People with disabilities have more opportunities on some modernised farms, but accessibility is still inconsistent.

In this 2050, agriculture reflects both the achievements and the missed opportunities of the twin transition. Progress is visible, but uneven adoption, infrastructure gaps, and persistent inequalities prevent the sector from fully realising its potential for inclusivity and sustainability.

Pessimistic scenario – Negative trends and landscape shocks (2050)

By 2050, Europe's agricultural sector is struggling to adapt to the demands of sustainability and digitalisation. Progress has been uneven and, in many places, slow. Small farms face rising compliance costs tied to environmental regulations and certification systems. For many, meeting the requirements for eco-schemes, carbon reporting, and traceability is financially and administratively overwhelming. Larger farms, with dedicated staff and greater resources, absorb the changes with less difficulty, widening the gap between big and small producers. Some smallholdings have closed, while others have been absorbed by

bigger operations, leading to reduced diversity in rural systems and the erosion of traditional farming knowledge.

Automation in farming has advanced rapidly, replacing a large proportion of manual crop work, especially among seasonal and migrant labour. While this has improved efficiency, it has also displaced many low-skilled workers. In regions that once depended on seasonal agriculture for employment, job losses have driven migration, economic decline, and social strain. Few retraining programs have been implemented, and those that exist are often mismatched to the local labour market.

Other positive developments that could have bridged the gap have failed to reach their potential. Cross-sector alliances remain small in scale and are mostly limited to high-tech farms. Cloud-based traceability systems exist but are inaccessible to many smallholders who lack the funds or digital infrastructure to join them. Irrigation systems using AI and remote sensing operate mainly in wealthy regions, leaving others vulnerable to worsening droughts. Cooperatively governed data hubs function in pockets of Europe, but in many areas, they have not been established or have collapsed due to lack of trust and funding.

Edge-computing coverage is far from universal. In many rural and mountainous areas, connectivity remains poor, preventing farmers from using autonomous machinery or real-time decision-making tools. This leaves them less productive and less competitive in increasingly globalised markets. Policies requiring regional customisation of digitalisation schemes have been poorly implemented, leading to tools that do not reflect local climate and crop needs, further discouraging adoption.

The consequences for disadvantaged subgroups are severe. Low-income rural households face shrinking income opportunities and higher production costs, pushing many out of farming entirely. Migrants, once a significant part of the agricultural workforce, have been displaced by automation without access to new employment pathways. Older farmers are disproportionately affected by digitalisation requirements, with many unable to adapt and therefore excluded from subsidy schemes. People with disabilities and those in remote areas find even fewer opportunities in agriculture, as most accessible roles are concentrated in regions with advanced infrastructure.

The sector’s vulnerability to climate change has worsened. Without widespread adoption of smart irrigation or precision agriculture, yields fluctuate with extreme weather. Droughts, floods, and heatwaves regularly disrupt production, reducing food security and increasing dependence on imports. Rural depopulation accelerates as young people leave for better prospects elsewhere, leaving behind ageing communities with limited capacity to maintain agricultural output.

In this 2050, the twin transition in agriculture has faltered. The promise of technology and sustainability has been realised only in isolated pockets, while large parts of the sector remain trapped in a cycle of high costs, low productivity, and exclusion. Inequalities have deepened, and the gap between the best-equipped farms and those left behind has become a defining feature of the rural landscape.

7.3 Energy

Classification of statements

Positive statements (supporting inclusive and sustainable twin transition)	Negative statements (opposing an inclusive and sustainable twin transition)
The transition towards wind and solar energy will continue accelerating	EU energy dependency on third countries for critical materials

	(e.g., lithium, rare earths) will be a strategic vulnerability
The main factors that will lead the energy transition will be the increase of prices, geopolitical aspects and environmental concerns	Grid resilience will face new challenges as illustrated by the solar-related blackout in Spain
The energy labor market will experience an increase in demand for highly skilled jobs, such as engineers	
The R&D&I on battery recycling will be a decisive factor in EU energy independence by 2050	
New financial models will arise due to the energy market incentives created by private companies and their technification	
Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumers rights	

Table 1: Energy Scenarios 2050 – Classification of Delphi survey #2 statements

Optimistic scenario – Twin transition takes place in the energy sector by 2050

By 2050, Europe's energy system is almost entirely powered by wind and solar. Offshore wind farms stretch along the coasts, producing large amounts of electricity. Solar panels cover rooftops, building walls, parking areas and farmland, often combined with crops in agrivoltaic systems. These renewable sources meet nearly all electricity demand across the continent. The challenges that once came with such high levels of renewables, like unstable voltage, low grid inertia or risk of blackouts, have been solved. Large-scale batteries, green hydrogen storage and smart grid technology keep the system stable. Automatic controls and real-time monitoring mean the grid can adapt before problems appear, even during extreme weather.

The shift to this system began decades ago. Rising fossil fuel prices made renewable energy more attractive. Political tensions over energy imports showed how risky dependency could be. People have become more aware of the need to protect the environment, pushing governments to act. Rules that used to delay renewable projects were changed, making it easier and faster to build. Clear plans were set to reduce and then stop the use of fossil fuels. The cost of renewable technology has dropped steadily, helping more people and businesses to invest.

Digital platforms are now a normal part of daily life. Every household and business can check where their energy comes from, how much they are using and how much it costs. Many store extra power in batteries and sell it back to the grid at times of high demand, earning extra income. These platforms are easy to use, and strict rules protect people's personal data. Because of this, trust in the system is high.

One of the biggest changes has been in battery recycling. Batteries are now made to be repaired, reused or taken apart so that their materials can be recovered. This has reduced Europe's need to import lithium,

rare earths and other critical materials. Large recycling centers have opened in towns that once depended on coal mines or heavy industry. These centers provide good jobs for former fossil-fuel workers, newly trained technicians and young graduates in engineering and manufacturing.

The benefits are shared widely. Low-income households live in well-insulated and energy-efficient homes. Many have their own solar panels or are part of community-owned energy projects that lower their bills and sometimes pay them a share of the profits. Rural communities run local wind farms and storage systems, using the income to improve services like schools, healthcare and internet access. People who rent their homes, once excluded from most of these benefits, now join through shared ownership programs, virtual net metering or subscription-based clean energy plans.

In this future, the system is designed for inclusion. Migrants and people who do not speak the local language well can access training and support to find jobs in the energy sector. Older people and those with disabilities use simple displays, voice controls or other adapted tools to manage their energy use. Energy and digital skills are taught in schools and offered to adults, making it possible for everyone to take part.

Europe in 2050 reached its climate goals and built an energy system that is clean, reliable and fair. Electricity is affordable for all, jobs are secure and spread across regions, and communities have more control over their energy future. The twin transition has brought environmental protection and social fairness together, showing that both can be achieved at the same time.

Neutral scenario – Partial achievement of the twin transition, coexistence of the new and the old (2050)

In 2050, Europe's energy landscape shows clear progress, but the transformation is uneven. Renewable energy from wind and solar provides a significant share of electricity, and in some regions it is the dominant source. Offshore wind farms operate efficiently along many coasts, and solar panels cover public buildings, homes, and farmland. Yet, fossil fuels still play an important role in the system. Gas plants are kept as backup for periods of high demand or low renewable generation, and some coal facilities remain active in countries that have struggled to replace them.

The advances that have made high renewable penetration possible are present, but not everywhere. In well-connected areas, large-scale batteries, hydrogen storage, and advanced control systems keep the grid stable. Elsewhere, infrastructure is older and less flexible, making it harder to manage fluctuations. Extreme weather events sometimes lead to temporary supply interruptions in these regions, reminding people that reliability is still uneven.

Digital tools have changed how many households and businesses interact with the energy system. In urban centers and connected rural areas, people monitor their energy use in real time, store extra power, and sell it back to the grid when prices are high. In these places, dynamic pricing schemes and automated systems help balance supply and demand. However, participation is not universal. Rural residents in areas with weak internet connections, older adults unfamiliar with digital platforms, and migrants facing language barriers often cannot use these services fully.

Battery recycling has improved and now provides a steady supply of materials like lithium and rare earth, reducing but not removing Europe's reliance on imports. Disruptions in supply still occur, and when they do, they can slow down new renewable projects and raise costs. Employment patterns also vary. Regions that embraced renewables early enjoy strong job markets in engineering, maintenance, and manufacturing. Other areas, particularly those that were slower to attract investment, continue to see higher unemployment and limited opportunities for workers from traditional energy sectors.

For specific groups, the picture is mixed. Low-income households in proactive municipalities benefit from energy-efficient homes, solar access, and stable bills. In less engaged regions, they face higher costs and fewer support programs. Renters gain access to shared solar projects and virtual net metering in some

areas, but elsewhere their landlords have not invested in energy improvements, leaving them with high bills. Rural communities in connected zones run profitable local projects and reinvest in public services, while others without such infrastructure miss out on these gains. Migrants in some regions find training and jobs in the energy sector, while in others they remain excluded by language and recognition barriers. Former fossil-fuel workers have transitioned successfully where retraining programs are linked to local jobs, but many in slower-moving regions work in lower-paid or insecure jobs.

This Europe has made important steps toward a clean and modern energy system, but progress is uneven. Renewables and digitalisation have improved efficiency and reduced emissions, yet fossil fuels still fill important gaps. The benefits of the transition are visible, but they do not reach all communities equally, leaving parts of society still waiting for the full promise of the twin transition.

Pessimistic scenario – Negative trends and landscape shocks (2050)

In 2050, Europe's energy system is under strain. The shift to renewables has slowed, and fossil fuels remain a large part of the mix. Some wind and solar projects have expanded, but their growth has been held back by delays in infrastructure, supply chain disruptions, and political disagreements over funding. Several countries still rely heavily on gas and coal to meet demand, especially during winter peaks. The result is a system that is less clean, less reliable, and more vulnerable to external shocks.

Import dependency on critical materials like lithium and rare earths remains high. Global tensions and export restrictions have led to repeated shortages, delaying battery production and renewable deployment. Costs have risen, making new projects harder to finance. Instead of achieving energy independence, Europe faces regular supply disruptions that force governments to restart or extend the life of fossil-fuel plants. These decisions are framed as necessary for stability, but they keep emissions high and slow the move toward climate goals.

Digitalisation has advanced unevenly. Some urban areas have smart grids and automated energy management, but much of the system still relies on outdated infrastructure. Large parts of the population cannot participate in dynamic pricing schemes or benefit from grid-connected storage. The lack of strong data governance in some countries has also led to privacy concerns and public mistrust, reducing the uptake of digital energy services.

The job market shows deep divides. High-skilled positions in advanced renewable technology are concentrated in a few economic hubs, while many regions have seen job losses from the closure of fossil-fuel industries without equivalent opportunities to replace them. Retraining programs have been patchy and often poorly matched to the needs of local economies. Former fossil fuel workers in these areas face long-term unemployment or accept lower-paid, insecure jobs.

The social impact of these failures is significant. Low-income households face higher and more unstable energy bills, as subsidies are inconsistent and fossil fuel prices fluctuate. Rural communities in less developed regions have limited access to renewable projects and remain dependent on expensive imported fuels. Renters are often left behind because landlords have not invested in energy efficiency or clean energy, keeping their bills high. Migrants and people with limited language skills face additional barriers to joining the energy workforce or accessing support programs. Older adults and people with disabilities struggle with poorly designed interfaces and a lack of tailored assistance, leaving them excluded from digital services that could reduce their costs.

Frequent extreme weather events have exposed the fragility of the system. Heatwaves, storms, and floods have caused repeated power outages in vulnerable areas. Without adequate investment in grid resilience, recovery from these events is slow, deepening inequalities between well-prepared regions and those left without resources.

This is a Europe where the twin transition has not delivered its promise. Renewable deployment has stalled, fossil fuel dependence persists, and the benefits of technology are concentrated in a few places. Many communities are locked in cycles of high costs, limited job opportunities, and vulnerability to climate impacts. The gap between those who can participate in and profit from the energy system and those who cannot has widened, turning the energy transition into a source of division rather than a driver of unity.

7.4 Mobility and transport

Classification of statements

Positive statements (supporting inclusive and sustainable twin transition)	Negative statements (opposing an inclusive and sustainable twin transition)
Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050	The automatization of the traditional automotive manufacturing jobs will impact blue-collar workers
New business models such as Mobilit-as-a-Service (MaaS) models will be a reality in 2050	By 2050, the long-haul road freight and aviation sectors will rely primarily on synthetic fuels and bio-derived alternatives, as battery technologies will remain unsuitable for those use cases
Battery reuse and recycling will enable circular business models in the automotive sector	
Pan-European reskilling programs will have retrained the combustion-engine workforce for roles in EV maintenance, smart mobility services, and battery recycling	
By 2050, Europe will have established a circular battery economy in which end-of-life batteries are systematically reclaimed and processed through urban-mining and recycling facilities	

Optimistic scenario – Twin transition takes place in the mobility and transport sector by 2050

By 2050, Europe's mobility system is an integrated, low-emission network that is both inclusive and sustainable. Electrification dominates urban and intercity travel, supported by diverse technologies that reflect strategic decisions made decades earlier. Electric vehicles account for nearly all private car sales, but the shift is not limited to ownership: subscription-based "car as a service" models and well-established Mobility-as-a-Service platforms allow access to shared, on-demand fleets in every major city. These platforms integrate booking and payment, as well as routing, across trains, buses, shared EVs,

micromobility, and demand-responsive shuttles, replacing much of the friction once found between modes. While private cars still exist, they are fewer and cleaner, and they are integrated in a system optimised for connectivity and efficiency.

Urban and rural transport fleets are fully electric, reducing both noise and air pollution. In non-urban areas, integrated rural mobility hubs have transformed access to work and education. These hubs combine shared electric vehicles, regular buses, on-demand mobility, and micromobility options. They are designed with universal accessibility in mind and have reduced transport poverty in areas that were once underserved.

Circular economy principles are characteristic of the automotive value chain. Domestic battery production, using diversified chemistries based on locally available materials, has reduced dependency on imported critical minerals. En-of-life batteries are reclaimed and processed through advanced recycling facilities, many located in towns that once depended on combustion-engine manufacturing. This has created stable jobs in battery recycling and second-life applications.

These shifts are supported by long-term investment in reskilling. Pan-European programs have retrained former blue-collar workers in EV maintenance, battery diagnostics, and smart-mobility service. Training is linked directly to hiring pipelines, ensuring that former fossil-fuel automotive workers and those from declining logistics subsectors move into stable employment without prolonged unemployment.

For low-income households, targeted subsidies and affordable leasing models remove upfront cost barriers. Shared EV schemes, supported by municipal and cooperative ownership, give them access to electric mobility without the cost of ownership. Rural communities benefit from the mobility hubs, with coordinated timetables reducing long travel times to jobs and services. Renters in dense urban areas, once excluded from EV incentives due to lack of private charging, gain from widespread charging in neighborhood-based shared fleets. Migrants and people with limited bureaucratic literacy are supported by one-stop mobility centers that provide multilingual assistance for subsidy applications and account set-up for MaaS platforms, reducing exclusions. Older adults enjoy barrier-free vehicles and priority, easy-to-use booking systems across fleets, while people with disabilities benefit from universal design standards included in every vehicle and hub.

Neutral scenario – Partial achievement of the twin transition, coexistence of the new and the old (2050)

In 2050, Europe's mobility landscape reflects both the successes and limits of the twin transition. Electrification and digitalisation have advanced substantially, but the pace and consistency of change vary widely depending on the market and geographical scope. Electric vehicles dominate new car sales in many urban areas, supported by robust charging networks and subscription-based "car as a service" models, but in smaller cities and rural zones, internal combustion vehicles (often running on hybrid or biofuel blends) remain common, coexisting with EVs. MaaS platforms are well-developed in major metropolitan regions, offering integrated booking for trains, buses, shared EVs, and micromobility. In other areas, fragmented ticketing systems and limited operator participation prevent full integration.

Public transport electrification is significant but incomplete. While most urban bus fleets are battery-powered and quiet, some interurban routes and regional lines still rely on diesel or LNG because of infrastructure and cost constraints. Rural mobility hubs exist in certain countries, but their coverage is uneven. In underserved regions, car dependence remains high, reinforcing disparities in access to services and jobs.

Battery recycling and domestic production have grown, but dependence on imported materials persists. Circular-economy initiatives recover a portion of lithium, cobalt, and nickel from end-of-life batteries yet

processing capacity is concentrated in a few industrial clusters. This creates regional imbalances in job creation, with some former automotive-manufacturing areas thriving while others face prolonged adjustment periods. Reskilling programmes have reached any, but participation rates are uneven, and training does not always match local job demand, leading to some workers accepting lower-paid roles in unrelated sectors.

For low-income households, targeted subsidies and affordable leasing exist in some member states, but eligibility criteria and application complexity, as well as budget limits, keep adoption rates modest. Many remain dependent on older, less efficient vehicles, exposing them to higher operating costs. Rural communities benefit when mobility hubs are funded, yet in many places public transport remains infrequent, forcing continued reliance on private cars. Renters without private parking see improved access to chargers in cities with strong policy frameworks, but in less regulated markets, charging availability lags EV growth. Migrants and people with limited bureaucratic literacy gain mobility when local authorities provide support for MaaS apps and subsidy applications, but in many regions such assistance is ad hoc or NGO-led. Older adults experience greater comfort and accessibility on modernised fleets where investment has been made, yet in other areas booking systems remain barriers. People with disabilities are well-served in flagship MaaS networks and urban transport upgrades, but universal design standards are inconsistently enforced outside major cities. They can access mobility systems through physical ticket offices and phone booking in some systems, but in many cases digital-only platforms create silent exclusion.

Pessimistic scenario – Negative trends and landscape shocks (2050)

By 2050, the promise of an inclusive and sustainable mobility transition has not been realised. Electrification has advanced in some wealthy urban sites, but large parts of Europe still rely on older combustion vehicles, often running on imported fossil fuels or low-grade biofuel blends. Fragmented regulation and insufficient investment have stalled the rollout of charging infrastructure, leaving many regions as “charging deserts”. The absence of a coordinated approach has locked in a two-speed transition: metropolitan centers enjoy modern fleets and digital integration, while rural and peri-urban areas remain dependent on outdated, polluting vehicles.

MaaS platforms exist in name but fail to deliver integration. Private operators guard data, and public systems lack the capacity to enforce interoperability. As a result, passengers must juggle multiple apps and payment systems. Public transport electrification is incomplete; diesel buses and ageing trains are still in service on many regional and rural routes. The few rural mobility hubs that were launched early in the transition have closed due to underfunding, and worsening transport poverty for low-density communities.

Battery production remains heavily dependent on imported critical minerals, making the sector vulnerable to price spikes and supply interruptions. Circular-economy goals for battery reuse and recycling were not met; most end-of-life batteries are processed abroad, losing both materials and job opportunities. Automation in vehicle manufacturing has accelerated without adequate reskilling programmes, leaving large numbers of former automotive and logistics workers unemployed or in precarious, low-paid jobs.

For low-income households, the high upfront cost of EVs remains prohibitive. Subsidies either no longer exist or are structured in ways that favour wealthier buyers who can afford initial payments and navigate complex applications. Operating older combustion cars exposes these households to volatile fuel prices and stricter emissions charges, deepening financial strain. Rural communities suffer from the collapse of mobility hubs and long gaps in public transport coverage, forcing continued dependence on expensive private vehicles. Renters without private parking face near-total exclusion from EV ownership, with charging spots limited to wealthier districts. Migrants and those with limited bureaucratic literacy are disproportionately excluded from incentives and subsidies, as assistance services have been cut. Older

adults can't use digital booking and payments systems that have replaced physical ticketing, and can't use MaaS services, which have shifted to digital-only formats as well, cutting them off from modern mobility options. People with disabilities face inconsistent or absent accessibility features in vehicles and stations.

8 Limitations

While the combined use of semi-structured interviews and the Delphi survey enabled broad sectoral and geographical coverage, several limitations should be acknowledged.

First, there were different rates of engagement across sectors and countries, resulting in uneven sample sizes. The housing and built environment sector showed the lowest engagement from experts compared to agriculture, energy, and mobility & transport. Also, in the case of this sector, no third round of interviews was possible due to the lack of experts. This limited the depth of the qualitative analysis for housing and built environment, as fewer perspectives were available to validate or contrast preliminary findings. Consequently, the insights from this sector should be interpreted with caution, as they may not fully capture the diversity of viewpoints present in other sectors.

At the country level, Hungary provided 2 experts and for the Delphi survey only. Despite the translation of the Delphi survey into Hungarian and the partner's efforts to facilitate participation, the political context of the country and its relationship with EU projects posed a significant obstacle. This situation limits the representativeness of perspectives from that national context.

Second, there was an over-representation of Spanish experts, with Spain contributing with a sample of 33 (38.37%). While this reflects strong stakeholder mobilisation in that country, it may have influenced the relative weight of Spanish perspectives in the aggregated findings.

Third, while the Delphi survey generated a substantial volume of qualitative feedback, the level of detail and quality of comments varied considerably between experts. Some provided rich, nuanced justifications for their responses, while others offered brief or minimal explanations, potentially affecting the depth of analysis.

Fourth, an expert bias within each sector was observed, as participants tended to emphasise issues and priorities linked to their own sub-field of expertise. This is a natural outcome of engaging specialised professionals but may lead to underrepresentation of other relevant dimensions within the same sector.

Fifth, researcher bias cannot be entirely excluded, as the interpretation and synthesis of qualitative data involve subjective judgement despite the application of a consistent coding framework by 2 different researchers.

Sixth, despite adapting to the DoA requirements, the co-creation of the scenarios with the expert's groups was not considered relevant due to the tight timing and quality of delivered results, as they were co-designed during the Delphi survey and interviews rounds.

Finally, in the context of scenario building, there is a potential optimism bias: experts may unconsciously describe what they would like to see happen rather than what they realistically expect to occur. This aspirational framing, while valuable for envisioning desirable futures, may skew the balance between exploratory and normative elements in the scenarios.

9 Conclusions

The findings from Deliverable D3.2 confirm that the twin transition will reshape sectoral systems in profound and interconnected ways across agriculture and food, energy, transport and mobility, and housing and the built environment. The scenarios developed for 2050 reveal that these transformations will be driven by a combination of technological innovation, regulatory evolution, environmental imperatives, and shifting socio-economic patterns. Together, these forces will determine not only the pace and direction of change but also their distributional impacts on different social groups.

A central conclusion is that anticipating these transformations through structured foresight exercises is essential for enabling proactive and inclusive policymaking. The co-designed scenarios offer a diverse range of plausible futures, each highlighting different opportunities, as well as distinct challenges. This diversity makes clear that no single trajectory is inevitable and that strategic decisions taken in the present can substantially influence long-term outcomes.

The identification of disadvantaged groups within each sector's scenario provides an important early step toward integrating equity considerations into policy design. Thanks to the mapping of how structural changes may affect specific populations, this analysis equips policymakers with actionable insights for developing targeted measures to mitigate exclusion risks and ensure that the benefits of the twin transition are broadly shared.

This work demonstrates the value of using participatory processes to create scenarios based on uncertainties. The scenarios developed in D3.2 are tools for reflection and strategic preparation, helping stakeholders to understand possible system evolutions and to design interventions that can guide transitions toward socially just and economically viable futures. This deliverable thus lays the groundwork for subsequent FITTER tasks that will deepen the analysis of disadvantaged groups and translate insights into practical policy recommendations.

10 Annexes

The open-ended questions for the semi-structure interviews are built upon D2.1 and the outcomes of T3.1.

Objective: To explore possible future scenarios for 4 sectors in 2030, 2040 and 2050 by identifying key drivers, uncertainties, disruptions and alternative development paths under the twin transition. These insights will inform Delphi survey questions.

General guidelines:

Need for open-ended question to stimulate imaginative and prospective answers

Encourage long-term thinking under uncertainty

Introduction to the interviewee:

Thank you for taking the time to participate in this interview. I am part of the FITTER-EU project, which aims to explore the potential future impacts of the twin transition—the combined digital and green transition—on key sectors, including agriculture, energy, transport & mobility, and housing & construction.

The objective of this interview is to gain insights from experts like you to anticipate key drivers, uncertainties, and disruptions that may shape these sectors by 2030, 2040, and 2050. Your expertise will help us construct possible future scenarios, identifying both opportunities and challenges that could arise.

This interview follows a foresight approach, meaning we are interested in forward-looking, open-ended discussions rather than definitive predictions. We encourage you to think broadly and imaginatively about how different factors—technological, economic, political, social, and environmental—may evolve and interact over time.

The interview will last approximately 60 minutes and consists of three main parts:

Horizon scanning – Identifying key factors and disruptions shaping the future.

Scenario exploration – Understanding how different drivers and uncertainties may interact.

Implications – Discussing the possible winners, losers, and policy responses.

Your responses will remain confidential and will be used exclusively for research purposes within the FITTER project. If at any point you have additional insights or would like to clarify anything, please feel free to share.

Blocks of questions for the interviews:

SECTOR X – Policy objective X

Horizon scanning [What?]:

Exploration of factors (assumptions and disruptions)

Objective: Identify key **disruptions and assumptions** as factors shaping the scenarios of the digitalization of agriculture.

GIVENS – General Assumptions

Taking the EU **current technological context** into account – can you list please - what factors may be still important with regard to the digitalisation of agriculture up to 2050?

Taking the EU **current political context** into account - what factors are important with regard to the digitalisation of agriculture up to 2050?

Taking the EU **current economic context** into account - what factors may be still important with regard to the digitalisation of agriculture up to 2050?

Taking the EU **current environmental context** into account - what factors may be still important with regard to the digitalisation of agriculture up to 2050?

Taking the EU **current social context** into account – what factors may be still important with regard to the digitalisation of agriculture up to 2050?

Of all of the mentioned factors - which do you believe that have a high influence on the digitalisation of agriculture and we can be **reasonably certain** they will occur or develop in predictable ways by 2050?

DRIVERS – Risks and uncertainties

Among the remaining factors mentioned, or any new ones you wish to introduce, which do you consider **most uncertain** for the digitalisation of agriculture? (DRIVERS)

Note to interviewer: cover again the PESTLE dimensions

In other words, what **assumptions** for the digitalisation of agriculture might no longer be valid in 2050

Scenario building [How?]

Objective: Connect **drivers and uncertainties**, exploring interactions and alternative development paths.

You have mentioned X factors, how do these X factors **relate** to each other? Would the **implications** of either of them be amplified if they co-occurred?

What **new opportunities** or challenges could arise from these interactions?

Implications [Who]

Objective: Identify winners, losers, and key policy responses in different future scenarios.

Who would be the **winers and losers** in a highly digitalized agricultural future (in other words; looking at multiple intersected socio-economic characteristics, which population cohorts will benefit the most and which ones will be negatively impacted)?

What specific policies, programs or initiatives would need to be newly created and/or adjusted if this scenario was to play out?

Note: Thank the interviewee for their time. Ask them if there is anything they would like to add.

Semi-structure interviews script

10.5 Round 1

[Common structure all iterations]

FITTER-EU Delphi survey on future scenarios for Europe by 2050: Insights from sectorial experts to achieve the twin transition objectives

This survey is conducted as part of the FITTER-EU project, funded by the European Commission under grant agreement 101132546. The project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition in Europe. Your input will provide valuable insights into future practices (within a timeframe of 25 years) in one or more of the following economic sectors: Energy, Transport and Mobility, Agriculture and Food, and Housing and the Building Environment. It will help us identify the level of expert consensus on potential future scenarios for Europe in 2050 in relation to those sectors. The statements that you will evaluate have been developed based on insights from semi-structures expert interviews. The results of this survey will be published in the form of project reports and peer-reviewed journal articles and presented at research conferences. This survey follows an iterative approach and participation will be across a number of cycles.

Instructions

How to respond: For each statement, please assess the following:

- The likelihood of the scenario happening, in Europe, in the next 25 years (by 2050)
- The impact that the scenario would have if it occurred to the European housing and building environment sector. (You can also provide qualitative comments to explain, nuance or justify your answers)
- Qualitative comment (Optional): Please briefly explain your reasoning, add nuances or suggest conditions under which this might (or might not) happen (write a dash "-" in case you do not want to add anything).

Estimated time: 7-15 minutes

Confidentiality: Your responses will be pseudoanonymized and used exclusively for research purposes. No individual will be identifiable from the research findings that are shared or published.

Participation: Your participation in this survey is voluntary. You may freely decline in answering any questions or end your participation at any time.

Contact: If you have any questions about this survey or what it entails, please contact the survey team (laura.armayones@eurecat.org and/or lluis.vine@eurecat.org) and we will respond within 3 working days.

Informed consent

By continuing to the first question, you are indicating your consent to taking part in this research and agreeing with the following points:

- I have read and fully understand the purpose of the current survey.
- I have been provided with the opportunity to contact researchers with any questions or concerns I may have before completing the survey.
- I understand that I can withdraw at any point during the survey (up to the completion of the survey).
- I am fully aware that my information and answers will be pseudoanonymized.
- I consent to my responses being used in the current research and future publications.
- I give my full consent to participate in the current study.
- I consent that my data can be used to contact me in relation to this research study and its outcomes

Please indicate whether you give your consent:

Demographic questions:**Email****Gender:**

Female

Male

Non-binary

Prefer not to say

Current role or position (e.g., Project Engineer, Researcher, Policy Advisor...)

Country of residence:

[Different for each iteration]

10.5.1 Energy

1. The transition towards wind and solar energy will continue accelerating.
 - a. **LIKELIHOOD:**
 - b. **IMPACT:**
 - i. Very low, Low, Neutral, High, Very high
 - c. **Qualitative comment**
2. The main factors that will lead the energy transition will be the increase of prices, geopolitical aspects and environmental concerns.
3. By 2050, more than 50% of European households will participate as prosumers (producing all their energy for self-sustaining and selling the excess).
4. Regulatory and bureaucratic barriers (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050.
5. The shortage of skilled labor (e.g., installers, technicians) will significantly hinder the energy transition.

6. By 2050, energy bills will be more difficult to understand for the average consumer, increasing the risk of individuals not understanding and potentially paying extra charges.
7. Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses.
8. Non-renewable energy producers and energy distributors will be key players in the renewable energy market, due to its diversifying strategy.
9. The future of the energy sector will be renewable 100% by 2050
10. The labor market will experience an increase in demand for highly skilled jobs, such as engineers.
11. The R&D&i on battery recycling will be a decisive factor in EU energy independence by 2050.
12. Activism in the energy sector will create obstacles for large-scale renewable projects, particularly wind farms.
13. New financial models will arise due to the energy market incentives created by private companies and its technification.
14. Highly specialized workers in technologies with high substitution potential (e.g., certain solar panel models) will be affected by innovations in the energy market.
15. Fossil fuel phase-out policies will generate long-term regional unemployment risks in areas historically dependent on extractive industries.
16. By 2050, decentralized energy systems will dominate the European grid, reducing reliance on large-scale centralized infrastructure.
17. Households without solar panels will face higher long-term energy costs, exacerbating inequality.
18. Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumer rights.
19. By 2050, energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs.
20. EU energy dependency on third countries for critical materials (e.g., lithium, rare earths) will be a strategic vulnerability.

10.5.2 Mobility and transport

1. By 2050, electric vehicles (EVs) will account for at least 90% of new private vehicles purchases in Europe.
2. Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption.
3. Fiscal incentives for EVs will be redesigned to better target low-income households.
4. Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050.
5. Urban areas citizens will face more challenges than rural areas citizens in the transition to EV.
6. Mechanics without proper knowledge to repair electric engines will be negatively impacted by the transition towards electric mobility sector.
7. Limited knowledge of bureaucratic processes and financial literacy constitutes a significant vulnerability related to EV acquisition.
8. The automatisisation of the traditional automotive manufacturing jobs will impact blue-collar workers.
9. Autonomous vehicle technology will remain marginal in urban transport until after 2050.
10. New business models such as Mobility-as-a-Service (MaaS) models will be a reality in 2050.
11. The challenges of battery recycling will hinder the growth of the EV market in 2050.
12. Multi-level governance will pose challenges for coherent policy implementation in the electrified mobility sector.
13. Fiscal incentives and subsidies will boost consumer adoption of EV by 2050.

14. The logistic sector will experience problems adopting EV on the road.
15. Regarding the batteries market, cobalt-based batteries will be left out
16. The lack of infrastructure to charge the EV vehicles will be a challenge for the transition.
17. The whole value chain of EU automotive factories will be disrupted by 2050, due to changes in consumption patterns and the transition to EV.
18. Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs.
19. Battery reuse and recycling will enable circular business models in the automotive sector.

10.5.3 Agriculture and food

Consider that the EU policy objective guiding this section is the **digitalization of agriculture**. Please assess the **likelihood** and **impact** of the following statements based on your expertise in the sector.

1. By 2050, fewer than 30% of small farms (below 10 hectares) will be able to independently adopt precision agriculture technologies without external support.
2. The twin transition will accelerate land concentration, reducing the number of family farms across Europe by 2050.
3. Data collected by farmers will increasingly be monetized by third parties unless regulatory frameworks ensure data ownership and value sharing.
4. Blockchain technology will significantly increase transparency and trust in agricultural supply chains by 2050.
5. By 2050, rural depopulation will be a reality in the EU.
6. Technological investment in agriculture (AI, biotech) will primarily benefit large agribusinesses. (i.e. non-adapted to local context)
7. Generalist digitalization policies (i.e. non-adapted to local context) in agriculture will be a paradigm by 2050.
8. Data-driven agriculture will be a reality in 2050, with efficient and interoperable data lakes and data hubs as key components of modern farming.
9. Small farmers will increasingly shift to fine food, local production and proximity-based distribution to stay competitive.
10. Large investment funds will own more than 65% of the fields in Europe by 2050.
11. Urban consumers' demand for high-quality, traceable food will be a major driver of small farmers' adoption of new business models.
12. Artificial intelligence will be widely used in decision-making processes for crop and livestock management in Europe by 2050.
13. Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems.
14. Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labor.
15. Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector.
16. Large companies buying land and retaining workers will become a key strategy to de-risk digital transitions for small farms.
17. By 2050, farmers older than 55 years old will lack the digital skills needed to comply with agricultural digitalization policies.

18. Displaced rural workers will face long-term unemployment and exclusion from the digital economy.
19. By 2050, the agriculture sector will experience a lack of generational renewal.
20. Smart farming technologies will become embedded in agricultural policy compliance frameworks, making tech adoption a condition for subsidies.

10.5.4 Housing and built environment

1. By 2050, retrofitting subsidies will require high bureaucratic knowledge and financial literacy.
2. Public housing will lead on the adoption of sustainable building materials and design compared to the private sector.
3. Buildings highly vulnerable to climate events (e.g., flood-prone zones) will increasingly be abandoned or relocated by 2050.
4. By 2050, concrete will remain the dominant construction material.
5. Technological complexity in smart housing systems (e.g., windows, energy dashboards) will create usability gaps among populations older than 55 years old and/or non-tech savvy population.
6. By 2050, the construction sector will be dominated by assembly and automation roles.
7. The demand for traditional low-skill labor (e.g. construction workers) will decrease by 2050.
8. By 2050, sustainable materials (e.g., ceramics, wood) will represent a significant share of materials used in construction.
9. Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households.
10. The shift to zero-emission buildings will be primarily driven by rising energy prices rather than climate awareness.
11. The majority of citizens of EU countries will be unable to participate in retrofitting initiatives.
12. Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050.
13. The New European Bauhaus framework from the European Commission will significantly influence urban regeneration and social housing design.
14. Energy prices will significantly influence incentives for adopting sustainable housing solutions.
15. Cultural attitudes (e.g. having the windows opened during cold times of the year) towards climate-adaptative housing will result in maladapted behavior, such as forcing opening the windows in automatic and self-regulated smart windows systems.
16. Sustainable architecture will be more scalable and affordable by 2050.
17. Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies.
18. The construction workforce will increasingly include profiles with less physical strength and more technical training, enabling greater inclusion of women and people with disabilities.
19. By 2050, climate-adaptive housing technologies will be underutilized by residents.
20. Urban development plans will increasingly include climate relocation strategies for housing built in high-risk areas where adaptation is no longer feasible.
21. Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation.

10.6 Round 2

10.6.1 Energy

1. The transition towards wind and solar energy will continue accelerating.
 - a. **LIKELIHOOD:**
 - b. **IMPACT:**
 - i. Very low, Low, Neutral, High, Very high
 - c. **Qualitative comment**
2. Regulatory and bureaucratic barriers (e.g., delays on permits) will remain the main bottleneck for renewable energy deployment across EU countries in 2050.
3. The shortage of skilled labor (e.g., installers, technicians) will significantly hinder the energy transition.
4. The main factors that will lead to the energy transition will be the increase of prices, geopolitical aspects and environmental concerns.
5. Low-skilled workers in extractive industries (coal, oil and gas sectors) will face job losses.
6. Non-renewable energy producers and energy distributors will be key players in the renewable energy market, due to its diversifying strategy.
7. The energy labor market will experience an increase in demand for highly skilled jobs, such as engineers.
8. The R&D&i on battery recycling will be a decisive factor in EU energy independence by 2050.
9. New financial models will arise due to the energy market incentives created by private companies and its technification.
10. Fossil fuel phase-out policies will generate long-term regional unemployment risks in areas historically dependent on extractive industries.
11. Decentralized energy systems will dominate the European grid, reducing reliance on large-scale centralized infrastructure.
12. Households without solar panels will face higher long-term energy costs, exacerbating inequality.
13. Energy systems will be increasingly integrated with digital platforms, demanding strong data governance to protect consumer rights.
14. Energy literacy will become essential for consumers to actively participate in dynamic pricing schemes and smart grid programs.
15. EU energy dependency on third countries for critical materials (e.g., lithium, rare earths) will be a strategic vulnerability.
16. By 2050, energy security will consistently be prioritized over sustainability in EU energy policy.
17. The acceleration of renewable energy deployment will have continued, and grid resilience will face new challenges as illustrated by the solar-related blackout in Spain.
18. The sharing of energy-sector knowledge and technology will have ceased between the EU and the rest of the world.

10.6.2 Mobility and transport

1. Individuals without access to private or public charging infrastructure will be structurally excluded from EV adoption.
2. Battery recycling and domestic production will become critical strategic sectors for EV autonomy by 2050.
3. The automatization of the traditional automotive manufacturing jobs will impact blue-collar workers.

4. New business models such as Mobility-as-a-Service (MaaS) models will be a reality in 2050.
5. Citizens without private garages or access to public charging points will be disadvantaged in the transition to EVs.
6. Battery reuse and recycling will enable circular business models in the automotive sector.
7. By 2050, retrofitting combustion vehicles into electric powertrains will be the dominant strategy for decarbonizing private transport across Europe.
8. European public transport fleets—urban and rural—will be fully electric.
9. By 2050, Europe will rely primarily on diversified battery chemistries based on domestically available materials, significantly reducing its dependency on imported critical minerals
10. Integrated rural mobility hubs combining shared electric vehicles, electric buses, and on-demand microtransit services will transform mobility in non-urban areas and reduce transport poverty.
11. Pan-European reskilling programs will have retrained the combustion-engine workforce for roles in EV maintenance, smart-mobility services, and battery recycling
12. By 2050, subscription-based “car as a service” models will be the default, instead of private cars.
13. By 2050, Europe will have established a circular battery economy in which end-of-life batteries are systematically reclaimed and processed through urban-mining and recycling facilities.
14. By 2050, the long-haul road freight and aviation sectors will rely primarily on synthetic fuels and bio-derived alternatives, as battery technologies will remain unsuitable for those use cases.

10.6.3 Agriculture and food

1. Small farms will be affected by rising compliance costs tied to environmental regulations and certification systems.
2. Automation in farming will significantly reduce the number of manual crop workers by 2050, especially among seasonal and migrant labor.
3. Cross-sector alliances (e.g., agriculture, clean energy or pharma) will drive innovation and accelerate digitalization in the agri-food sector.
4. Large companies buying land and retaining workers will become a key strategy to de-risk digital transitions for small farms.
5. By 2050, farmers older than 55 years old will lack the digital skills needed to comply with agricultural digitalization policies.
6. Displaced rural workers will face long-term unemployment and exclusion from the digital economy.
7. By 2050, the agriculture sector will experience a lack of generational renewal.
8. Smart farming technologies will become embedded in agricultural policy compliance frameworks, making tech adoption a condition for subsidies.
9. Cloud-based platforms will deliver end-to-end “farm-to-fork” traceability visible to consumers
10. Irrigation networks will use remote sensing and AI to schedule watering autonomously.
11. By 2050, most family farms will join cooperatively governed data hubs.
12. Small farms will earn regular income from digitally managed ecosystem services (carbon credits, pollination).
13. By 2050, edge-computing nodes (e.g. 5G base stations) will cover nearly all cultivated land, enabling real-time AI on remote farms.
14. By 2050, farm management systems will bundle weather-indexed insurance via smart contracts.
15. Most small farmers will join pan-European digital cooperatives to coordinate production and market access across Member States.

16. By 2050, public agencies and private ag-tech firms will jointly fund regional centers that deliver continuous digital literacy and business training.
17. By 2050, receipt of any Common Agriculture Policy (CAP) subsidy will require farmers to complete a certified digital-skills programme.
18. No farm under 10 ha will be able to adopt precision technologies without joining a formal support network of service providers or cooperatives.
19. EU and national agricultural digitalization schemes will include mandatory regional customization clauses to reflect local climate, soil and crop differences.
20. National agricultural advisory services will integrate peer-coaching networks—pairing digitally proficient farmers with those needing upskilling.
21. All public funding for farm digitalization will flow through farmer-led cooperatives or community hubs to ensure equitable distribution and cost-sharing.

10.6.4 Housing and built environment

1. Buildings highly vulnerable to climate events (e.g., flood-prone zones) will increasingly be abandoned or relocated by 2050.
2. By 2050, concrete will remain the dominant construction material.
3. Technological complexity in smart housing systems (e.g., windows, energy dashboards) will create usability gaps among populations older than 55 years old and/or non-tech savvy population.
4. By 2050, the construction sector will be dominated by assembly and automation roles.
5. Regional inequalities in climate vulnerability (e.g., flood zones) will increasingly displace low-income households.
6. Access to climate-adapted housing will be strongly stratified by income, with wealthier households better protected from climate risks by 2050.
7. Energy prices will significantly influence incentives for adopting sustainable housing solutions.
8. Sustainable architecture will be more scalable and affordable by 2050.
9. Public tender processes will increasingly favor construction companies that demonstrate climate adaptive design strategies.
10. The construction workforce will increasingly include profiles with less physical strength and more technical training, enabling greater inclusion of women and people with disabilities.
11. Retrofitting practices will increasingly rely on prefabricated and modular construction techniques to reduce costs and speed implementation.
12. By 2050, no publicly funded housing will remain in high-risk flood or wildfire zones without certified adaptation measures or financed relocation plans.
13. All smart-home control systems will include an independently certified “easy mode” accessible to users aged 65 + and those with low digital literacy.
14. More than the half of construction workers and/or companies will hold an accredited certificate in climate-adaptive building techniques.
15. The majority of public retrofit subsidies will be paid before construction starts, removing liquidity barriers for households.
16. By 2050, deep-retrofit uptake will be similar among households regardless of income level.
17. By 2050, digital-twin modelling will enable residents of historic buildings to explore and approve retrofit options virtually, reducing resistance and speeding up decision-making.
18. By 2050, national renovation strategies will offer a graduated menu of upgrade bundles—from quick window replacement to full-envelope insulation—so households can choose a depth of retrofit that matches their means.

19. By 2050, time-of-use tariffs will automatically reward homes that achieve zero-emission performance, making energy savings a stronger driver than up-front subsidies.
20. By 2050, climate-aware behaviour programmes will be embedded in schools and community centers, shifting everyday practices—such as habitual winter window-opening—toward energy-efficient norms.

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